# Bibliometric study of EPFL (2014–2023)

An open bibliometric analysis of the institutions of the ETH Domain

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**Report for the ETH Board**


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**General parameters of the bibliometric report**

| | |
|---|---|
| Database | OpenAlex (Articles, Reviews and Proceedings) |
| Version | v03_eth_domain |
| Classification system | Publication-level classification system (approximately 4,516 topics nested within 252 subfields, 26 fields and four domains) |
| Publication window | 2014-2023 |
| Temporal aggregation | Four-year overlapping intervals |
| Counting method | Fractional counting at the level of institution for citation impact and performance measurement |
| Citation window | 2014-2025 |
| Citation normalisation level | Subfields |
| Author self-citations | Included |

## Executive summary

As set out in the mandate for the intermediate evaluation of the ETH Domain, the ETH Board commissioned a bibliometric study of the six institutions of the ETH Domain. For the first time, the ETH Board mandated an internal team composed of data scientists from ETH Zurich and EPFL to carry out this study, relying entirely on open data, transparent workflows and reproducible code. This study thus aims not only to fulfil the requests set for the intermediate evaluation, but also to demonstrate the ETH Domain's commitment to the promotion and development of open science and open research information. In line with the Swiss National Open Science Strategy, this marks an important step for the ETH Domain and the Swiss science and education system towards a more open, transparent and participatory research information landscape.

The *Bibliometric study of EPFL (2014–2023)* assesses the bibliometric performance of EPFL as one of the six institutions of the ETH Domain by analysing articles, reviews and conference proceedings published between 2014 and 2023 that are indexed in *OpenAlex*.[1] Compared with previous bibliometric reports of the ETH Domain performed in the context of the intermediate evaluation of the ETH Domain in 2019 and 2023 [1], it uses OpenAlex rather than Web of Science as its main bibliometric data source. Additionally, the report considers all citations recorded up to and including August 2025, as well as self-citations to provide a more comprehensive view of scholarly work and uptake. This means that results are not directly comparable with those previous reports of the ETH Domain.[2] Nevertheless, overall patterns and trends between and within institutions remain highly similar.

While the bibliometric report provides systematic evidence on publication output, citation impact, collaboration patterns, open-access publishing and gender-related authorship characteristics, it covers only selected dimensions of research activity and only research that is published in international (i.e., English-language) journals and proceedings. As such, it cannot provide a complete assessment of research quality, relevance or societal contribution. The results should therefore be interpreted with care and complemented by qualitative information, institutional context and expert assessment.

[1] OpenAlex is an open-source alternative to proprietary bibliometric databases such as Web of Science, which was developed as a successor to Microsoft Academic Graph. It offers broader coverage of scholarly work, but remains comparatively young and still evolves.

[2] In particular, the overall citation impact tends to be higher.

## Indicators and methodology

As the first implementation of an open bibliometric assessment for the ETH Domain, this report uses OpenAlex to support a transparent, reproducible and openly documented analysis of scientific output, impact and collaboration. At the same time, bibliographic databases, whether open or proprietary, provide only partial and evolving representations of research activity, with limitations in coverage, metadata quality and classification. However, because open data can be shared, reviewed and improved for the benefit of the wider scientific community, these limitations are documented explicitly as part of the methodological transparency of this report. By combining institutional publication data, in-house expertise and an open research information approach, the report also helps identify and improve the representation of ETH Domain institutions in open bibliometric data. It should therefore be understood not only as a bibliometric assessment, but also as a contribution to the long-term development of open research information infrastructure in Switzerland.

The report offers a variety of indicators aimed at measuring the scientific output of the ETH Domain institutions, as well as the impact of the publications they produce.

**Output indicators** measure the number of publications published during a given period. To account for differences in collaboration intensity across fields and institutions, publication output is reported using both full and fractional counting. Under full counting, each publication is fully assigned to every involved institution ($P_{full}$), which provides a straightforward measure of output. Under fractional counting, publication output is divided across the involved institutions ($P_{frac}$), which improves comparability across times, fields and institutions with different collaboration profiles. For example, a publication involving three institutions counts as one publication for each institution under full counting, whereas it counts as one-third of a publication for each institution under fractional counting.[3]

**Citation impact indicators** are used to measure the impact of publications based on the number of citations they received. The two most relevant impact indicators for this report are the *Mean Normalised Citation Score* ($MNCS_{frac}$) and the proportion of publications among the top 10% most-cited publications worldwide ($PP[top\,10\%]_{frac}$). To account for differences in citation practices, both indicators compare publications with similar publications in the same subfield, publication year and document type. An $MNCS_{frac}$ of 2 indicates that the institution's publications have been cited twice as often as the average publication (1 corresponds to the world average). A $PP[top\,10\%]_{frac}$ of 20% indicates that the institution has twice as many publications among the top 10% most-cited publications as the world average (within their respective research area). $MNCS_{frac}$ is based on average citations and is sensitive to extreme citation outliers. By contrast, $PP[top\,10\%]_{frac}$ is not influenced by outliers, but creates an arbitrary separation between what is considered a top publication. Thus, $MNCS_{frac}$ and $PP[top\,10\%]_{frac}$ have complementary strengths and weaknesses and are used in combination for assessing the impact of an institution.

**Research profiles** are analysed using fractional publication output and citation impact within OpenAlex's topic classification. For the institutional reports, this analysis is carried out at the level of subfields; for the ETH Domain report, it is reported at the broader field level. A complementary perspective is provided by the specialisation index (Spec.), which compares the institution's publication activity in a given subfield with the corresponding worldwide publication activity. A specialisation index of 2 indicates that a publication from the institution is twice as likely to belong to that subfield as a random (world average) publication. Together, both indicators reflect the absolute and relative research activity within subfields. Developments within the respective subfields are captured using the *global field growth* indicator. This indicator compares the number of publications worldwide in a given subfield during the most recent four-year period (2020–2023) with the number of publications in the preceding period

[3] The notation full or frac may also appear in other indicators, indicating whether they are calculated using full or fractional counting.

(2014–2019). It therefore shows whether publication activity in a subfield has increased, remained stable or decreased over time. The interdisciplinarity of publications is measured with InterDisc. This indicator looks at the references used in publications and assesses how broadly they are spread across different research areas. Publications receive higher values when they cite work from several different fields and lower values when most references come from the same field.

**Collaborations and partners** are assessed based on the affiliations of co-authors. Three to four mutually exclusive types of collaboration are identified: (i) *no institutional collaborations*, referring to publications without institutional collaborations; (ii) *Swiss or national collaborations*, referring to publications produced exclusively by different institutions in Switzerland; and (iii) *international collaborations*, referring to publications with at least one institution outside Switzerland. International collaborations are further decomposed into *European collaborations*, referring to publications with at least one non-Swiss European partner, but no partner outside Europe; and *global collaborations*, referring to publications with at least one partner institution outside Europe. Publications are classified according to the most geographically extensive category involved (i.e., if publications involve both national and international institutions, they are classified as international). Collaborations with industry are also analysed and correspond to publications for which at least one (co)-author is affiliated with a company. Changes in collaborations are assessed by comparing the shares of publications across collaboration types during the most recent four-year period (2020–2023) with the corresponding shares in the preceding period (2014–2019).

**Research accessibility** is assessed based on whether publications are classified as open-access (OA) or closed-access. Three types of OA are distinguished: (i) *Gold/Diamond* (publications in journals that make all articles and related content freely available immediately on the journal's website); (ii) *Green* (publications in paid-access journals that are self-archived in an OA repository); and (iii) *Hybrid* (publications in subscription journals that are immediately free to read under an open licence, for example after payment of an article processing charge). Each OA publication is assigned to only one type. If a publication is both Green and Gold, it is counted as Gold. Bronze OA publications are free to read on the publisher's platform, but have no identifiable open licence, making long-term availability and reuse rights unclear. As such, they are disregarded as OA and identified as closed-access publications.

**Gender indicators** describe the composition of the institution's authors by inferred gender based on first name. Two measures are used. The first counts each author once, regardless of how many publications they appeared on, and describes the gender composition of the underlying unique author populations. The second counts authors separately for each publication in which they appear, capturing the gender composition of authorship as it is distributed across the publication output. In both cases, gender shares are computed among authors with an inferred gender; those classified as unknown are excluded from the calculation. For a given publication, the indicators of gender diversity (RPA[F]) correspond to the proportion of female authors from the institution under study compared to the proportion of females among all co-authors of that publication (i.e., from all institutions). A value above 1 indicates that the institution under study contributes a higher share of female authors than the overall share of female authors for the same publication (or publication set).

## Main findings

**Output and impact of publications:** EPFL has a large and internationally visible publication portfolio. Across 2014–2023, it produced 43,772 publications. Citation impact is above international reference levels, with 19.3% of output among the global top 10% most-cited publications ($PP[top\,10\%]_{frac}$) and publications on average cited 1.72 times more often in their respective fields ($MNCS_{frac}$). Absolute output increased by 5.7% from 2014–2017 to 2020–2023, while the output adjusted for the number of involved institutions declined (-4.6%), pointing to increasingly collaborative publication structures. Incoming citations show broad international visibility and a strong domestic and European citation environment.

**Research profile:** EPFL's largest publication areas in terms of output are Electrical and Electronic Engineering, Biomedical Engineering, Atomic and Molecular Physics, and Optics, Materials Chemistry and Artificial Intelligence. Together, they account for around 30% of total fractional output. EPFL is especially specialised in Instrumentation, where the subfield's share in EPFL's publication profile is about 9 times as high as that subfield's share in the global publication profile. The citation impact of the major publication areas is consistently high and above the global benchmark ($PP[top\,10\%]_{frac} > 16.8\%$, $MNCS_{frac} > 1.49$). Most of these areas have shown worldwide output growth during the most recent four-year period, with Atomic and Molecular Physics and Optics as the main exception.

**Collaboration and partners:** EPFL has a highly collaborative publication profile with a substantial domestic and European base and broad international partnerships. Across all publications produced between 2014 and 2023, 79% of publications involved cross-institutional collaboration, 68% involved international partners, and 39.2% involved at least one partner outside Europe; these global collaborations (at least one partner outside Europe) have grown the fastest in the most recent four-year period and are associated with a higher citation impact. National collaborations remain important, with 31% of total output involving national partners and the University of Lausanne standing out as the largest Swiss partner outside the ETH Domain. Within the ETH Domain, EPFL collaborates most with ETH Zurich (2,742 co-publications), which is also the ETH Domain partner associated with the highest citation impact ($MNCS_{frac}$: 2.06). Outside Switzerland, EPFL's main partners are located in the USA and European countries, especially Germany, France, Italy and the United Kingdom; China also emerges as an increasingly important collaboration partner.

**Research accessibility:** Across the full evaluation period, 61.7% of EPFL publications are classified as open-access, with the share increasing from 51% in 2014–2017 to 73% in 2020–2023. Open-access publications also show higher citation impact than closed-access publications and perform clearly above the international benchmark ($PP[top\,10\%]_{frac}$: 21.3%, $MNCS_{frac}$: 1.8). Open-access adoption is high across the relevant subfields, exceeding the corresponding subfield average by at least 26%.

**Gender distribution of authors:** Female authors account for around 29% of EPFL's gender-inferred unique authors, with their share increasing from 27.3% in 2014–2017 to 29.8% in 2020–2023. This share is close to the benchmark share of all female authors on the same publications (i.e., from all contributing institutions, RPA[F]: 0.98). This indicates that, although women are substantially less numerous than men among EPFL authors in absolute terms, they are roughly as well represented in EPFL's affiliated author base as among all contributing authors on the same publications. However, the corresponding publication-level ratios are lower, at 0.91 under author weighting and 0.96 under equal publication weighting, with the relative gap more pronounced in larger author teams. Across subfields, female-author shares vary substantially, with higher shares on EPFL publications in Organic Chemistry (33%), Molecular Biology (31%) or Building and Construction (29%), and lower shares in Mechanics of Materials (11%), Astronomy and Astrophysics (11%) or Computational Mechanics (13%).

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## List of indicators

Indicator symbols in this report follow a compact and mostly coherent notation grammar. The rightmost symbol denotes the main object being measured: P for publications, C for citations and A for authors. P before another symbol denotes a proportion or share, as in PP or PA, for the proportions of publications or authors. R denotes a ratio between two indicators and often involves comparison with a reference population or benchmark, as in RPA for the ratio of proportions of authors.

Square brackets identify subsets or categories, for example A[F] for female authors, P[OA] for open-access publications, P[top 10%] for publications among the top 10% most-cited and C[self-cits] for self-citations. Superscripts indicate contextual stratification, such as $^{\text{pub}}$, $^{\text{by inst}}$ or $^{\text{by country}}$. Subscripts indicate the relevant population or counting approach, for example $_{\text{aff}}$, $_{\text{all}}$, $_{\text{full}}$ and $_{\text{frac}}$. A bar over a symbol denotes an average, as in $\overline{\text{A}}$ for an average over authors or $\overline{\text{PA}}$ for the average over proportions of authors.

In addition, several multi-letter base indicators denote established named metrics, including TCS, MCS, MedCS, MNCS, Cov and InterDisc. A complete list of indicator definitions, together with short explanations, is provided below.

$A_{aff}$ The number of unique affiliated authors on the institution's publications.

$\overline{A}_{aff}^{pub}$ The average number of affiliated authors per publication, including female, male and unknown gender categories.

$A_{all}$ The number of unique contributing authors on the institution's publications.

$A[F]_{aff}$ The number of female authors affiliated with the institution.

$A[F]_{all}$ The number of all female authors contributing to the institution's publications, including authors from other contributing institutions.

$\overline{A}[F]_{aff}^{pub}$ The average number of affiliated authors inferred as female per publication.

$\overline{A}[F]_{all}^{pub}$ The average number of all contributing authors inferred as female per publication, regardless of institutional affiliation.

$A[M]_{aff}$ The number of male authors affiliated with the institution.

$A[M]_{all}$ The number of all male authors contributing to the institution's publications, including authors from other contributing institutions.

$\overline{A}_{all}^{pub}$ The average number of all contributing authors per publication, regardless of institutional affiliation and including female, male and unknown gender categories.

$C[inst]_{full}^{by\ country}$ The number of citations to publications of the institution under evaluation by the country of the citing institutions, using full counting on the citing side.

$C[inst]_{full}^{by\ inst}$ The number of citations to publications of the institution under evaluation by the citing institution, using full counting on the citing side.

$C[self\ cits]_{full}$ The number of incoming citations that are self-citations, where the citing and cited works share at least one author, computed using full counting.

Cov The share of unique matched works that are included in the analysis period after scope filtering.

InterDisc The interdisciplinarity of publications, measured using Rao-Stirling diversity by the extent to

which their references cite publications from distant subfields in the topic hierarchy.

$MCS_{full}$ The average number of citations received by an institution's publications, irrespectively of the number of co-affiliated institutions. Computed using full counting.

$MNCS^{collab}_{frac}$ The mean normalised citation score (MNCS) of collaborative publications with a partner institution. Computed using fractional counting.

$MNCS_{frac}$ The mean citation impact of an institution's publications, normalised by research field, publication year and publication type. Computed using fractional counting.

$P\bar{A}[F]^{pub}_{aff}$ The author-weighted proportion of female authors among the institution's affiliated authors with inferred gender, with publications weighted by the number of gender-inferred affiliated author appearances.

$P\bar{A}[F]^{pub}_{all}$ The author-weighted proportion of female authors among all contributing authors with inferred gender, with publications weighted by the number of gender-inferred contributing author appearances.

$P[collab]_{frac}$ The number of publications with at least one collaborating institution, computed using fractional counting across affiliated institutions.

$P[collab]_{full}$ The number of publications with at least one collaborating institution. Computed using full counting.

$P_{frac}$ The number of publications of an institution, counted fractionally. Each publication is weighted by the inverse of the number of co-affiliated institutions.

$P_{full}$ The number of publications affiliated with an institution, counted in full. Each publication is counted as one, irrespectively of the number of co-affiliated institutions.

$P[OA]_{full}$ The number of open-access (or closed-access) publications affiliated with an institution, computed using full counting.

$P[top\ 10\%]_{frac}$ The number of publications affiliated with an institution (fractional counting) that belongs to the 10% most frequently cited publications, relative to other publications in the same research field, publication year and publication type. Each publication is weighted by the inverse of the number of co-affiliated institutions. Computed using fractional counting.

$P[uncited]_{full}$ The number of an institution's publications with zero citations, restricted to reliable citation cohorts, computed using full counting.

$PA[F]_{aff}$ The proportion of female authors among the institution's affiliated authors with inferred gender.

$PA[F]_{all}$ The proportion of female authors among all contributing authors with inferred gender.

$\overline{PA[F]}^{pub}_{aff}$ The average publication-level proportion of female authors among the institution's affiliated authors with inferred gender, with each publication weighted equally.

$\overline{PA[F]}^{pub}_{all}$ The average publication-level proportion of female authors among all contributing authors with inferred gender, with each publication weighted equally.

$PC[inst]^{by\ country}_{full}$ The proportion of incoming citations in which the citing country is represented.

$PC[inst]^{by\ inst}_{full}$ The proportion of incoming citations in which the citing institution is represented.

$PC[self\ cits]_{full}$ The proportion of incoming citations that are self-citations, where the citing and cited

works share at least one author, computed using full counting.

$PC[world]^{by\ country}_{full}$ The proportion of global citations attributable to citing institutions located in the citing country.

$PC[world]^{by\ inst}_{full}$ The proportion of global citations in which the citing institution is represented.

$PP[collab]_{full}$ The proportion of publications with at least one collaborating institution. Computed using full counting.

$PP[industry\ collab]_{full}$ The proportion of publications with at least one collaborating institution classified as industry/company. Computed using full counting.

$PP[int\ collab]_{full}$ The proportion of publications with at least one collaborating institution outside Switzerland. Computed using full counting.

$PP[OA]_{full}$ The proportion of an institution's total publication output that is considered open-access (or closed-access). Computed using full counting.

$PP[top\ 10\%]^{collab}_{frac}$ The proportion of collaborative publications with a partner institution that belongs to the top 10% most-cited publications. Computed using fractional counting.

$PP[top\ 10\%]_{frac}$ The proportion of an institution's total publication output that belongs to the top 10% most-cited publications in the same research field, publication year and publication type. Computed using fractional counting.

$PP[uncited]_{full}$ The proportion of an institution's publications with zero citations, restricted to reliable citation cohorts, computed using full counting.

$R\overline{PA}[F]^{pub}$ The ratio of the average publication-level proportion of female authors among the institution's affiliated authors to the average publication-level proportion of female authors among all contributing authors (excluding authors with unknown gender), with each publication weighted equally.

$RP\overline{A}[F]^{pub}$ The ratio of the author-weighted proportion of female authors among the institution's affiliated authors to the author-weighted proportion of female authors among all contributing authors (excluding authors with unknown gender).

RPA[F] The ratio of the proportion of female authors among affiliated authors to the proportion of female authors among all contributing authors on the institution's publications. Values above 1 indicate overrepresentation of female affiliated authors, while values below 1 indicate underrepresentation.

$RPC^{by\ country}_{full}$ The ratio between the evaluated institution's share of citations from the citing country and that country's share of global citations. Values above 1 indicate that the citing country is overrepresented among citations to the evaluated institution, while values below 1 indicate underrepresentation.

$RPC^{by\ inst}_{full}$ The ratio between the evaluated institution's share of citations from a citing institution and the citing institution's share of global citations. Values above 1 indicate that the citing institution is overrepresented among citations to the evaluated institution, while values below 1 indicate underrepresentation.

RPP[OA] The ratio of the institution's open-access publication proportion to the benchmark open-access proportion for the same field or subfield and period.

Spec. The specialisation index is the ratio of an institution's fractional publication-output share in a given

field or subfield to that field's or subfield's share of global publication output. A value of 2 indicates that the field or subfield is represented twice as strongly in the institution's publication profile as in global publication output.

$TCS_{full}$ The total number of citations received by an institution's publications, irrespectively of the number of co-affiliated institutions. Computed using full counting.

## Definitions, abbreviations and acronyms

**CC** Country code.

**CWTS** Centre for Science and Technology Studies at Leiden University.

**DOI** Digital Object Identifier, a persistent identifier for publications.

**Eawag** Swiss Federal Institute of Aquatic Science and Technology.

**Empa** Swiss Federal Laboratories for Materials Science and Technology.

**EPFL** Swiss Federal Institute of Technology Lausanne.

**ETH Domain** Federal Institutes of Technology Domain comprising ETH Zurich, EPFL, PSI, WSL, Empa and Eawag.

**ETH Zurich** Swiss Federal Institute of Technology Zurich.

**EU** European Union.

**OA** Open access.

**pp** Percentage points.

**PSI** Paul Scherrer Institute.

**ROR** Research Organization Registry, an open registry of persistent identifiers for research organisations and funders.

**WSL** Swiss Federal Institute for Forest, Snow and Landscape Research.

# 1. Introduction

The ETH Domain consists of two Federal Institutes of Technology, ETH Zurich and EPFL, and four research institutes: PSI, WSL, Empa and Eawag. Together, they play a central role in the Swiss science system in education, research and the transfer of knowledge and technology.

Every four years, the ETH Domain undergoes an intermediate evaluation mandated by the Head of the Federal Department of Economic Affairs, Education and Research (EAER). As part of their mandate, the ETH Domain commissions a bibliometric study of the ETH Domain and its institutions to support the intermediate evaluation and strategic reflection on the ETH Domain's research activities. For the first time, the ETH Board commissioned ETH Zurich and EPFL to develop a fully methodologically transparent report of the ETH Domain that relies exclusively on open data sources and open code, aiming at more transparent, reusable and insightful bibliometric analyses.

While bibliometric reports provide relevant insights and evidence as part of a broader self-assessment, the evaluation of institutional research quality is inherently multidimensional and cannot be reduced to quantitative indicators alone. Bibliometric measures provide structured and comparable evidence on scientific production, collaboration and impact, but only for research output visible in international publication channels such as journals and conference proceedings. Quantitative results therefore require careful interpretation and should be complemented by qualitative information, including institutional missions, research objectives and expert assessments.

This report presents the results of a bibliometric analysis of the scientific output of EPFL covering the period 2014–2023. The analysis is based on harmonised institutional publication data matched to OpenAlex and applies subfield-, year- and document-type-normalised indicators aligned with established bibliometric methodologies. While the overall methodological framework aligns with earlier bibliometric studies of the ETH Domain, the present analysis differs in two important methodological respects: It relies exclusively on open data sources and code, and, unlike previous reports, it does not restrict the citation window to four years or exclude citations between publications that share authors. These extensions strengthen transparency, improve coverage and provide a more comprehensive view of scholarly uptake, but also mean that results are not directly comparable with earlier reports.[4] Nevertheless, overall patterns and trends between and within institutions remain highly similar.

As the first implementation of an open bibliometric assessment for the ETH Domain, the report also transparently acknowledges the challenges associated with the shift from closed to open bibliographic data. These challenges are an expected part of this transition that are made explicit within the report. By combining in-house institutional expertise with the transparent open science approach, the report strengthens the capacity to identify limitations in the use of open research data and creates opportunities to address them in the future. The report should therefore be understood not only as a bibliometric assessment, but also as a contribution to the long-term development of open research infrastructure within the ETH Domain and the Swiss science and education system.

Data and methods are described in Chapter 2 of this report. Chapter 3 presents the data, results and interpretation in detail. Each section begins with a summary of its main findings, followed by a detailed presentation of the underlying data and analyses. The results are organised into five sections:

1. Output and impact of publications
2. Research profile
3. Collaboration and partners
4. Research accessibility
5. Gender distribution of authors

[4] In particular, the overall citation impact tends to be slightly higher and should be interpreted in light of these changes.

# 2. Indicators and methods

## 2.1. Indicators

This section defines the indicators used to analyse EPFL's publication output and citation impact, collaboration patterns, open-access publishing and gender-related authorship characteristics. The indicators draw on internationally established bibliometric practices and are constructed to support comparison across disciplines, publication types and years. The paragraphs below describe the main counting, normalisation and classification choices needed to interpret the results. A complete list of all indicator symbols and their definitions is provided in the front matter (page 8).

### 2.1.1. Counting methods and normalisation

Two counting methods are used throughout this report. Under *full counting*, each publication (or outgoing citation) is fully attributed to every affiliated institution; a publication co-authored by three institutions counts as one full publication for each institution involved. Under *fractional counting*, each publication is attributed to the involved institutions as a fraction, defined as the inverse of the number of distinct institutions on the publication ($\frac{1}{n}$, where $n$ is the number of co-affiliated institutions). A publication that lists three distinct affiliated institutions would count as one publication for each institution under full counting and as $\frac{1}{3}$ of a publication for each institution under fractional counting. Note that the fractional counting method used in this report does not take into account the number of authors per institution, but only the number of distinct institutions per publication. Thus, irrespectively of the number of authors per institution on a publication, each institution receives the same fractional weight. Accordingly, a single-author publication may also be fractionally assigned to several institutions if the author lists multiple institutions as affiliations.

Fractional counting avoids the systematic inflation of output in highly collaborative environments and improves comparability across disciplines with different collaboration and authorship profiles. At the same time, fractional counting can be perceived as disincentivising cross-institutional collaboration. Both methods are applied in parallel where informative; the subscripts *full* and *frac* in indicator symbols denote the counting method used. To enhance readability and interpretation, full counting is used for high-level descriptive statistics. To enhance comparability across disciplines and collaboration profiles, fractional counting is used for comparative indicators, including rankings and citation impact.

Citation-based impact indicators are normalised by research subfield (based on the OpenAlex classification), publication year and document type (journal publications or conference proceedings) to account for differences between research areas and publication types. The mean normalised citation score ($\text{MNCS}_{\text{frac}}$) expresses the citations received by each publication relative to the average number of citations received by publications in the same subfield, year and document type. A value of 1.0 indicates citation impact equal to the corresponding reference average; a value of 2.0 indicates twice as many citations as that average. $\text{PP[top 10\%]}_{\text{frac}}$ corresponds to the proportion of publications among the top 10% most-cited publications worldwide (accounting for subfield, year and document type). Values above 10% are above the worldwide average. For all normalised impact indicators, fractional counting is applied to ensure consistent attribution of institutional contributions. All citations are considered, including author self-citations (i.e., citations for which the citing and cited publications share at least one author; see Section 2.3.1).

### 2.1.2. Operationalisation

**Output and impact of publications**

Publication output is measured through full ($P_{full}$) and fractional ($P_{frac}$) publication counts. Absolute citation impact is described by total citations ($TCS_{full}$) and average citations ($MNCS_{full}$). Field-normalised citation impact is captured by $MNCS_{frac}$ and the top-cited share $PP[top\,10\%]_{frac}$, as defined above. Citation flows are analysed by identifying leading citing institutions ($C[inst]_{full}^{by\ inst}$) and citing countries ($C[inst]_{full}^{by\ country}$). For each citing institution or country, its share of citations to EPFL is compared with its share of all citations in the global reference set. Relative citation presence (RPC) therefore indicates whether a citing institution or country is over- or underrepresented among citations to EPFL relative to its overall citation activity.

To assess how stable each impact measure is, we report bootstrapped 95% stability intervals for moving averages of citation impact indicators in Figure 3. We build these intervals by repeatedly drawing random samples from the same publication set (sampling with replacement) and recalculating the indicators for each sample. The intervals show how much $MNCS_{frac}$ and $PP[top\,10\%]_{frac}$ may change due to small changes in the underlying publications induced by the resampling (i.e., they give the range in which the indicator falls for 95% of randomly sampled publication sets). Narrow stability intervals suggest more robust indicator estimates, while wide intervals suggest that a small number of publications may strongly influence the indicator.

To assess geographic citation uptake, each citation is assigned to the location of the institution that produced it. We count how many of the received citations come from each location to measure absolute citation uptake (using full counting). Relative and disproportional citation uptake is measured by comparing the observed number of citations from each location to what would be expected based on that location's overall citation activity. Values above 1 indicate that a citing institution or location allocates a higher share of its citations to EPFL than EPFL's share of global incoming citations. In this sense, citations to EPFL are overrepresented among the citation links of a given institution or location (i.e., they show disproportional citation uptake). The institution-level and region-level output use this measure to identify citing institutions and connected regional clusters where citation uptake is stronger than the global baseline.

**Research profile**

Research profiles are analysed using the OpenAlex topic hierarchy, mainly at the level of fields and subfields. Publication output is reported in full and fractional counting and profile tables combine these measures with field-normalised citation impact indicators where relevant. The output perspective identifies where most of EPFL's publication activity is located.

To capture disproportionate contributions to a given field, we use a specialisation index (Spec.) that compares the share of EPFL's fractional publication output in a field or subfield with the corresponding share of that field or subfield in global publication output. A value of 1 indicates that the field or subfield has the same relative weight in EPFL's profile as it has globally; values above 1 indicate relative overrepresentation.

Interdisciplinarity is measured with InterDisc, a reference-based Rao–Stirling integration index. For each publication with at least five references that can be assigned to known OpenAlex subfields, the indicator evaluates how widely the publication's references are distributed across the topic hierarchy. References within the same subfield contribute least to the score, while references reaching across fields or domains contribute more. Aggregate values are calculated as fractionally weighted means across publications. In the research-profile section, subfield-level values are compared with the corresponding worldwide mean for the same subfield, so positive differences indicate that EPFL publications draw on a broader range of cited subfields than the global subfield baseline. The indicator is therefore a descriptor of reference integration, not a direct measure of research quality.

### Collaboration and partners

Collaboration indicators are based on the institutional affiliations of authors. A publication is considered to involve an institutional collaboration if at least one additional institutional partner is present. Most collaboration indicators are presence-based: They measure whether at least one partner of a given type is involved, rather than the number or extent of such partners. The main indicators report the proportion of total output involving at least one institutional partner (PP[collab]$_{full}$) and at least one international partner (PP[int collab]$_{full}$). For selected analyses, international collaborations are further decomposed into European collaborations and global collaborations. European collaborations are defined as publications with at least one partner outside Switzerland, but none outside Europe. Global collaborations are defined as publications with at least one partner outside Europe. In mutually exclusive decompositions, publications involving multiple geographic scopes are assigned to the geographically most extensive category. Collaborations with industry are analysed separately and refer to publications for which at least one author is affiliated with a company. Most recent collaboration trends are assessed by comparing the shares of publications across collaboration types between the most recent four-year period (2020–2023) and the preceding period (2014–2019).

### Research accessibility

Output is disaggregated into open-access and closed-access categories. Bronze open access is treated as closed access throughout all analyses, consistent with established bibliometric practice. Citation impact is evaluated separately for open-access and closed-access publications using mean normalised citation score and top-cited share by access type. The analysis distinguishes four access categories: Gold/Diamond (publications in journals that make all articles and related content freely available immediately on the journal's website), Green (publications in paid-access journals that are self-archived in an OA repository), Hybrid (publications in subscription journals that are immediately free to read under an open licence, for example after payment of an article processing charge) and Closed (including Bronze). For each category, output, shares and citation impact are calculated.

To place open-access adoption in a disciplinary context, field and subfield open-access shares are compared with a global benchmark for the same field or subfield and period. The relative open-access adoption ratio (RPP[OA]) divides EPFL's open-access share by this benchmark; values above 1 indicate above-benchmark open-access adoption. Geographic citation uptake by access status is summarised through an open-versus-closed citation-origin ratio, which compares where incoming citations to open-access publications are concentrated relative to citations to closed-access publications.

### Gender distribution of authors

The gender analysis describes the composition of the institution's authors by first-name inferred gender.[5] The proportion of female authors (PA[F]) is defined by the number of female authors divided by the total number of female and male authors (i.e., authors with unknown gender are excluded from the analysis). This proportion is calculated separately for EPFL-affiliated authors (PA[F]$_{aff}$) and for all contributing authors on the same publications (PA[F]$_{all}$). The all-author proportion provides the comparison value. The relative female representation ratio (RPA[F]) is defined as the affiliated-author proportion divided by the all-author proportion. A value above 1 indicates that female authors are more strongly represented among EPFL-affiliated authors than among all contributing authors on the same publications.

There are two major approaches to calculating these metrics. The author-base approach counts

[5] Based on a test sample with known genders, the name-based inference service https://genderize.io has been used. A gender classification is accepted only when the service returns a sufficiently confident prediction; authors whose names cannot be resolved with adequate confidence or who are recorded only by initials are classified as *unknown*. Each author is identified through OpenAlex's own author identifiers, without additional name-based or institution-based clustering. Limitations of this approach are discussed in Section 2.3.2.

each author once, regardless of how many publications they contributed to, and describes the gender composition of the underlying unique author population (Section 3.5.1). As the same weight is given to authors with one publication and one hundred publications, it is more prone to reflect the gender composition of non-tenured authors, as staff turnover can add many unique authors to the institution's author base. The publication-level approach counts authors separately for each publication in which they appear, capturing the gender composition of authorship as it is distributed across the publication output (Section 3.5.3). This distinction is relevant because the latter also takes publication frequency into account and therefore reflects gender composition at the level of an institution's publication output.

For the aggregation of the publication-level proportion of female authors, there are two different weighting logics. $\overline{\text{PA}}[\text{F}]^{\text{pub}}$: The female author share is determined for each publication and these shares are then averaged across all publications. Each publication contributes equally, regardless of the size of its author team. $\bar{\text{PA}}[\text{F}]^{\text{pub}}$: The average number of female authors per publication is divided by the average total number of female and male authors per publication. As a result, publications with larger author teams have a greater influence on the result. The corresponding relative representation ratios ($\text{R}\overline{\text{PA}}[\text{F}]^{\text{pub}}$ and $\text{R}\bar{\text{P}}\bar{\text{A}}[\text{F}]^{\text{pub}}$) compare EPFL-affiliated authors with all contributing authors under the same weighting logic (serving as benchmark). The difference between the two measures is informative because it shows whether gender imbalance is more pronounced in smaller or larger author teams.

## 2.2. Data collection

The bibliometric indicators reported in this study are generated through a single, reproducible workflow that connects curated institutional publication lists with a dated OpenAlex snapshot. Using a dated local snapshot ensures that the evidence base remains stable across all analytical steps, so results can be checked, repeated and compared on a consistent basis. The complete implementation is available at [public code repository (forthcoming)].

From that common data foundation, the pipeline transforms raw input into harmonised, analysis-ready output through consistent treatment across institutions and full traceability from submitted records to reported indicators. The resulting data warehouse integrates source data, processing outcomes and derived metrics in one auditable structure that supports transparent reporting and reuse. Detailed information on the pipeline architecture is provided in Annex A.

### 2.2.1. Matching

EPFL provided 59,617 publication records from its internal repository to be included in the bibliometric processing. These records were matched with OpenAlex (Articles, Reviews and Proceedings) using a two-step linkage procedure: Direct identifier matching (primarily DOI) followed by indirect evidence matching based on title similarity, author overlap, source information and institutional affiliation consistency. Of the submitted works, 51,864 fall within 2014–2023 (evaluation period). From the works in the evaluation period, 49,979 were matched and found in OpenAlex. After deduplication, 49,894 unique works remained. Applying the scope criteria (see Section 2.2.2) retained 43,772 *in-scope publications* (i.e., publications used for the analysis of this report).

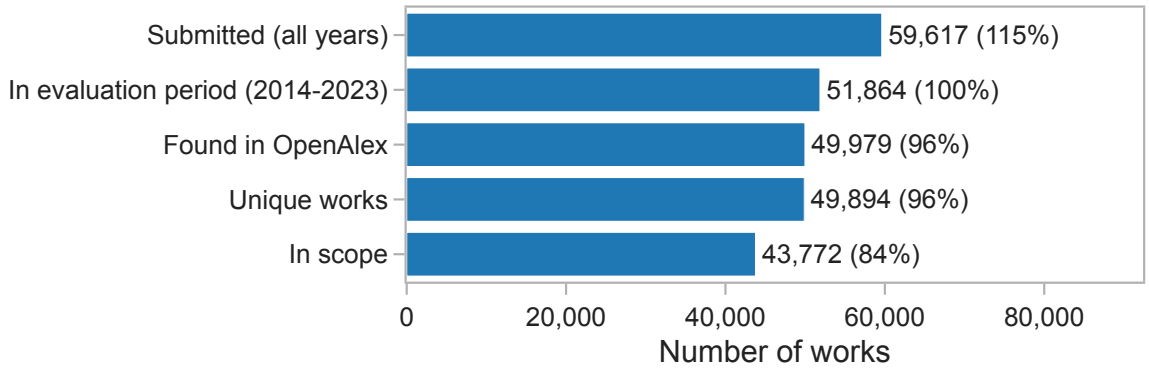



**Figure 1.** Publication coverage funnel for EPFL, showing the number of works retained at each stage of the data pipeline: total submissions, restriction to the evaluation period (2014–2023), matching against OpenAlex (Articles, Reviews and Proceedings), deduplication and application of scope criteria. Percentages are relative to the number of works in the evaluation period.

#### 2.2.2. Scope criteria

Not all matched publications are suitable for bibliometric analysis. To ensure reliable and comparable indicators, a uniform set of inclusion criteria is applied across all institutions. A publication is retained as *in-scope* when it is a substantive, English-language scientific work published in an established international journal or conference proceedings, with at least one identified author, at least one reference and an assigned research subfield. Retracted publications and non-substantive items such as front matter or tables of contents are excluded (see Figure 2).

The same criteria apply to citing works: A citation is counted only when the citing publication is itself in scope. Further details on the scoping approach and its relationship to the CWTS Leiden Ranking methodology are provided in Annex A.3.

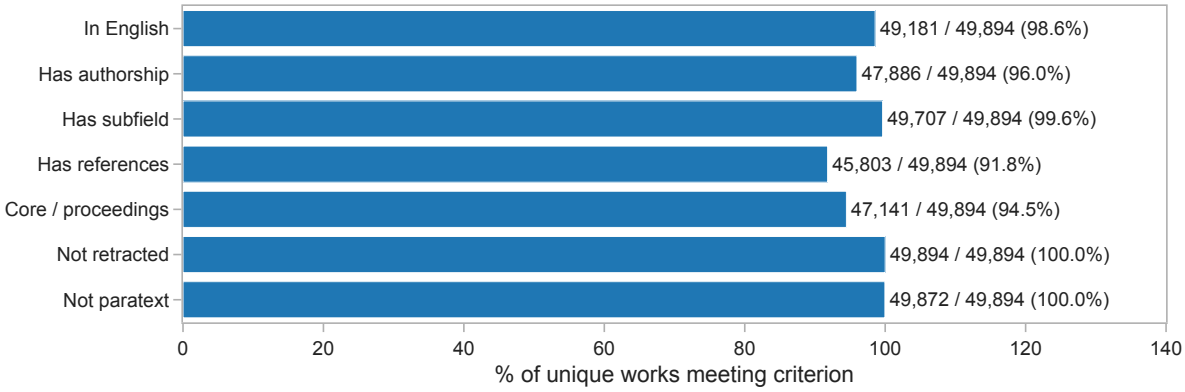



**Figure 2.** In-scope inclusion criteria. The figure shows the share and number of unique matched publications meeting each criterion used to define the in-scope publication set.

### 2.3. Additional notes

#### 2.3.1. Differences compared to previous bibliometric analyses of the ETH Domain

Compared to earlier reports performed by CWTS in 2019 and 2023 [1], the present study differs in

several methodological respects that affect the comparability of numerical results. First, this study uses OpenAlex as its primary data source rather than Web of Science. OpenAlex enables a fully open, transparent and reproducible assessment, provides broader coverage of scholarly output and allows the underlying data, code and analytical procedures to be published and reused. At the same time, some limitations in OpenAlex metadata are addressed through validated institutional publication lists, harmonised matching procedures and a transparently defined analytical scope.[6]

Second, the present study refines the measurement of citation impact. Citation counts are calculated over the full available citation window and include author self-citations, whereas earlier reports used a maximum four-year citation window and excluded self-citations. This provides a more comprehensive account of citation uptake, especially in fields with longer citation cycles or delayed recognition, and treats self-citations as part of the documented continuity of research programmes. These choices increase absolute citation counts by design, but have only limited effects on field-normalised citation indicators, since they are applied consistently to all publications and to the corresponding field averages. Field-normalised indicators are calculated using OpenAlex-provided publication-based subfields rather than the journal-based micro-field classification used in earlier reports. Annex B.1 provides a short empirical comparison between the Web of Science categories used in earlier reporting contexts and the OpenAlex subfields used here. Overall, the approach provides a more complete and internally consistent OpenAlex-based measurement of citation impact, but it also means that absolute and normalised impact indicators are not directly comparable with those reported in earlier studies.

Third, a further difference concerns gender-related indicators. Earlier reports drew on multiple gender-inference services and additional country-specific or name-based signals, whereas the present study relies on a single name-based service with a uniform confidence threshold. Authors whose names cannot be resolved with sufficient confidence or who are recorded only by initials are classified as unknown. Author identity is also resolved through OpenAlex author identifiers rather than custom disambiguation algorithms. The resulting gender indicators may therefore not be directly comparable to earlier values.

Together, these differences mean that changes relative to earlier reports may be methodological rather than substantive. The present approach offers important advantages in transparency, reproducibility and the more complete representation of scholarly uptake, but direct numerical comparison with earlier reports should be avoided.

### 2.3.2. Data limitations

The data-quality limitations described below are not necessarily specific to open bibliometric data sources; similar issues may also occur in closed or provider-curated databases. They become visible in this report because the analyses are carried out directly by the project team, combining inspectable source data with institutional expertise. This makes such limitations part of the interpretation of the report and, in the case of open data sources, creates opportunities to document and correct them.

Open institutional affiliation data can be inaccurate. OpenAlex assigns publications to institutions by processing affiliation strings and matching them to institutions. When affiliation strings contain ambiguous or very similar organisation names, publications may be matched to organisations that were not substantively involved, while relevant organisations may be omitted. This issue is particularly relevant for ETH Domain institutions, where similar or nested affiliation names may lead publications to be falsely assigned to multiple ETH Domain institutions. To mitigate this issue, all institution-based analyses are

[6] In both earlier reports and the present study, validated publication lists can introduce biases in normalised citation impact indicators. Although unwanted document types were filtered for all institutions using the same metadata-based procedures, some records may have remained in the publication sets of external institutions. If low-cited auxiliary document types remain in the publication sets of other institutions, the average citation impact within fields may be depressed, increasing the relative citation impact of ETH Domain publications with more thoroughly cleaned publication lists. This highlights the need for reliable institutional representation in open research data and, where possible, correction at the source.

seeded exclusively from the publication lists provided and approved by the respective institutions; only works included in these approved lists enter the analysis for the corresponding institution. However, some indicators still depend on OpenAlex institutional affiliation data after this restriction is applied, for example when identifying how many other institutions are linked to a publication. As a result, affiliation-based indicators such as the share of collaborative works may be partially inflated and fractional publication counts may be underestimated if OpenAlex incorrectly assigns additional institutions to a publication. Identified cases are reported transparently and flagged in the corresponding results.[7]

For collaboration analyses, we therefore distinguish between internal ETH Domain collaborations and external partner collaborations. External collaborations are identified by taking the approved publication set of the focal institution and using OpenAlex institutional affiliation data to identify partner institutions on those publications. Internal ETH Domain collaborations are treated more cautiously: A collaboration between two ETH Domain institutions is counted only when the publication is present in the approved publication lists provided by the institutions. This reduces the risk that ambiguous affiliation metadata create false internal or asymmetric collaboration links. As a result, internal and external collaboration counts are both informative, but they are based on different measurement approaches and should not be interpreted as fully comparable. The difference in the identification of collaborations affects not only Section 3.3.2 and Table 8, but also the overall collaboration share ($PP[collab]_{full}$) reported in Tables 1 and 7. This is because some publications are not recognised as collaborations in the OpenAlex metadata, although they are identified as collaborative based on the publication lists provided by the institutions.[8] Given the nested structure and partly similar institutional names within the ETH Domain, future data-quality work directly in OpenAlex may help ensure that these institutions are represented as accurately as possible in open bibliometric data sources.

A related limitation concerns the classification of institutions' organisational types. Metadata on institutions' organisational types in open affiliation sources can be incomplete or inconsistent, making strict type-based inclusion rules vulnerable to classification error. For the collaboration and partner analyses (Section 3.3), we therefore do not filter partners to specific organisational types, such as research institutions. Partner tables and maps should consequently be interpreted as showing collaboration-linked institutions, which, depending on how institutions are represented in the source metadata, may include universities, hospitals, public agencies, foundations, companies and other organisational types. The same limitation also affects indicators that rely directly on organisational-type metadata, such as the share of industry collaborations ($PP[industry\ collab]_{full}$); these indicators should therefore be treated with particular caution.

A further limitation concerns the author-gender inference used in Section 3.5. Gender is inferred from author first names using a name-based service (see Section 2.1.2 for the procedure). Coverage and accuracy are lower for authors whose names originate from naming traditions where first names are less predictive of gender or where authors are recorded only by initials. Because affiliated authors tend to have more complete name records than the broader set of all contributing authors, the share of unknown-gender authors can differ between the two groups; this asymmetry reflects differences in name availability rather than a substantive finding about gender composition.

[7] Examples include publications by the Institute for Chemical and Bioengineering at ETH Zurich that are incorrectly also assigned to Bioengineering (Switzerland) or publications from the Oeschger Centre for Climate Change Research, which is not recognised as a child institution of the University of Bern due to missing ROR data.

[8] Manual reviews suggest that several unmatched cases are due to non-standard institutional affiliation names. For example, one publication [2], listed by both Eawag and ETH Zurich, gives the affiliation as "Institute of Neuroinformatics and Institute of Computational Science UZH/ETHZ, Irchel Campus 8057 Zurich, Switzerland". This affiliation refers to ETH Zurich; however, ETH Zurich is not recognised as an affiliated institution in the bibliometric metadata.

# 3. Results

This chapter presents a bibliometric overview of publication-related activity at EPFL over the evaluation period 2014–2023, based on the OpenAlex database and the defined publication set. Trend analyses are reported using overlapping four-year intervals to increase the number of publications per interval, stabilise estimates and reduce the risk of over-interpreting short-term fluctuations, while retaining sufficient temporal resolution to assess developments over the evaluation period. The results cover publication output and citation impact (Section 3.1), research profile (Section 3.2), collaboration patterns (Section 3.3), research accessibility through open-access publishing (Section 3.4) and author gender representation (Section 3.5). Each section begins with a main finding summary and then provides an overview followed by more detailed analyses. Tables 1 and 2 in the first section provide an overview of key bibliometric indicators across sections.

## 3.1. Output and impact of publications

*Main findings*

EPFL combines a large publication portfolio with consistently strong citation visibility. Across 2014–2023, EPFL produced 43,772 publications, corresponding to 20,558 publications under fractional counting. Over time, publication output under full counting remained broadly stable, increasing only slightly from 16,887 publications in 2014–2017 to 17,853 (+5.7%) in 2020–2023, while fractionally counted output declined modestly (-4.6%), suggesting that the stable overall output volume is increasingly distributed across publications co-authored by more institutions. Citation impact is consistently above international benchmarks, with 19.3% of output among the global top 10% most-cited publications and an $MNCS_{frac}$ of 1.72. The share of highly-cited publications remained high, but eased slightly from around 19.6% in 2014–2017 to 18.3% in 2020–2023, while the mean-based $MNCS_{frac}$ declined from 1.79 to 1.59, remaining well above the world average throughout the period. Incoming citations show broad international visibility, with the largest citation volumes coming from major science systems such as China, the USA and Switzerland, followed by major European research systems. Taken together, the results point to a large, internationally visible publication profile with stable output volume, increasing collaboration intensity and sustained above-average citation performance.

### 3.1.1. Overall performance

Table 1[9] presents the overall bibliometric statistics for EPFL for the evaluation period 2014–2023. Over this period, EPFL produced 43,772 publications, comprising 37,452 journal publications (85.6% of total output) and 6,320 proceedings publications (14.4%). Under fractional counting, this corresponds to a total of 20,558 publications. Around 88% of the publications provided by EPFL fall within the analytical coverage (Cov), i.e., they are English-language articles or proceedings from the evaluation period with a unique OpenAlex match and at least one recorded reference. This suggests that EPFL's publication output is reasonably well covered by the database used for this study.

Around 62% of EPFL publications are available in some form of open access. The open-access share is substantially higher for journal publications than for proceedings publications (64% for journal publications vs. 46% for proceedings publications).

Most of EPFL's publications are collaborative. Overall, 79% of publications involve collaboration between institutions, 68% involve international collaboration, and 8% involve collaboration with industry partners. Journal publications show higher rates of institutional (81% vs. 66%) and international collaboration (70% vs. 54%) than proceedings publications, while proceedings publications show a slightly higher rate of industry collaboration (8% vs. 10%). This suggests that institutional and international collaboration levels are high overall and particularly pronounced in EPFL's journal-based output.

[9] An extended version of Table 1, including additional indicators discussed in the text, is provided in Annex C.1.

**Table 1.** Main bibliometric performance statistics of EPFL for 2014–2023, covering output, open access, collaboration, citation impact and gender-related author indicators, with results reported for journals, proceedings and overall totals.

| Indicator | Journals | Proceedings | Overall |
|---|---|---|---|
| | Output | | |
| $P_{full}$ | 37,452 | 6,320 | 43,772 |
| $P_{frac}$ | 16,643 | 3,915 | 20,558 |
| | Open Access | | |
| $PP[OA]_{full}$ | 64% | 46% | 62% |
| | Collaborations and Partners | | |
| $PP[collab]_{full}$ | 81% | 66% | 79% |
| $PP[int\ collab]_{full}$ | 70% | 54% | 68% |
| $PP[industry\ collab]_{full}$ | 8% | 10% | 8% |
| | Citation Impact | | |
| $PP[top\ 10\%]_{frac}$ | 18.7% | 21.7% | 19.3% |
| $MNCS_{frac}$ | 1.69 | 1.84 | 1.72 |
| | Author and Gender Distribution | | |
| $A_{aff}$ | 17,074 | 4,593 | 18,727 |
| $PA[F]_{aff}$ | 30% | 21% | 29% |
| RPA[F] | 0.99 | 1.05 | 0.98 |

*Note:* Authors with unknown gender represent 3,370 (18.0%) among EPFL-affiliated authors and 80,182 (45.0%) among all authors contributing to EPFL publications.

EPFL publications received 1,622,579 citations in total, including self-citations, with the large majority concentrated in journal publications. Journal publications account for 1,547,393 citations, compared with 75,186 citations for proceedings publications. Self-citations ($C[self\ cits]_{full}$) account for 259,118 citations, corresponding to 16% of all citations ($PC[self\ cits]_{full}$). The mean citation score ($MCS_{full}$), that is, the average number of citations per publication, is 37.07 overall, with a substantial difference between journal publications and proceedings publications (41.32 vs. 11.9). This lower average citation rate for proceedings publications is consistent with the comparatively high share of uncited proceedings publications of 21% (compared with 4% for journal publications). Overall, 6% of EPFL's publications remain uncited ($PP[uncited]_{full}$).

When considering field-normalised citation impact metrics, EPFL shows high citation impact relative to the international baseline. EPFL's overall mean normalised citation score ($MNCS_{frac}$) is 1.72, meaning that its publications are cited about 72% above the world average for publications of the same field, year and document type. Both journal and proceedings publications perform well above the world average, with an $MNCS_{frac}$ of 1.69 for journal publications and 1.84 for proceedings publications.

A similar pattern is visible for highly cited output. In total, EPFL has 3,902 fractionally counted top 10% most-cited publications in the respective fields ($P[top\ 10\%]_{frac}$), corresponding to 19.3% of total output ($PP[top\ 10\%]_{frac}$, global benchmark is 10%). The share is somewhat higher for proceedings publications than for journal publications (21.7% vs. 18.7%). Overall, EPFL's publication output has a high citation impact, indicating strong visibility and use in subsequent research that clearly exceeds the international baseline for comparable publications.

The author and gender indicators point to a gender imbalance among affiliated authors, while the share of women among EPFL-affiliated authors is close to the corresponding share among all authors contributing to EPFL's publications. Over the period, EPFL recorded 18,727 unique affiliated authors, including 4,471 women, 10,886 men and 3,370 authors with unknown gender.[10] Among those with inferred gender, women account for 29% of affiliated authors overall. This share is moderately lower for proceedings publications than for journal publications (21% for proceedings publications; 30% for journal publications). With 29%, the overall proportion of female authors affiliated with EPFL ($PA[F]_{aff}$) is almost equal to the share of women among all authors contributing to EPFL's publications ($PA[F]_{all}$) serving as a benchmark with 30%. The ratio between the two shares (RPA[F]) is close to parity overall, at 0.98, with values that are slightly higher for proceedings publications (1.05) than for journal publications (0.99). This indicates that within the observed (field-dependent) authorship base, EPFL's female authors are similarly represented as female authors among all contributing authors on the same publication set, even though their overall representation compared to men remains lower.

### Trends

Table 2 and Figure 3 show how EPFL's bibliometric profile developed over the reporting period using overlapping four-year moving intervals.

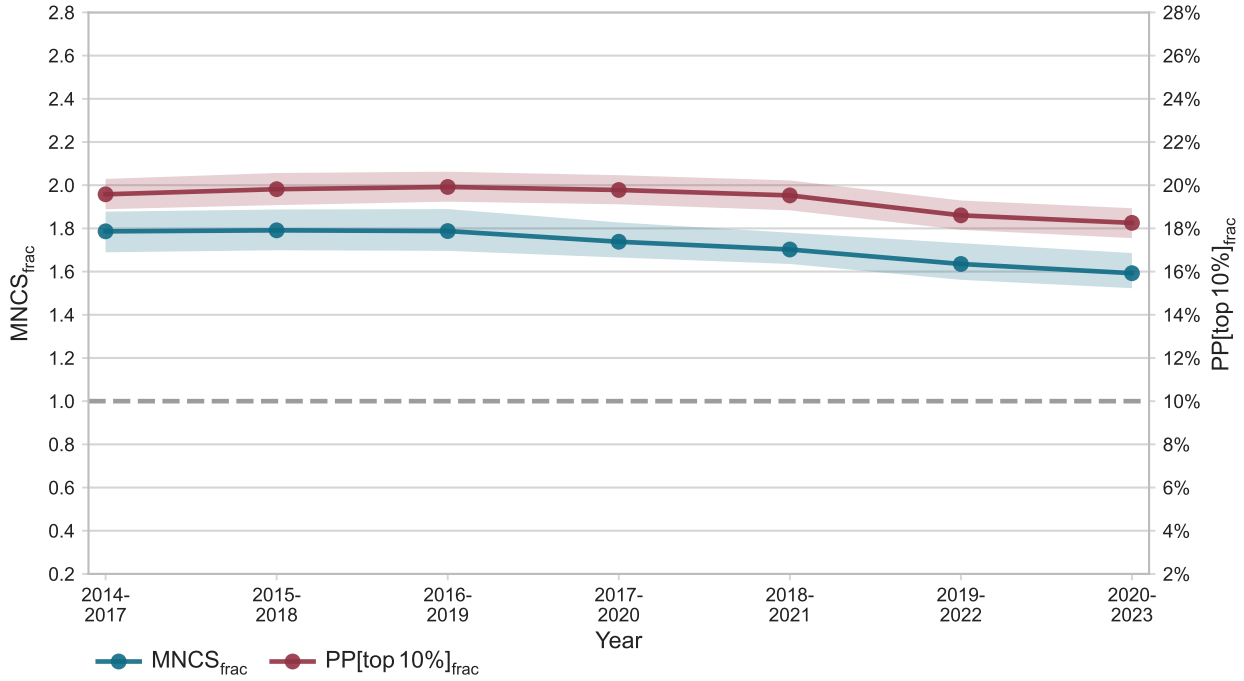



**Figure 3.** Development of EPFL's impact indicators over time. Shown are field-normalised citation impact ($MNCS_{frac}$) and the output share belonging to the global top 10% of most-cited publications ($PP[top\ 10\%]_{frac}$) together with bootstrapped 95% stability intervals. The dashed reference line represents the corresponding world averages ($MNCS_{frac}$ = 1 and $PP[top\ 10\%]_{frac}$ = 10%). A full-count version of the figure can be found in Annex C.2.

Publication output under full counting remained broadly stable, increasing only slightly from 16,887 publications in 2014–2017 to 17,853 in 2020–2023 (+5.7%). Fractionally counted output declined modestly, from 8,317 to 7,935 publications (-4.6%). The slight rise under full counting combined with the modest decline under fractional counting indicates that EPFL's stable publication output is increasingly distributed across publications co-authored by a larger number of institutions, consistent with growing

[10] These and the following results should be interpreted with caution, as gender is inferred based on first names only and the share of authors with unknown gender is relatively high.

collaboration intensity per publication.

**Table 2.** EPFL's main bibliometric indicators across the reporting period, shown as developments in overlapping four-year moving intervals. The table summarises changes in publication output, citation impact, collaboration, openness and selected author-related descriptive indicators over the reporting period.

| Indicator | 2014-2017 | 2015-2018 | 2016-2019 | 2017-2020 | 2018-2021 | 2019-2022 | 2020-2023 | Overall |
|---|---|---|---|---|---|---|---|---|
| **Output** | | | | | | | | |
| $P_{full}$ | 16,887 | 17,272 | 17,632 | 17,996 | 18,150 | 17,995 | 17,853 | 43,772 |
| $P_{frac}$ | 8,317 | 8,387 | 8,464 | 8,483 | 8,392 | 8,176 | 7,935 | 20,558 |
| **Open Access** | | | | | | | | |
| $PP[OA]_{full}$ | 51% | 53% | 56% | 61% | 65% | 70% | 73% | 62% |
| **Collaborations and Partners** | | | | | | | | |
| $PP[collab]_{full}$ | 77% | 78% | 78% | 79% | 79% | 80% | 81% | 79% |
| $PP[int\ collab]_{full}$ | 66% | 66% | 67% | 67% | 68% | 69% | 70% | 68% |
| $PP[industry\ collab]_{full}$ | 8% | 8% | 9% | 9% | 9% | 9% | 9% | 8% |
| **Citation Impact** | | | | | | | | |
| $PP[top\ 10\%]_{frac}$ | 19.6% | 19.8% | 19.9% | 19.8% | 19.5% | 18.6% | 18.3% | 19.3% |
| $MNCS_{frac}$ | 1.79 | 1.79 | 1.79 | 1.74 | 1.70 | 1.64 | 1.59 | 1.72 |
| **Author and Gender Distribution** | | | | | | | | |
| $A_{aff}$ | 9,352 | 9,569 | 9,801 | 10,027 | 10,045 | 10,310 | 10,596 | 18,727 |
| $PA[F]_{aff}$ | 27% | 28% | 28% | 29% | 29% | 30% | 30% | 29% |
| RPA[F] | 1.01 | 1.02 | 1.02 | 1.02 | 0.99 | 0.99 | 0.97 | 0.98 |

The collaboration indicators point to a consistently highly collaborative profile that has continued to intensify over time. The share of collaborative publications rose from 77% in the earliest interval to 81% in the most recent one, while the share of internationally collaborative publications increased from 66% to 70%. Industry collaboration remained broadly stable at around 8%.

Open-access publishing shows the clearest upward development over the period. The corresponding share rose from 51% in 2014–2017 to 73% in 2020–2023. This marks a clear shift in the accessibility profile of EPFL's publication output over time.

The citation-impact indicators show sustained high performance throughout the reporting period, with a moderate softening towards the end. As shown in Figure 3, the proportion of publications belonging to the top 10% most-cited remains well above the 10% benchmark in all moving intervals, ranging from 18.3% to 19.9%, with values gradually easing in the most recent intervals while still remaining clearly above the international baseline. The $MNCS_{frac}$ also declined gradually, from 1.79 in 2014–2017 to 1.59 in 2020–2023, while remaining well above the world average throughout the period. Taken together, these patterns indicate that EPFL's citation impact stays clearly above international benchmarks across the entire evaluation period, even as it converges somewhat towards those benchmarks in the most recent intervals.

The author-related indicators point to a gradual expansion of EPFL's affiliated author base and a

modest increase in the share of women among affiliated authors. The number of unique affiliated authors, measured over the four-year intervals, increased from 9,352 in 2014–2017 to 10,596 in 2020–2023, outpacing the modest growth in full-counted publication output and consistent with the parallel intensification of collaboration per publication. Over the same period, the share of female affiliated authors rose from 27% to 30%. The relative representation indicator for female authors (RPA[F]) remained close to parity throughout the period, moving from 1.01 in 2014–2017 to 0.97 in 2020–2023. This indicates that the share of women among EPFL-affiliated authors is broadly comparable to the share of women among all authors contributing to EPFL's publication set, with a slight movement towards somewhat lower relative representation in the most recent interval.

Taken together, these developments suggest a stable, highly collaborative publication profile that has progressively intensified its multi-institutional and international character, accompanied by sustained, but moderately easing citation performance and a clear structural shift towards open-access dissemination. Full-counted output increased only slightly while fractionally counted output declined modestly, with the divergence between the two indicating that growth in publication activity per institution was outpaced by growing participation in multi-institutional publications. Presence-based collaboration indicators[11] were already very high at the beginning of the period and increased further, particularly for international collaboration. Citation impact remained consistently above the international baseline throughout the period, although it eased somewhat in the most recent intervals. At the same time, the strong rise in open-access publishing and the modest increase in the share of female affiliated authors point to broader structural changes in EPFL's publication profile beyond output and citation impact alone.

### 3.1.2. Knowledge use

This section shifts the focus from publication-level citation impact to the sources and broader uptake of citations received by EPFL's publication output. It first considers the geographical distribution of incoming citations, then examines which institutions cite EPFL's publications most frequently overall, before narrowing the focus to the domestic citation environment and the countries from which citations originate most often.

Figure 4 shows where citations to EPFL publications come from, based on the locations of the citing institutions. The coloured contours show absolute citation volume, highlighting where citing institutions are most concentrated. Overall, EPFL's citation footprint is broad and international. The highest concentrations are in western Europe and North America, with further strong citation activity across East and South Asia, especially the United States, China, the United Kingdom, Germany, France, Italy, Japan, India and South Korea. The European inset shows the densest area in more detail, with a particularly strong concentration in western Switzerland, eastern France and northern Italy, as well as further citation activity across several parts of Europe.

The black contour lines add a relative perspective. Unlike the coloured contours, they do not show where the largest number of citations originate in absolute terms. Instead, they highlight areas where citations to EPFL are more frequent than expected, given the overall outgoing citation activity of the citing institutions located there.[12] The three contour levels mark areas where the observed citation share is at least 1.25-, 1.5- and 2-fold above this baseline. This layer should therefore be read as a spatial concentration of above-baseline citation activity, not as a ranking of individual institutions. Complementary institution-level and cluster-level results are reported below and in Annex C.3 and additional information can be found in B.2.

[11] The indicator measures whether at least one partner institution is present on a publication and is insensitive to changes in the number of partner institutions within a given publication.

[12] For each citing institution, the observed share is the share of its outgoing citations that point to EPFL's publications; the expected share is its share of outgoing citations to the global publication set.

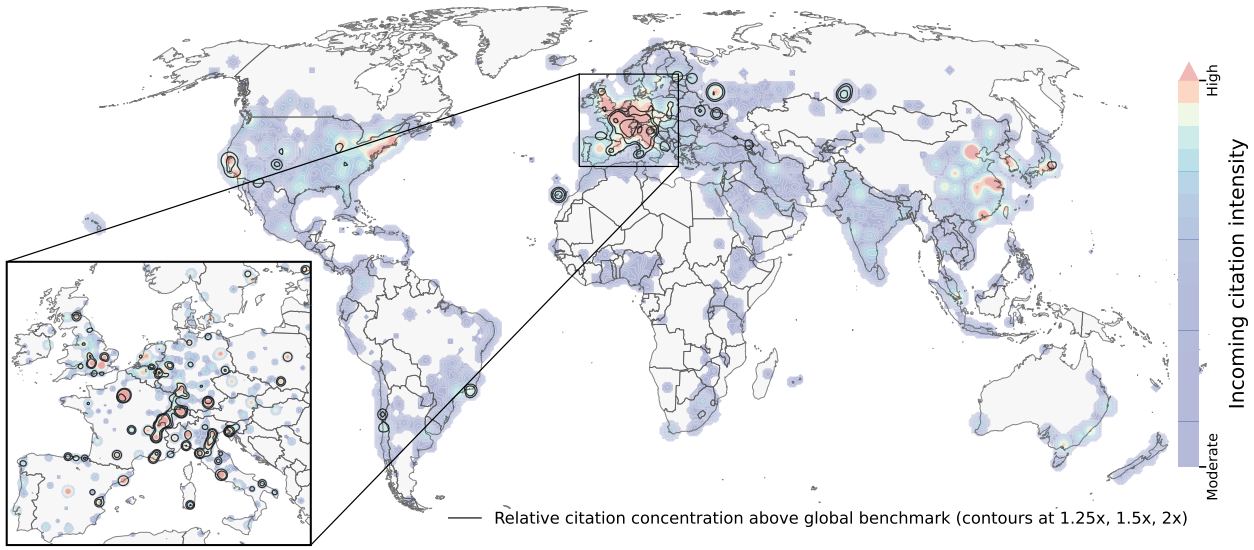



**Figure 4.** Geographic distribution of institutions citing EPFL's publications. Filled contours show the smoothed spatial intensity of incoming citations from citing-institution locations, using full citation counts. Black contour lines highlight locations where citations to EPFL are stronger than expected given the citing institutions' overall outgoing citation activity. The expected level is derived from outgoing citations to the global publication set; contour levels mark areas where the observed citation share is approximately 1.25-, 1.5- and 2-fold above this baseline.

For EPFL, the strongest above-baseline concentration is centred on a France–Switzerland–Italy cluster around Lake Geneva and extending into eastern France, western Switzerland and northern Italy. This suggests that EPFL's incoming citation links are not only shaped by the size of large publication systems, but are also relatively concentrated in a European research environment that appears closely aligned with EPFL's thematic profile. The highlighted area includes institutions such as the European Organization for Nuclear Research (CERN), the Commissariat à l'Énergie Atomique et aux Énergies Alternatives (CEA), the University of Geneva, ETH Zurich, the Institut National de Physique Nucléaire et de Physique des Particules (IN2P3), the University of Lausanne, Université Paris-Saclay, Sorbonne Université and several Istituto Nazionale di Fisica Nucleare (INFN) sections. Taken together, these institutions appear to be active in areas broadly related to high-energy and particle physics, accelerator-based research, materials and engineering and adjacent quantitative sciences. This pattern should therefore be interpreted cautiously as an indication of regional embeddedness and comparatively strong citation uptake within specialised European research communities close to EPFL's engineering, physics and life-sciences profile.

Above-baseline citation concentrations are also visible outside this dominant European area. These patterns are more selective than comprehensive national effects and appear to reflect regional pockets of citation activity in research environments related to EPFL's profile. In the western United States, the highlighted area appears to be associated with major research universities and national laboratories, including the Lawrence Berkeley National Laboratory, Stanford University, the University of California, Berkeley, the California Institute of Technology and the SLAC National Accelerator Laboratory, which are active in physics, engineering and computer-science research thematically adjacent to EPFL. Further smaller clusters in Asia point in a similar direction, with above-baseline activity around Moscow and Tokyo anchored by Lomonosov Moscow State University and The University of Tokyo, respectively. Overall, the relative layer therefore suggests that EPFL receives above-baseline citation attention not only near Switzerland, but also from selected international research communities working in related fields.

The two map layers therefore answer different questions. The filled contours show where large numbers of incoming citations to EPFL originate in absolute terms, which naturally emphasises large and research-

intensive science systems in Europe, North America and East Asia. The black contours show where EPFL receives a disproportionately high share of citations relative to the citing institutions' overall outgoing citation activity. Some regions, especially the France–Switzerland–Italy cluster around Lake Geneva, are prominent in both layers. Others, including parts of North America and Asia, account for substantial absolute citation volume, but appear more selectively in the above-baseline layer. Read together, the figure suggests a broad global citation footprint, combined with particularly strong relative citation uptake in a Lake Geneva-centred European research environment and selected international research communities active in physics, engineering, materials and related quantitative sciences.

**Table 3.** Institutions citing EPFL's publications. Shown are the 20 leading citing institutions by full-count citation volume, together with their country codes, the proportion of EPFL's incoming citations in which the citing institution is present, the corresponding proportion among all citations worldwide, and the resulting relative presence ratio, which indicates over- or underrepresentation of that institution among EPFL's incoming citations.

| Rank | Institution | CC | $C[inst]^{by\ inst}_{full}$ | $PC[inst]^{by\ inst}_{full}$ | $PC[world]^{by\ inst}_{full}$ | $RPC^{by\ inst}_{full}$ |
|---|---|---|---|---|---|---|
| 1 | EPFL | CH | 127,834 | 7.88% | 0.20% | 39.90 |
| 2 | CNRS | FR | 65,589 | 4.04% | 1.49% | 2.71 |
| 3 | Chinese Acad Sci | CN | 60,866 | 3.75% | 2.44% | 1.54 |
| 4 | Univ Chinese Acad Sci | CN | 33,657 | 2.07% | 1.23% | 1.69 |
| 5 | Univ Cambridge | GB | 25,840 | 1.59% | 0.50% | 3.17 |
| 6 | Tsinghua Univ | CN | 23,810 | 1.47% | 0.75% | 1.97 |
| 7 | ETH Zurich | CH | 23,609 | 1.46% | 0.34% | 4.28 |
| 8 | Univ Oxford | GB | 22,256 | 1.37% | 0.60% | 2.29 |
| 9 | Peking Univ | CN | 20,807 | 1.28% | 0.73% | 1.76 |
| 10 | Sorbonne Université | FR | 20,049 | 1.24% | 0.41% | 3.05 |
| 11 | Imp Coll London | GB | 19,483 | 1.20% | 0.50% | 2.41 |
| 12 | MIT | US | 19,243 | 1.19% | 0.32% | 3.71 |
| 13 | Univ Coll London | GB | 18,405 | 1.13% | 0.71% | 1.61 |
| 14 | Shanghai Jiao Tong Univ | CN | 18,357 | 1.13% | 0.84% | 1.35 |
| 15 | Stanford Univ | US | 17,909 | 1.10% | 0.57% | 1.94 |
| 16 | Univ Tokyo | JP | 17,331 | 1.07% | 0.42% | 2.56 |
| 17 | Zhejiang Univ | CN | 17,266 | 1.06% | 0.85% | 1.25 |
| 18 | Univ Calif Berkeley | US | 16,839 | 1.04% | 0.34% | 3.07 |
| 19 | CEA | FR | 16,758 | 1.03% | 0.21% | 5.03 |
| 20 | Harvard Univ | US | 16,542 | 1.02% | 1.05% | 0.97 |

Table 3 shows the institutions that cited EPFL's publications most frequently. EPFL ranks first with a presence among 127,834 incoming citations. This corresponds to 7.88% of EPFL's total incoming citations, which is 39.9 times higher ($RPC^{by\ inst}_{full}$) than EPFL's presence among all citations worldwide (0.2%), indicating a comparatively strong citation-based knowledge flow. EPFL is followed by Centre National de la Recherche Scientifique (65,589 citations, RPC: 2.71) and Chinese Academy of Sciences (60,866 citations, RPC: 1.54). EPFL itself, CNRS and large Chinese institutions account for the largest citation volumes, indicating broad uptake across Swiss, French and Chinese institutions. In relative terms, EPFL remains dominant, while ETH Zurich, Cambridge, Sorbonne Université and MIT appear more prominently than their overall citation activity would suggest. The relative pattern therefore keeps the broad international picture, but gives more weight to selected Swiss, French, UK and US institutions.

**Table 4.** Swiss institutions citing EPFL's publications. Shown are the 20 leading domestic citing institutions by full-count citation volume, together with the proportion of EPFL's incoming citations in which the citing institution is present, the corresponding proportion among all citations worldwide, and the resulting relative presence ratio, which indicates over- or underrepresentation of that institution among EPFL's incoming citations.

| Rank | Institution | $C[inst]^{by\ inst}_{full}$ | $PC[inst]^{by\ inst}_{full}$ | $PC[world]^{by\ inst}_{full}$ | $RPC^{by\ inst}_{full}$ |
|---|---|---|---|---|---|
| 1 | EPFL | 127,834 | 7.88% | 0.20% | 39.90 |
| 2 | ETH Zurich | 23,609 | 1.46% | 0.34% | 4.28 |
| 3 | Univ Geneva | 15,729 | 0.97% | 0.17% | 5.84 |
| 4 | Univ Zurich | 13,587 | 0.84% | 0.27% | 3.08 |
| 5 | CERN | 12,073 | 0.74% | 0.05% | 15.30 |
| 6 | Univ Lausanne | 11,326 | 0.70% | 0.15% | 4.76 |
| 7 | PSI | 8,594 | 0.53% | 0.06% | 9.61 |
| 8 | Univ Bern | 6,152 | 0.38% | 0.22% | 1.73 |
| 9 | Empa | 5,868 | 0.36% | 0.04% | 8.79 |
| 10 | Univ Basel | 5,223 | 0.32% | 0.15% | 2.22 |
| 11 | Eawag | 3,427 | 0.21% | 0.02% | 8.75 |
| 12 | Univ Fribourg | 2,841 | 0.18% | 0.04% | 4.21 |
| 13 | Univ Hosp Geneva | 2,595 | 0.16% | 0.05% | 2.95 |
| 14 | Univ Hosp Lausanne | 2,464 | 0.15% | 0.03% | 5.39 |
| 15 | SIB Swiss Inst Bioinf | 2,262 | 0.14% | 0.03% | 5.01 |
| 16 | Ctr d'Imagerie BioMedicale | 1,972 | 0.12% | 0.00% | 28.17 |
| 17 | WSL | 1,712 | 0.11% | 0.03% | 3.37 |
| 18 | HES-SO Univ Applied Sci Arts Western Switzerland | 1,691 | 0.10% | 0.02% | 6.24 |
| 19 | Swiss Ctr Electronics Microtechnology (Switzerland) | 1,587 | 0.10% | 0.00% | 29.93 |
| 20 | Univ Hosp Zurich | 1,572 | 0.10% | 0.10% | 0.97 |

Table 4 narrows the focus to institutions located within Switzerland that cited EPFL's publications most frequently. Among Swiss institutions, EPFL ranks first with a presence among 127,834 incoming citations. This corresponds to 7.88% of EPFL's total incoming citations, which is 39.9 times higher ($RPC^{by\ inst}_{full}$) than EPFL's presence among all citations worldwide (0.2%), indicating a comparatively strong citation-based knowledge flow. EPFL is followed by ETH Zurich (23,609 citations, RPC: 4.28) and University of Geneva (15,729 citations, RPC: 5.84). In relative terms, CERN, PSI, Empa, Eawag and the Centre d'Imagerie BioMédicale become more prominent. This suggests that EPFL's work is especially visible within the Lake Geneva research environment and selected ETH Domain or infrastructure-related institutions.

**Table 5.** Countries of institutions citing EPFL's publications. Shown are the 20 leading countries by full-count citation volume aggregated across institutions located in those countries, together with the proportion of EPFL's incoming citations in which institutions from the citing country are present, the corresponding proportion among all citations worldwide, and the resulting relative presence ratio, which indicates over- or underrepresentation of that country among EPFL's incoming citations.

| Rank | Country | CC | $C[inst]_{full}^{by\ country}$ | $PC[inst]_{full}^{by\ country}$ | $PC[world]_{full}^{by\ country}$ | $RPC_{full}^{by\ country}$ |
|---|---|---|---|---|---|---|
| 1 | China | CN | 474,111 | 29.2% | 30.6% | 0.95 |
| 2 | USA | US | 400,905 | 24.7% | 22.3% | 1.11 |
| 3 | Switzerland | CH | 192,012 | 11.8% | 1.9% | 6.24 |
| 4 | Germany | DE | 178,309 | 11.0% | 6.1% | 1.81 |
| 5 | United Kingdom | GB | 159,652 | 9.8% | 7.5% | 1.31 |
| 6 | France | FR | 119,271 | 7.4% | 3.8% | 1.93 |
| 7 | Italy | IT | 118,538 | 7.3% | 4.9% | 1.50 |
| 8 | Spain | ES | 84,073 | 5.2% | 3.6% | 1.42 |
| 9 | India | IN | 78,433 | 4.8% | 6.3% | 0.77 |
| 10 | Japan | JP | 76,573 | 4.7% | 3.5% | 1.35 |
| 11 | Australia | AU | 67,637 | 4.2% | 4.3% | 0.98 |
| 12 | Canada | CA | 67,314 | 4.1% | 4.1% | 1.01 |
| 13 | South Korea | KR | 64,415 | 4.0% | 3.1% | 1.27 |
| 14 | Netherlands | NL | 59,237 | 3.7% | 2.4% | 1.51 |
| 15 | Sweden | SE | 41,246 | 2.5% | 1.7% | 1.53 |
| 16 | Russia | RU | 38,137 | 2.4% | 1.4% | 1.65 |
| 17 | Brazil | BR | 33,767 | 2.1% | 2.4% | 0.86 |
| 18 | Poland | PL | 32,778 | 2.0% | 1.7% | 1.19 |
| 19 | Hong Kong | HK | 30,875 | 1.9% | 1.5% | 1.28 |
| 20 | Denmark | DK | 29,068 | 1.8% | 1.2% | 1.48 |

Table 5 reports incoming citations by the countries of institutions on citing publications, aggregated at the country level. It shifts the focus from citing institutions to the countries in which citing institutions are located. By aggregating citation counts across institutions within each country, the table provides a compact view of the geographical distribution of citation uptake and highlights the countries most strongly represented in EPFL's incoming citations. The ranking is led by institutions from China with 474,111 citations, corresponding to a presence of 29.2% among EPFL's incoming citations and an $RPC_{full}^{by\ country}$ of 0.95 (i.e., China's share among EPFL's incoming citations is roughly in line with China's share among all citations worldwide). The next two countries in the ranking are the USA (400,905 citations and an RPC of 1.11) and Switzerland (192,012 citations and an RPC of 6.24). China and the USA account for the largest absolute citation volumes, indicating broad international visibility. Relative to countries' global citation shares, however, China is slightly below expected levels, while citation uptake from the USA is slightly higher than expected. Switzerland shows by far the strongest relative citation uptake, followed by neighbouring countries such as France, Germany and Italy. The overrepresentation of several other European countries further points to a strong regional and European citation footprint alongside substantial international reach.

## 3.2. Research profile

*Main findings*

EPFL shows a broad and engineering-oriented research profile, with publication activity distributed across a wide range of subfields and centred on engineering, physics-related and computer-science areas. The largest publication areas include Electrical and Electronic Engineering ($PP[top\ 10\%]_{frac}$: 20.1%, $MNCS_{frac}$: 1.96), Biomedical Engineering (16.9%, 1.49), Atomic and Molecular Physics, and Optics (21.4%, 1.93), Materials Chemistry (19%, 1.87) and Artificial Intelligence (18.4%, 1.77). Together, these areas represent 30% of EPFL's total fractionally counted output. Beyond these large publication areas, EPFL also shows pronounced relative strengths in smaller specialised subfields, including Instrumentation, with a specialisation index of 9.1 times its global share, as well as Biophysics (4.9) and Computer Graphics and Computer-Aided Design (4.7), while Atomic and Molecular Physics and Optics combines high absolute output with a high specialisation index of 4.6. Most of these output areas are situated in expanding global publication contexts, with the strongest growth in Building and Construction, Civil and Structural Engineering and Artificial Intelligence; some large physics-related areas, such as Nuclear and High Energy Physics and Atomic and Molecular Physics and Optics, are located in globally stable or slightly declining subfields, while big-physics areas such as Astronomy and Astrophysics and Nuclear and High Energy Physics show full-counted output more than four times their fractional counts, reflecting strong involvement in large international collaborations. Citation impact is consistently above the global benchmark across the displayed subfields, indicating broad visibility in subsequent research across both core output areas and adjacent specialised fields. Exploratory citation-based interdisciplinarity indicators suggest above-baseline reference diversity in several engineering, applied physics and materials subfields, near-baseline patterns in the largest engineering and computer-science areas and below-baseline diversity in Molecular Biology and Biomedical Engineering. Taken together, EPFL's research profile is characterised by broad publication activity in engineering and physical sciences, selected niches of pronounced relative specialisation, predominantly growing global field environments and consistently above-average citation visibility.

### 3.2.1. Overview

This section examines EPFL's research profile in more detail by analysing its publication output across the 252 subfield categories provided by OpenAlex. Subfields are part of a four-level topical hierarchy consisting of four domains, 26 fields, 252 subfields and more than 4,500 topics that publications are assigned to, which was developed together with CWTS. This provides a structured view of EPFL's publication portfolio, while the results should be interpreted as a limited bibliometric classification of publication activity rather than as a complete representation of the institution's organisational or strategic research structure.

Figure 5 shows EPFL's largest subfields by fractional publication output and situates them in relation to full-count output, recent global field growth over the last four years and citation impact (see also Figure 14 in Annex C.4 for the corresponding aggregation at the higher field level). The largest subfield is Electrical and Electronic Engineering, which accounts for 10% of EPFL's fractional publication output, followed by Biomedical Engineering (6%), Atomic and Molecular Physics, and Optics (5%), Materials Chemistry (4%) and Artificial Intelligence (4%). Together, these five subfields represent 30% of EPFL's

total output. This top-five concentration points to a broad publication profile spanning engineering, biomedical engineering, applied physics, materials chemistry and computer-science subfields, without a single subfield dominating the portfolio.

At the same time, the difference between full and fractional counts highlights pronounced variation in collaboration intensity across subfields. Subfields with large multi-institutional publication structures can appear less prominent under fractional counting than under full counting, because publication credit is distributed across all participating institutions. This is most visible in Astronomy and Astrophysics, where the full count is more than four times as high as the fractional count, reflecting EPFL's participation in large international observational collaborations. A similarly large gap is observed in Nuclear and High Energy Physics, where full-counted output is also more than four times as high as the fractional count, consistent with EPFL's involvement in major particle-physics experiments. The gap is smaller, but still notable in Condensed Matter Physics, where the full count is more than twice as high as the fractional count. These gaps should be interpreted primarily as indicators of collaboration and authorship structures, rather than as a weaker role of these areas within EPFL's publication portfolio.

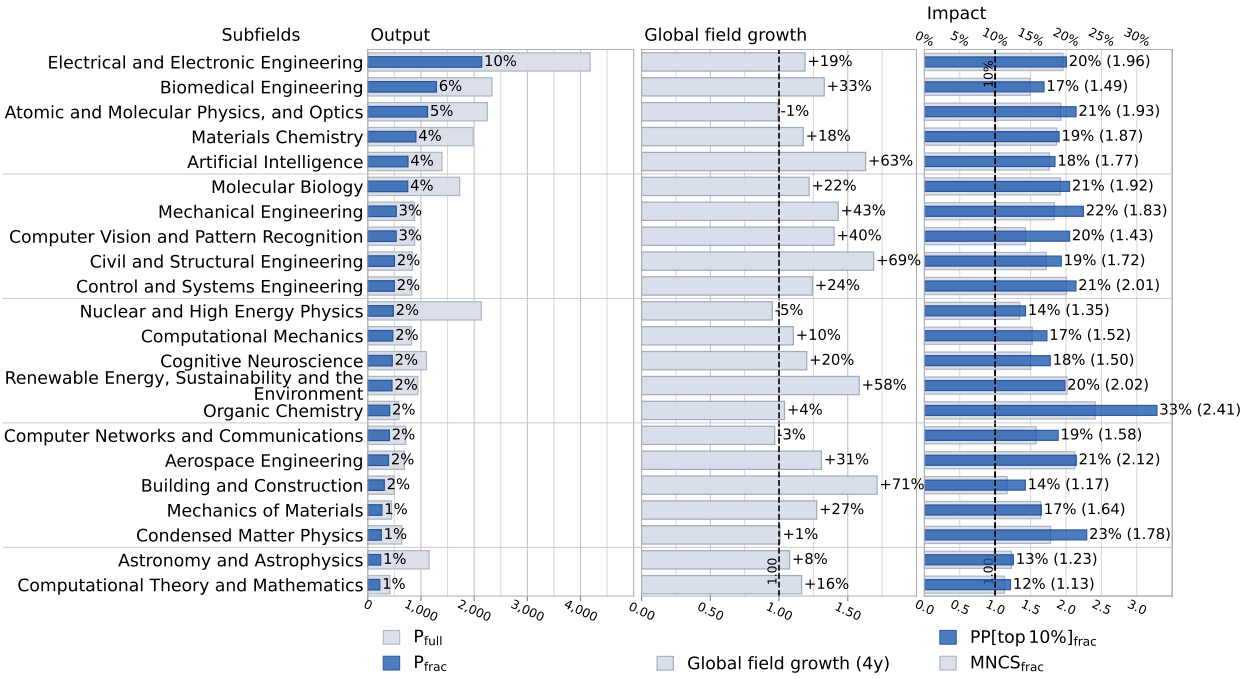



**Figure 5.** Subfield research profile of EPFL by output, global field growth and citation impact. Shown are subfields that account for at least 1% of EPFL's total fractional publication output and have at least 50 fractionally counted publications. **Left** (Output): Publication output in full counting (light blue) and fractional counting (dark blue), with labels indicating each subfield's share of total fractional output. **Middle** (Global field growth): Global subfield growth is measured as the ratio of mean annual global publication output in 2020–2023 relative to 2014–2019, using full counting; the dashed line marks a ratio of 1, indicating no absolute growth. **Right** (Impact): Citation impact is reported as the share of publications among the top 10% most-cited publications ($PP[top\ 10\%]_{frac}$, dark blue, in %), followed by the mean normalised citation score ($MNCS_{frac}$, light blue, in brackets); dashed lines indicate the global benchmark values (10% and 1).

Most of these output areas are embedded in expanding global publication contexts. Global publication output increased particularly strongly during 2020–2023 in Building and Construction (+71%), Civil and Structural Engineering (+69%) and Artificial Intelligence (+63%). At the same time, several large areas in EPFL's portfolio are located in globally stable or slightly declining subfields, including Atomic and Molecular Physics, and Optics (-1%), Computer Networks and Communications (-3%) and Nuclear and High Energy Physics (-5%). These growth values do not describe EPFL's own output growth, but they

help contextualise where its publication activity is situated relative to global field dynamics. Overall, EPFL's major publication activity spans a mix of strongly growing engineering and computer-science subfields and globally mature physics-related areas.

Citation impact is consistently above the global benchmark across all displayed subfields, although with some variation between areas. The highest shares of top-10% cited publications are observed in Organic Chemistry (32.8%, $MNCS_{frac}$: 2.41), Condensed Matter Physics (22.9%, $MNCS_{frac}$: 1.78) and Mechanical Engineering (22.5%, $MNCS_{frac}$: 1.83). Citation impact is also high in areas in which EPFL is most active, such as Electrical and Electronic Engineering, where 20.1% of publications are among the top 10% most-cited and the mean normalised citation score ($MNCS_{frac}$) reaches 1.96, suggesting that citations to EPFL's publications in that field are roughly twice as high as the field average. The citation profile therefore indicates that high citation impact is broadly distributed across EPFL's subfields, including both core output areas in engineering and physics and selected areas with fewer publications in chemistry and applied physics. The overall pattern suggests that EPFL combines a broad and engineering-oriented publication profile with consistently above-average citation visibility across both its largest output areas and adjacent specialised fields.

### 3.2.2. Specialisation

The preceding overview identifies EPFL's main research areas by their share of the institution's publication output. This output-based perspective shows where EPFL publishes most, but it does not relate the output to the size of global research activity in the same areas. The specialisation perspective presented in this section adds this relative dimension. It is less about absolute publication output and more about relative contribution to or prominence within a field. In this sense, specialisation highlights fields in which EPFL is disproportionately active compared with the global publication output.

Table 6 complements the output-based research profile by ranking subfields in which EPFL is disproportionately active. To do so, subfields are ordered by their specialisation index (Spec.), defined as the ratio between the share of EPFL's fractional publication output in a given subfield and the corresponding share of that subfield in global publication output. A value above 1 indicates that a subfield accounts for a larger share of EPFL's publication output than it does globally. Higher values therefore point to areas of pronounced relative specialisation rather than simply large publication output. Only subfields with more than 50 fractional publications and a specialisation index of at least 1.5 are shown, meaning that the subfield's share in EPFL's publication portfolio is at least 50% higher than its share in the global publication profile.

This relative perspective changes which subfields appear most distinctive in EPFL's research profile. Several subfields that appear most distinctive under the specialisation index are not necessarily the largest publication areas. Instrumentation has the highest specialisation index, at 9.1, meaning that its share in EPFL's publication profile is 9.1 times as high as its share in the global publication profile, despite a comparatively limited absolute publication output. Similar patterns are visible for Biophysics (4.9), Computer Graphics and Computer-Aided Design (4.7), Atomic and Molecular Physics, and Optics (4.6) and Discrete Mathematics and Combinatorics (4.2). These areas are partly less prominent in a purely output-based ordering, but become more visible when the profile is assessed relative to global field structure, pointing to selected niches in physics-related instrumentation, biological physics, computer graphics and discrete mathematics.

**Table 6.** Research subfields with the highest EPFL reference-based specialisation index. Shown are subfields ranked by this index (Spec.), together with publication output in full counting ($P_{full}$), the field-normalised mean citation score ($MNCS_{frac}$) and the fractionally counted share of publications among the top-10% most-cited publications in their subfield, publication year and document type ($PP[top\ 10\%]_{frac}$). Background colour indicates magnitude.

| | Subfield | Spec. | $P_{full}$ | $MNCS_{frac}$ | $PP[top\ 10\%]_{frac}$ |
|---|---|---|---|---|---|
| 1 | Instrumentation | 9.1 | 138 | 2.33 | 36.8% |
| 2 | Biophysics | 4.9 | 275 | 1.83 | 19.3% |
| 3 | Computer Graphics and Computer-Aided Design | 4.7 | 100 | 4.55 | 41.1% |
| 4 | Atomic and Molecular Physics, and Optics | 4.6 | 2249 | 1.93 | 21.4% |
| 5 | Discrete Mathematics and Combinatorics | 4.2 | 107 | 1.10 | 12.8% |
| 6 | Condensed Matter Physics | 4.1 | 639 | 1.78 | 22.9% |
| 7 | Hardware and Architecture | 4.0 | 240 | 1.78 | 22.8% |
| 8 | Nuclear and High Energy Physics | 3.6 | 2135 | 1.35 | 14.3% |
| 9 | Mathematical Physics | 3.3 | 256 | 1.13 | 12.3% |
| 10 | Computational Mechanics | 3.2 | 821 | 1.52 | 17.3% |
| 11 | Building and Construction | 2.7 | 497 | 1.17 | 14.3% |
| 12 | Spectroscopy | 2.6 | 410 | 1.11 | 13.9% |
| 13 | Biomedical Engineering | 2.5 | 2335 | 1.49 | 16.9% |
| 14 | Cognitive Neuroscience | 2.4 | 1103 | 1.50 | 17.8% |
| 15 | Computer Science Applications | 2.3 | 165 | 1.66 | 22.6% |
| 16 | Civil and Structural Engineering | 2.3 | 839 | 1.72 | 19.3% |
| 17 | Renewable Energy, Sustainability and the Environment | 2.3 | 942 | 2.02 | 19.8% |
| 18 | Electrical and Electronic Engineering | 2.3 | 4181 | 1.96 | 20.1% |
| 19 | Cellular and Molecular Neuroscience | 2.2 | 423 | 1.71 | 22.4% |
| 20 | Radiation | 2.1 | 246 | 1.55 | 16.0% |
| 21 | Statistical and Nonlinear Physics | 2.0 | 334 | 1.42 | 15.7% |
| 22 | Control and Systems Engineering | 2.0 | 829 | 2.01 | 21.4% |
| 23 | Computational Theory and Mathematics | 2.0 | 416 | 1.13 | 12.2% |
| 24 | Organic Chemistry | 2.0 | 584 | 2.41 | 32.8% |
| 25 | Geometry and Topology | 2.0 | 189 | 1.16 | 13.1% |
| 26 | Human-Computer Interaction | 1.9 | 156 | 1.74 | 20.9% |
| 27 | Transportation | 1.9 | 129 | 1.33 | 21.2% |
| 28 | Surfaces, Coatings and Films | 1.9 | 110 | 1.39 | 5.4% |
| 29 | Artificial Intelligence | 1.8 | 1396 | 1.77 | 18.4% |
| 30 | Computer Vision and Pattern Recognition | 1.8 | 882 | 1.43 | 20.5% |
| 31 | Inorganic Chemistry | 1.7 | 283 | 1.90 | 23.9% |
| 32 | Materials Chemistry | 1.7 | 1979 | 1.87 | 19.0% |
| 33 | Computer Networks and Communications | 1.7 | 713 | 1.58 | 18.9% |
| 34 | Aerospace Engineering | 1.6 | 689 | 2.12 | 21.4% |
| 35 | Statistics, Probability and Uncertainty | 1.6 | 119 | 0.69 | 8.0% |
| 36 | Signal Processing | 1.6 | 242 | 1.75 | 22.5% |
| 37 | Mechanics of Materials | 1.5 | 447 | 1.64 | 16.5% |

While some highly specialised subfields are comparatively small in absolute publication terms, others combine substantial publication activity with clear relative overrepresentation in EPFL's portfolio, including Electrical and Electronic Engineering (2.3), Biomedical Engineering (2.5), Atomic and Molecular Physics, and Optics (4.6), Materials Chemistry (1.7), Artificial Intelligence (1.8) and Computer Vision and Pattern Recognition (1.8). The specialisation perspective therefore separates two aspects that are partly conflated in an output-based profile: the absolute scale of activity and the distinctiveness of that activity relative to the global distribution of publications. Overall, the table shows that EPFL's visible

research profile is shaped both by a backbone of large engineering, computer-science and materials subfields with moderate, but consistent specialisation and by a set of smaller subfields in physics-related instrumentation, biophysics, computer graphics and applied mathematics in which EPFL has a particularly pronounced relative presence.

### 3.2.3. Interdisciplinarity

After the preceding output- and specialisation-based views of EPFL's research profile, this section adds a structural perspective on how publications draw on knowledge across subject areas. Interdisciplinarity is measured with a publication-level Rao–Stirling index that captures how broadly and how far across subfields a publication's references are distributed in the topic hierarchy. Scores are lower when references stay within the same subfield and higher when they extend across fields or even domains; only works with at least five references mapped to known subfields are included. This indicator is an exploratory profile characteristic, not a direct performance metric.

Figure 6 shows that interdisciplinarity among the displayed subfields is not uniform, but differs both by the general interdisciplinarity of the subfield worldwide and by EPFL's position within that subfield. The horizontal axis shows whether subfields are, on average, less or more interdisciplinary globally, while the vertical axis shows whether EPFL publishes in a more or less interdisciplinary way than the worldwide average in the same subfield. Positive values therefore indicate that EPFL draws on more diverse references than the worldwide average in that subfield, while negative values indicate a less interdisciplinary reference profile.

A first group of subfields lies above the horizontal reference line, indicating above-average interdisciplinarity relative to their respective worldwide subfield baselines. This is particularly visible for Computational Mechanics, Mechanical Engineering, Computer Vision and Pattern Recognition, Atomic and Molecular Physics and Optics and Materials Chemistry. These areas suggest that, within several engineering, applied physics and materials subfields, EPFL publications tend to draw on a broader range of subfields than the corresponding global publication set.

A second group is located close to the horizontal reference line, indicating that EPFL's interdisciplinarity is broadly similar to the worldwide average in these subfields. This applies especially to Electrical and Electronic Engineering, Control and Systems Engineering, Civil and Structural Engineering, Nuclear and High Energy Physics and Artificial Intelligence. These subfields are central to EPFL's profile, but the figure suggests that the reference diversity of its publications is not strongly different from the global pattern within the same subfields.

A third group lies below the horizontal reference line, indicating lower interdisciplinarity than the corresponding worldwide subfield average. Among the displayed subfields, this applies to Molecular Biology and Biomedical Engineering. Both are positioned to the right of the vertical reference line, meaning that they are globally more interdisciplinary than the displayed subfield average, but EPFL's publications in these areas draw on a narrower reference mix than the corresponding global subfield baselines.

Figure 6 therefore adds a different perspective to the research profile. It does not simply show whether EPFL is active in more or less interdisciplinary subfields, but whether its publications within these subfields are more or less interdisciplinary than comparable publications worldwide. Overall, the profile combines engineering-, applied-physics- and materials-related subfields with above-baseline interdisciplinarity, several large engineering and physics subfields close to their global baseline and Molecular Biology and Biomedical Engineering with below-baseline reference diversity relative to their worldwide subfield averages. These indicators should be interpreted as exploratory descriptors of referencing behaviour rather than as performance measures.




**Figure 6.** Interdisciplinarity profile of EPFL's high-output subfields. Shown are up to the 12 largest subfields by output with at least 1% of EPFL's total output and at least 50 publications, using fractional counting. The horizontal position of a subfield shows that subfield's global mean interdisciplinarity (measured by the citation-based Rao-Stirling integration index); the vertical position shows whether EPFL's publications within that subfield are more or less interdisciplinary than the corresponding global subfield mean. Thus, subfields displayed on the right (left) side are generally more (less) interdisciplinary; subfields at the top (bottom) indicate that EPFL's publications within these subfields are more (less) interdisciplinary than the global average. Reference lines mark the median global interdisciplinarity of the displayed subfields and no deviation from the subfield mean. Marker size reflects EPFL's output; values in parentheses report $MNCS_{frac}$. Only publications with at least five references that can be assigned to known classfication are considered (i.e., MNCS values can differ slightly). The indicator is purely based on the narrowness (less interdisciplinary) or broadness (more interdisciplinary) of citation profiles.

## 3.3. Collaboration and partners

*Main findings*

EPFL's publication profile is highly collaborative and combines a substantial domestic and European base with a strong and growing orientation towards global collaborations (at least one partner outside Europe). Across the evaluation period, 79% of EPFL's publications involve at least one external institutional partner, with a moderate 8% share of industry collaboration. 68% of EPFL's publications involved international collaboration, and 39.2% involved at least one partner institution outside Europe. The strongest recent increase in publication share is visible in these global collaborations that involve at least one partner outside Europe (+5.14pp), which are also associated with high citation impact ($PP[top\ 10\%]^{collab}_{frac}$ = 23.5%, $MNCS^{collab}_{frac}$ = 2.08). Within these global collaborations (at least one partner outside Europe), publications that combine Swiss, European and non-European partners are particularly impactful (25.5%, 2.25), and publications involving European and non-European partners showed the strongest recent growth within this group (+2.24pp). At the same time, national collaboration remains a meaningful component, present in 31% of EPFL's full-counted output, with University of Lausanne as the largest external Swiss partner. Within the ETH Domain, EPFL collaborates most with ETH Zurich (2,742 co-publications), which is also the ETH Domain partner associated with the highest citation impact ($MNCS_{frac}$ = 2.06). Outside Switzerland, EPFL's main partner countries are the USA, Germany, France, Italy, the United Kingdom and China. The strongest recent country-level increase is visible for collaborations involving China, whose co-authored share of EPFL's total output rose from 5.92% in 2014–2019 to 9.81% in 2020–2023.

### 3.3.1. Overview and trends

This section examines EPFL's collaboration profile over the evaluation period. It first describes the overall level and development of collaborative publishing, distinguishing between all collaborations, national collaborations, international collaborations, collaborations within the ETH Domain and industry collaborations. It then decomposes EPFL's publication output into mutually exclusive collaboration scopes to show how different collaboration patterns contribute to output and citation impact. The section subsequently turns to the main partner institutions within the ETH Domain, the national partner network outside the ETH Domain, international partner institutions and country-level collaboration patterns. Together, these perspectives describe both the scale and the structure of EPFL's collaboration system, while recognising that collaboration categories differ in definition and should not always be compared one-to-one.

Table 7 shows that EPFL's publication output remained broadly stable over the evaluation period under full counting, while collaborative output grew more clearly. In full counting, the overlapping four-year output rises from 16,887 publications in 2014–2017 to 17,853 in 2020–2023 (5.72%). Collaborative output grows more strongly over the same intervals, from 13,007 to 14,457 publications (11.15%). Fractional output, by contrast, declines modestly from 8,317 to 7,935 publications (-4.6%), while fractionally counted collaborative output remains broadly stable, edging up from 4,437 to 4,539 (2.29%). Non-collaborative fractional output declines markedly from 3,880 to 3,396 (-12.47%). Together, the stable full-counted output and the declining fractional output indicate that EPFL's overall publication output is increasingly distributed across larger multi-institutional collaborations rather than reflecting a proportional expansion of its own publication contributions.

The collaboration shares confirm a high and increasing level of collaborative publishing. Across the full evaluation period, 79% of EPFL's publications involve at least one external institutional partner. This share rises steadily across the intervals, from 77% in 2014–2017 to 81% in 2020–2023. International collaboration is by far the largest specific collaboration scope, increasing from 66% in 2014–2017 to 70% in 2020–2023. National collaboration also grows mildly, from 30% in 2014–2017 to 32% in 2020–2023, while collaboration within the ETH Domain remains a comparatively small, but rising component, increasing from 10% in 2014–2017 to 11% in 2020–2023. National collaboration outside the ETH Domain rises somewhat more clearly (23% to 25%). Industry collaboration is moderate and broadly stable at around 8%.

Overall, Table 7 points to a publication profile that is high and rising in its collaborative orientation, while EPFL's own fractional contribution per publication declines. The main temporal change is not in the general propensity to collaborate, which is already high throughout the period, but a gradual strengthening of all collaboration scopes, with international collaboration as the dominant component. National and ETH Domain collaboration shares grow alongside this shift, indicating that the increasing internationalisation does not displace EPFL's domestic collaboration base, but is layered on top of it.[13]

**Table 7.** Collaborative output over time. Publication output and collaborative output of EPFL by overlapping four-year intervals and for the full evaluation period. The upper panel distinguishes total output from collaborative output and reports both full and fractional publication counts. The lower panel expresses collaborative output as a share of total full-counted output, separately for different collaboration scopes. Since collaboration scopes are not mutually exclusive, the shares should be read as separate dimensions of collaboration rather than as components of a total.

| Indicator | 2014-2017 | 2015-2018 | 2016-2019 | 2017-2020 | 2018-2021 | 2019-2022 | 2020-2023 | Overall |
|---|---|---|---|---|---|---|---|---|
| Output | | | | | | | | |
| $P_{full}$ | 16,887 | 17,272 | 17,632 | 17,996 | 18,150 | 17,995 | 17,853 | 43,772 |
| $P[collab]_{full}$ | 13,007 | 13,405 | 13,777 | 14,156 | 14,418 | 14,428 | 14,457 | 34,527 |
| $P_{frac}$ | 8,317 | 8,387 | 8,464 | 8,483 | 8,392 | 8,176 | 7,935 | 20,558 |
| $P[collab]_{frac}$ | 4,437 | 4,520 | 4,609 | 4,643 | 4,660 | 4,609 | 4,539 | 11,313 |
| Share of collaborative output as % of total output ($PP[collab]_{full}$) by collaboration scope | | | | | | | | |
| All collaborations | 77% | 78% | 78% | 79% | 79% | 80% | 81% | 79% |
| Within ETH Domain | 10% | 10% | 10% | 10% | 11% | 11% | 11% | 11% |
| National (outside ETH Domain) | 23% | 23% | 23% | 24% | 24% | 25% | 25% | 24% |
| National (including within ETH Domain) | 30% | 30% | 30% | 31% | 31% | 32% | 32% | 31% |
| International | 66% | 66% | 67% | 67% | 68% | 69% | 70% | 68% |
| Industry collaboration | 8% | 8% | 9% | 9% | 9% | 9% | 9% | 8% |

Table 8 decomposes EPFL's publication output into mutually exclusive collaboration scopes, reporting publication output, output share, recent 4-year change, 10-year yearly development and citation impact for each scope. These scopes approximate the geographical reach of co-publication partnerships: publications without institutional collaborations (*No institutional collaborations*), publications with partners only within Switzerland (*Swiss collaborations*), publications with at least one non-Swiss European partner, but no partner outside Europe (*European collaborations*), and publications with at least one partner outside Europe (*Global collaborations*).

Each publication by EPFL is assigned to exactly one of these scopes, with more detailed subcategories

[13] For interpretation, it is important to note that within-ETH-Domain and external-partner collaboration indicators are built from different matching approaches because open institutional affiliation data and submitted institutional publication lists are not always perfectly aligned. Their levels should therefore not be read as directly comparable one-to-one (see Section 2.3.2).

highlighted below. The Swiss and European scopes should therefore be read as exclusive categories, not as total Swiss or European collaboration footprints, since publications involving at least one partner outside Europe are assigned to global collaborations even if they also include Swiss or European partners.[14]

Overall, EPFL's publication output is shaped by collaboration across a broad range of partnerships, with a clear orientation toward global and European partnerships. The largest collaboration scope is global collaborations, which accounts for 39.2% of output, followed by European collaborations with 28.5%. By contrast, purely Swiss collaborations account for a markedly smaller share of output (11.1%), while 21.2% of publications are published without institutional collaborations. This indicates that EPFL's collaboration profile is oriented primarily toward global and European partnerships rather than toward purely Swiss collaborations.

**Table 8.** Decomposition of EPFL's publication output by mutually exclusive, geographically expanding collaboration scopes. For each scope, the table reports full-counted publication output and output share, the change in that share from 2014–2019 to 2020–2023, annual full-counted output over 2014–2023 and citation impact. Subcategories are based on the full set of partner institutions listed on each publication. Swiss and European collaborations include only publications whose partner set does not extend beyond the respective geography and thus do not represent total Swiss or European collaboration footprints.

| Collaboration Scope | Output | | | | Impact | |
|---|---|---|---|---|---|---|
| Scope | $P_{full}$ | Share | Change (4y) | Trend | $MNCS_{frac}^{collab}$ | $PP[top\ 10\%]_{frac}^{collab}$ |
| **All publications** | 43,772 | 100% | +0.00pp | | 1.72 | 19.3% |
| **No institutional collaborations** | 9,279 | 21.2% | -3.62pp | | 1.58 | 17.7% |
| **Swiss collaborations** | 4,860 | 11.1% | +0.12pp | | 1.52 | 18.1% |
| ETH Domain only | 1,115 | 2.5% | +0.12pp | | 1.61 | 19% |
| Swiss (outside ETH Domain) only | 3,217 | 7.3% | -0.19pp | | 1.48 | 17.4% |
| Swiss (ETH Domain and others) | 528 | 1.2% | +0.20pp | | 1.58 | 20.9% |
| **European collaborations** | 12,476 | 28.5% | -1.64pp | | 1.70 | 18.7% |
| European only | 9,050 | 20.7% | -2.05pp | | 1.70 | 18.3% |
| Swiss + European | 3,426 | 7.8% | +0.41pp | | 1.70 | 20.4% |
| **Global collaborations** | 17,157 | 39.2% | +5.14pp | | 2.08 | 23.5% |
| Non-European only | 6,583 | 15% | +1.14pp | | 2.02 | 22.2% |
| European + non-European | 5,284 | 12.1% | +2.24pp | | 2.09 | 25% |
| Swiss + European + non-European | 5,290 | 12.1% | +1.76pp | | 2.25 | 25.5% |

*Note:* "European" refers to partners in Europe outside Switzerland. "Non-European" refers to partners outside Europe and Switzerland. Thus, "European + non-European" excludes other Swiss partners, whereas "Swiss + European + non-European" includes all three collaboration scopes.

The recent change indicators further suggest that EPFL's collaboration profile has shifted away from publications without institutional collaborations and from European collaborations (without partners outside Europe) toward global collaborations (at least one partner outside Europe). Publications without institutional collaborations declined by -3.62 percentage points over the recent 4-year period, while European collaborations recorded a smaller decline of -1.64 percentage points. Swiss collaborations remained essentially flat, changing by +0.12 percentage points overall. By contrast, global collaborations increased markedly by +5.14 percentage points. Within this broad category, non-European only collaborations account for the largest share of output (15%), while European + non-European collaborations showed the strongest recent increase (+2.24 percentage points). Taken together, this points to a growing

[14] Note that the scopes capture only partner presence, defined as at least one partner at a given scope level being present, rather than the depth or intensity of collaboration, such as the number of collaborating institutions involved from that scope. It is therefore possible that collaboration intensity within a scope may decline without this being visible in the scope decomposition, as long as at least one partner from that scope remains present on each publication.

importance of geographically extensive collaboration structures in EPFL's publication output.

Citation impact also varies across collaboration scopes. Publications without institutional collaborations and those in purely Swiss collaborations show the lowest impact values, with a $MNCS_{frac}^{collab}$ of 1.58 and $PP[top\,10\%]_{frac}^{collab}$ of 17.7% for publications without institutional collaborations and 1.52 and 18.1% for Swiss collaborations. European collaborations reach a slightly higher $MNCS_{frac}^{collab}$ of 1.7 and $PP[top\,10\%]_{frac}^{collab}$ of 18.7%. Collaboration scopes involving non-European partners reach clearly higher impact values. Global collaborations record a $MNCS_{frac}^{collab}$ of 2.08 and a $PP[top\,10\%]_{frac}^{collab}$ of 23.5%. Within this group, collaborations involving partners across all three geographic levels—within Switzerland, elsewhere in Europe and outside Europe (Swiss + European + non-European)—record the highest citation impact values ($MNCS_{frac}^{collab}$ = 2.25; $PP[top\,10\%]_{frac}^{collab}$ = 25.5%). This suggests that EPFL's most geographically extensive collaborations tend to be associated with higher citation visibility and are also among the collaborations that have shown the strongest recent growth.

### 3.3.2. Collaborations within the ETH Domain

After this broad decomposition of collaboration scopes, the following subsections examine the concrete partner structures behind these aggregate patterns, beginning with collaborations within the ETH Domain. Table 9 shows EPFL's co-authorships within the ETH Domain by full-count collaboration output and the average field-normalised citation impact of the corresponding joint publications.

**Table 9.** Collaboration and impact within the ETH Domain. The highlighted column reports EPFL's overall publication output and field-normalised citation impact across all in-scope publications, while the remaining columns report the corresponding joint publication output in full counting and the $MNCS_{frac}$ of publications co-authored with the respective ETH Domain institutions.

| Indicator | **EPFL** | **ETH Zurich** | **PSI** | **WSL** | **Empa** | **Eawag** |
|---|---|---|---|---|---|---|
| $P_{full}$ | 43,772 | 2,742 | 1,359 | 386 | 722 | 514 |
| $MNCS_{frac}$ | 1.72 | 2.06 | 1.31 | 1.34 | 1.81 | 1.72 |

Among the other ETH Domain institutions, EPFL collaborates most frequently with ETH Zurich, comprising 2,742 joint publications, while the smallest extent of collaboration is observed with WSL at 386 joint publications. The highest $MNCS_{frac}$ among these ETH Domain collaboration links is also observed for joint publications with ETH Zurich (2.06), while even the lowest mean normalised citation impact, found for collaborations with PSI (1.31), remains above global benchmarks. Overall, this indicates that EPFL's collaborations within the ETH Domain are characterised by a particularly strong link to ETH Zurich, both in terms of co-authored publications and citation impact, complemented by smaller, but consistently above-average links to the other ETH Domain institutions.

### 3.3.3. National collaborations (outside the ETH Domain)

Understanding where EPFL collaborates within Switzerland provides context for the institution's national partner network. Figure 7 provides a geographical overview of EPFL's Swiss partners by location, with highlighted points marking the top 10 national partners (by fractionally counted publication output and including ETH Domain).[15] Table 10 complements this perspective by showing the most important national and international partner institutions outside the ETH Domain, together with their joint publication output and absolute trend, their relative presence within EPFL's total collaborative output, how this has changed over the last four years compared with the preceding period and the corresponding impact indicators.

[15] Non-research institutions are included as collaboration partners throughout this section to provide a more complete picture of EPFL collaboration patterns. See Section 2.3.2 for limitations related to institution-type metadata.

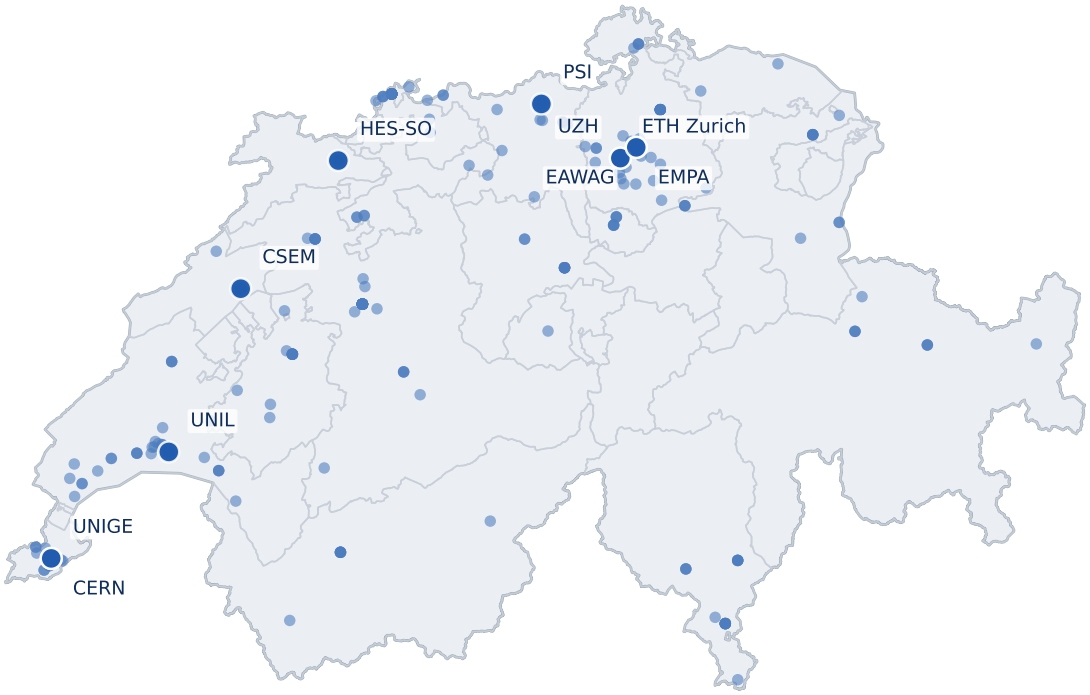



**Figure 7.** National collaborations of EPFL. Light-blue points mark collaborating Swiss institutions; dark-blue highlighted points and labels mark the top 10 Swiss partner institutions by collaborative output, based on fractional counting and including ETH Domain members and non-research institutions.

In terms of co-authored publication output, the ranking of national partners (outside the ETH Domain) is led by University of Lausanne with 550 fractionally counted joint publications, corresponding to 4.9% of EPFL's total collaborative output, followed by University of Geneva (3.3%) and Idiap Research Institute (2%).

Looking at the change in collaborations over time, Table 10 shows that the composition of EPFL's national partner portfolio is not static. Over the final four years of the evaluation period, the strongest increase in collaborative output among the top national partners is observed for University of Lausanne (4.9%) with an increase of 0.8 percentage points, while a notable decline among substantively identified partners[16] is found for University Hospital of Lausanne with a decline of -0.52 percentage points (overall at 1.1%). These shifts should be read as changes in relative weight within EPFL's overall collaboration portfolio rather than as direct measures of absolute collaboration growth or decline.

When it comes to citation impact, publications co-authored with IBM Research - Zurich and University of Zurich are among the highest-impact national collaborations, with these collaborative publications being cited on average more than 112% above field averages (MNCS>2.12) and with at least 24.4% of these publications belonging to the top 10% most-cited in their respective fields. Overall, this indicates that EPFL's national collaboration profile combines a strong anchor at the two large neighbouring Swiss universities, the University of Lausanne and the University of Geneva, with a broader set of public research institutes, university hospitals and non-academic partners that differ in relative weight, recent development and citation impact.

[16]The table also includes one OpenAlex institution record (id 4210091874, labelled by an address-style name) whose partner share declines to zero by 2023; this pattern is consistent with an upstream re-allocation of publications to a parent Swiss institution rather than a substantive change in collaboration and the entry is therefore not discussed individually in the prose.

### 3.3.4. International collaborations

Turning to EPFL's international collaboration partners, Figure 8 provides a geographical overview of partner institutions outside Switzerland. The figure shows that EPFL's international collaborations are broadly distributed across major global science regions, with the strongest visible concentration in Europe and additional dense clusters along the US East and West Coasts, in eastern Canada, as well as across East Asia.

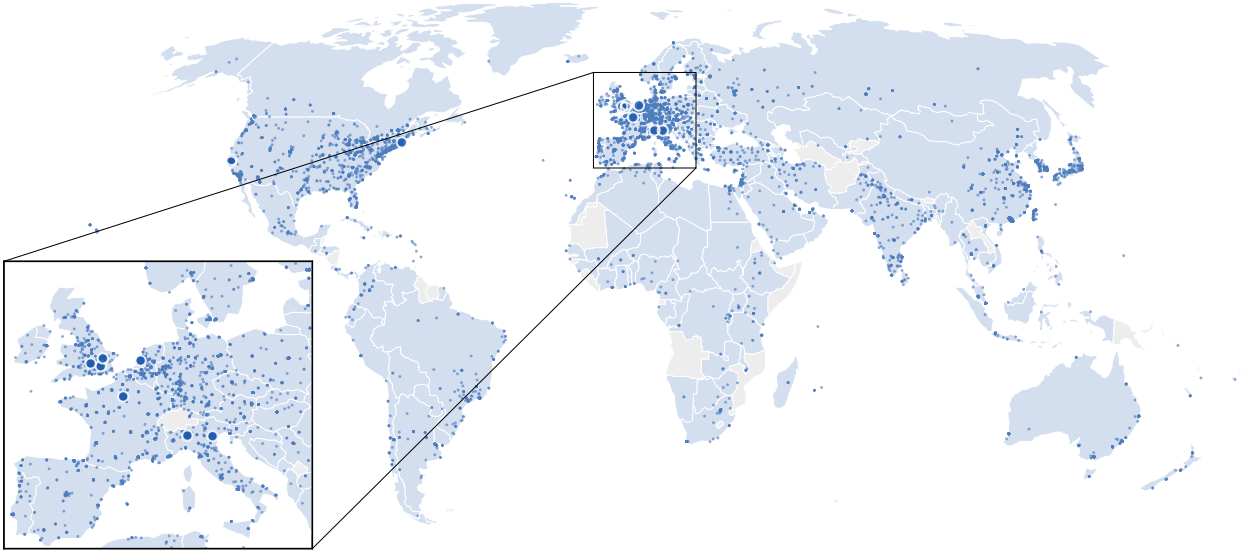


**Figure 8.** International collaborations of EPFL at institution level outside Switzerland. Light-blue points mark collaborating institutions, dark-blue points highlight the 10 partner institutions with the highest collaborative output, and the background shading indicates countries with at least one collaborating institution.

Table 10 complements this geographical perspective by identifying the most important partner institutions outside the ETH Domain.[17] Restricting the interpretation here to partners outside Switzerland, the largest international partner is Centre National de la Recherche Scientifique, with 318 fractionally counted joint publications, corresponding to 2.9% of EPFL's total collaborative output. It is followed by Politecnico di Milano (135; 1.2%), University of Cambridge (127; 1.1%), Imperial College London (121; 1.1%) and Delft University of Technology (120; 1.1%). Further major partners include Massachusetts Institute of Technology (119; 1.1%), University of California, Berkeley (114; 1%) and Stanford University (100; 0.9%). Overall, the leading international partner institutions combine a strong concentration in major European science systems—particularly France, the United Kingdom, Italy and the Netherlands—with prominent links to leading research universities in the United States.

The recent change indicators suggest moderate shifts in the relative weight of these international partnerships. Among the listed international partners, increases are visible for institutions such as Imperial College London (+0.53 percentage points), University of Cambridge (+0.32 points), Technical University of Denmark (+0.28 points), Harvard University (+0.24 points) and Politecnico di Milano (+0.21 points). Declines are observed for École Normale Supérieure - PSL (-0.71 points), University of California, Berkeley (-0.53 points) and University of Lisbon (-0.35 percentage points). These changes should be interpreted as shifts in relative weight within EPFL's overall collaboration portfolio rather than as direct measures of absolute growth or decline.

[17] When interpreting Table 10, the size of the underlying collaboration sets should be taken into account. Partner shares, recent percentage-point changes and citation impact indicators may be sensitive to small changes in the underlying co-authored publication set, especially where it contains only a limited number of publications.

**Table 10.** Top partner institutions outside the ETH Domain for EPFL. The table ranks partner institutions by fractional collaboration output. It reports the number of co-authored publications, the yearly development of this number, the partner's relative weight within EPFL's collaborative output, the recent change (last four years) in that relative weight and citation impact indicators for the corresponding joint publications.

| Partner | | | Output | | Relative Weight | | Impact | |
|---|---|---|---|---|---|---|---|---|
| Rank | Institution | CC | $P_{frac}$ | Trend | Share | Change (4y) | $MNCS_{frac}^{collab}$ | $PP[top\ 10\%]_{frac}^{collab}$ |
| 1 | Univ Lausanne | CH | 550 | | 4.9% | +0.80pp | 1.63 | 19.4% |
| 2 | Univ Geneva | CH | 371 | | 3.3% | +0.17pp | 1.73 | 20.6% |
| 3 | CNRS | FR | 318 | | 2.9% | -0.05pp | 1.84 | 21.2% |
| 4 | Idiap Res Inst | CH | 222 | | 2% | +0.38pp | 1.72 | 21.3% |
| 5 | Univ Zurich | CH | 140 | | 1.3% | +0.37pp | 2.12 | 24.4% |
| 6 | Swiss Ctr Electronics Microtechnology (Switzerland) | CH | 137 | | 1.2% | -0.42pp | 1.75 | 18% |
| 7 | Politecnico Milano | IT | 135 | | 1.2% | +0.21pp | 1.70 | 23.4% |
| 8 | Univ Cambridge | GB | 127 | | 1.1% | +0.32pp | 2.71 | 30.4% |
| 9 | HES-SO Univ Applied Sci Arts Western Switzerland | CH | 126 | | 1.1% | +0.62pp | 0.94 | 11.3% |
| 10 | CERN | CH | 125 | | 1.1% | -0.01pp | 1.53 | 17.5% |
| 11 | Univ Hosp Lausanne | CH | 121 | | 1.1% | -0.52pp | 1.70 | 20.7% |
| 12 | SIB Swiss Inst Bioinf | CH | 121 | | 1.1% | -0.30pp | 2.05 | 27.8% |
| 13 | Imp Coll London | GB | 121 | | 1.1% | +0.53pp | 2.58 | 29.7% |
| 14 | Delft Univ Technol | NL | 120 | | 1.1% | -0.11pp | 2.62 | 27.5% |
| 15 | MIT | US | 119 | | 1.1% | +0.06pp | 2.71 | 29.3% |
| 16 | Univ Calif Berkeley | US | 114 | | 1% | -0.53pp | 2.59 | 30.8% |
| 17 | Univ Bern | CH | 108 | | 1% | +0.04pp | 1.44 | 16.9% |
| 18 | Univ Hosp Geneva | CH | 101 | | 0.9% | +0.15pp | 1.43 | 17.7% |
| 19 | Stanford Univ | US | 100 | | 0.9% | -0.15pp | 3.72 | 34.3% |
| 20 | Ctr d'Imagerie BioMedicale | CH | 95 | | 0.9% | +0.47pp | 1.48 | 16.4% |
| 21 | Univ Oxford | GB | 94 | | 0.8% | -0.03pp | 2.52 | 28.5% |
| 22 | Univ Padua | IT | 93 | | 0.8% | +0.21pp | 1.62 | 18.9% |
| 23 | Univ Basel | CH | 92 | | 0.8% | +0.55pp | 1.62 | 23.9% |
| 24 | Harvard Univ | US | 89 | | 0.8% | +0.24pp | 3.47 | 35.5% |
| 25 | Univ Coll London | GB | 86 | | 0.8% | +0.15pp | 2.34 | 27.9% |
| 26 | CEA | FR | 83 | | 0.7% | +0.07pp | 1.87 | 16.3% |
| 27 | IBM Res - Zurich | CH | 80 | | 0.7% | +0.14pp | 2.73 | 32.8% |
| 28 | TU Munich | DE | 79 | | 0.7% | +0.19pp | 2.28 | 28.7% |
| 29 | Karlsruhe Inst Technol | DE | 78 | | 0.7% | -0.24pp | 2.07 | 20.2% |
| 30 | Univ Fribourg | CH | 78 | | 0.7% | +0.19pp | 1.78 | 16.2% |
| 31 | Scuola Superiore Sant'Anna | IT | 75 | | 0.7% | +0.16pp | 1.57 | 18.5% |
| 32 | Univ Lisbon | PT | 74 | | 0.7% | -0.35pp | 1.38 | 17.4% |
| 33 | Uppsala Univ | SE | 74 | | 0.7% | +0.00pp | 2.04 | 23.4% |
| 34 | Sorbonne Université | FR | 72 | | 0.6% | +0.13pp | 2.16 | 25% |
| 35 | Charles Humbert 8 | CH | 71 | | 0.6% | -0.66pp | 2.12 | 22.1% |
| 36 | KTH Royal Inst Technol | SE | 71 | | 0.6% | +0.02pp | 2.04 | 25.8% |
| 37 | Polytechnic Univ Turin | IT | 70 | | 0.6% | +0.04pp | 1.26 | 16% |
| 38 | Tech Univ Denmark | DK | 69 | | 0.6% | +0.28pp | 1.79 | 22% |
| 39 | Univ Bologna | IT | 68 | | 0.6% | -0.08pp | 1.38 | 16.7% |
| 40 | École Normale Supérieure - PSL | FR | 65 | | 0.6% | -0.71pp | 1.75 | 25.2% |

Citation impact is high for many of the major international partnerships. Among the international institutions listed in Table 10, particularly high citation impact is observed for Harvard University with a top 10% most-cited share ($PP[top\ 10\%]_{frac}^{collab}$) of 35.5% and mean normalised citation impact ($MNCS_{frac}^{collab}$) of 3.47, Stanford University (34.3%, 3.72), University of California, Berkeley (30.8%, 2.59), University of Cambridge (30.4%, 2.71), Imperial College London (29.7%, 2.58) and Massachusetts Institute of Technology (29.3%, 2.71). Taken together, the international partner profile combines consistently high citation impact with extensive collaboration links concentrated in major European research systems and leading research universities in the United States. Several US partners that contribute a relatively modest number of publications are nevertheless associated with particularly high citation impact, while the individual partnerships are anchored in continental Europe and the United Kingdom.

### 3.3.5. Country-level collaborations

This section complements the preceding institution-level partner analyses by aggregating collaborating institutions according to the countries in which they are located. The unit of analysis is therefore not the nationality or location of individual researchers, but the location of partner institutions appearing on EPFL publications. The results should be interpreted as the geographical distribution of institutional collaboration partnerships.

**Table 11.** Top countries by location of EPFL's collaborating institutions. The table reports, for each country, the number of EPFL collaborative works involving at least one partner institution located there and the corresponding share of EPFL's total output, using full counting. Collaboration impact is reported as ($MNCS_{frac}^{collab}$ and $PP[top\ 10\%]_{frac}^{collab}$) computed with focal-institutions fractional weighting over the collaborative works involving institutions located in each country.

| Country | | | Output | | | | Impact | |
|---|---|---|---|---|---|---|---|---|
| Rank | Country | CC | $P_{full}$ | Share | Change (4y) | Trend | $MNCS_{frac}^{collab}$ | $PP[top\ 10\%]_{frac}^{collab}$ |
| 1 | Switzerland | CH | 13,102 | 29.93% | +2.54pp | | 1.71 | 20.3% |
| 2 | USA | US | 9,213 | 21.05% | +0.72pp | | 2.39 | 27% |
| 3 | Germany | DE | 6,189 | 14.14% | +1.96pp | | 2.28 | 24.1% |
| 4 | France | FR | 6,152 | 14.05% | +0.81pp | | 1.69 | 19.3% |
| 5 | Italy | IT | 5,177 | 11.83% | +1.61pp | | 1.61 | 19.3% |
| 6 | United Kingdom | GB | 5,158 | 11.78% | +2.37pp | | 2.19 | 25.1% |
| 7 | China | CN | 3,286 | 7.51% | +3.90pp | | 2.38 | 27.7% |
| 8 | Spain | ES | 3,164 | 7.23% | +0.84pp | | 1.85 | 20.3% |
| 9 | Netherlands | NL | 2,515 | 5.75% | +1.09pp | | 2.18 | 25.6% |
| 10 | Australia | AU | 1,637 | 3.74% | +1.62pp | | 2.30 | 27.4% |
| 11 | Canada | CA | 1,603 | 3.66% | +1.31pp | | 1.88 | 20.4% |
| 12 | Sweden | SE | 1,541 | 3.52% | +1.32pp | | 2.07 | 25.1% |
| 13 | Japan | JP | 1,519 | 3.47% | -0.08pp | | 2.38 | 23.5% |
| 14 | Russia | RU | 1,482 | 3.39% | -1.56pp | | 1.60 | 17.1% |
| 15 | Poland | PL | 1,328 | 3.03% | -0.61pp | | 1.56 | 16.3% |
| 16 | Brazil | BR | 1,182 | 2.7% | +0.35pp | | 1.76 | 15% |
| 17 | Belgium | BE | 1,152 | 2.63% | +1.20pp | | 1.93 | 24.6% |
| 18 | Austria | AT | 1,025 | 2.34% | +0.79pp | | 2.59 | 26.9% |
| 19 | South Korea | KR | 1,023 | 2.34% | +0.64pp | | 3.06 | 25.5% |
| 20 | Denmark | DK | 970 | 2.22% | +1.11pp | | 1.98 | 25.3% |
| 21 | India | IN | 934 | 2.13% | +0.22pp | | 1.78 | 20.5% |
| 22 | Czechia | CZ | 803 | 1.83% | +0.43pp | | 2.04 | 23% |
| 23 | Portugal | PT | 803 | 1.83% | +0.75pp | | 1.57 | 20.6% |
| 24 | Ireland | IE | 796 | 1.82% | +0.09pp | | 1.83 | 18.4% |
| 25 | Finland | FI | 757 | 1.73% | +1.00pp | | 1.82 | 19.2% |
| 26 | Romania | RO | 678 | 1.55% | +0.14pp | | 1.25 | 9.3% |
| 27 | Ukraine | UA | 664 | 1.52% | +0.41pp | | 1.77 | 18.5% |
| 28 | Colombia | CO | 618 | 1.41% | +0.60pp | | 1.64 | 14.7% |
| 29 | Iran | IR | 606 | 1.38% | +0.81pp | | 1.36 | 16.7% |
| 30 | Greece | GR | 603 | 1.38% | +0.92pp | | 1.39 | 18.3% |

Table 11 highlights the countries of the institutions with which EPFL researchers most frequently collaborate. This profile is led by Switzerland, which here represents collaborations with Swiss institutions, including partners within the ETH Domain. Switzerland appears on 13,102 collaborative publications,

corresponding to 29.93% of EPFL's total output. It is followed by the USA (21.05%) and Germany (14.14%). Overall, the ranking points to a collaboration structure that combines a strong domestic component with dense links to the United States and to the major neighbouring European science systems—Germany, France, Italy and the United Kingdom—followed by sizeable collaboration with China and Spain.

The recent changes in share of total output show that this partner-country profile is not fully static, with both gains and declines across the listed countries (see also Figure 9). The strongest recent increase is visible for China (3.9 percentage points), rising from 5.92% in 2014–2019 to 9.81% in 2020–2023 of EPFL's total output. This is followed by Switzerland (2.54 percentage points) and the United Kingdom (2.37 percentage points). At the same time, the strongest decrease in share among the listed countries is observed for the Russian Federation (-1.56 percentage points of EPFL's total output). Taken together, this suggests a collaboration profile that remains anchored in a stable core of institutions located in established partner countries while gradually strengthening links to institutions in selected faster-rising countries, particularly China.

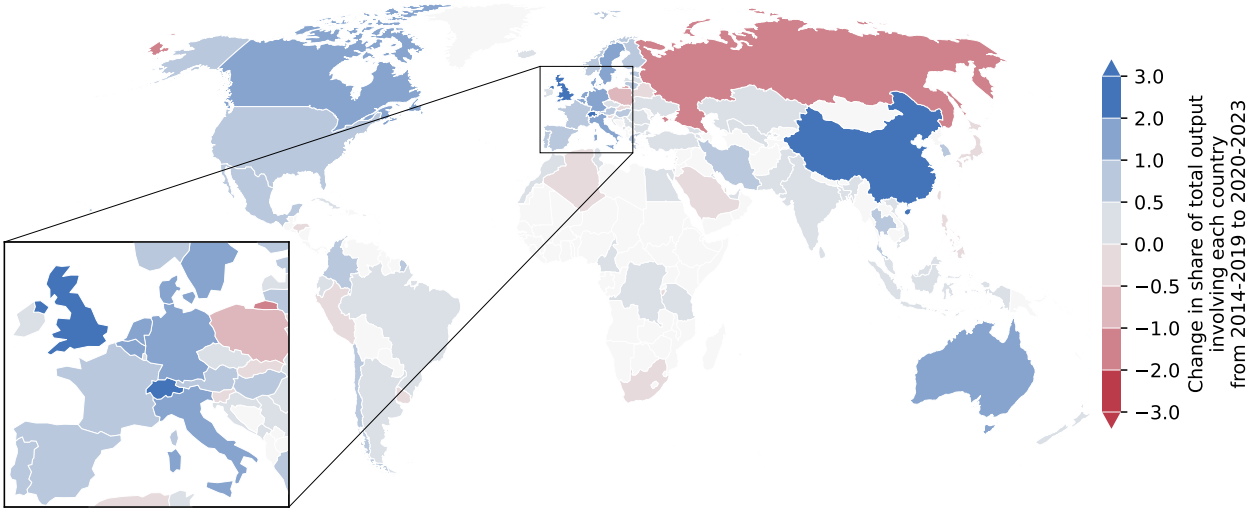



**Figure 9.** Change in the share of EPFL's total output involving each country, using full counting, comparing 2020–2023 with 2014–2019. Blue shades indicate an increasing share, red shades indicate a decreasing share, and stronger colour intensity denotes larger percentage-point changes.

The citation impact of EPFL's collaborative publications remains high across the listed countries. The highest $\text{MNCS}_{\text{frac}}^{\text{collab}}$ values are observed for collaborations involving South Korea (3.06), Austria (2.59) and the USA (2.39). A similar pattern is visible for $\text{PP[top 10\%]}_{\text{frac}}^{\text{collab}}$, where 27.7% of collaborative output involving China are among the top 10% most-cited publications. Overall, the results suggest that collaborations with institutions located outside Switzerland play a central role for EPFL, both in terms of publication output—where the United States and major European partners dominate—and in terms of citation impact, with several large international partner countries reaching mean normalised citation values well above two.

## 3.4. Research accessibility

*Main findings*

Open access has become the dominant publication mode at EPFL. Across the full evaluation period, 61.7% of EPFL's publications are published as open access, and this share increased strongly from 51% in 2014–2017 to 73% in 2020–2023. Open-access publications show higher overall citation impact than closed-access publications, with an overall PP[top 10%]$_{frac}$ of 21.3% compared with 16.3% for closed-access publications, and an MNCS$_{frac}$ of 1.8 compared with 1.59, with green open access showing the highest citation-impact values among the open-access categories. Open-access adoption is high across the relevant subfields, exceeding the corresponding subfield average by at least 26%. The level of adoption nevertheless varies by subfield, ranging from 47% in Civil and Structural Engineering to 83% in Nuclear and High Energy Physics. If citations to open-access publications are considered as benchmark, closed-access publications are underrepresented among incoming citations from many Western science systems, whereas closed-access publications appear to be less associated with reduced citation visibility within Switzerland and regions such as China.

### 3.4.1. Overview and trends

This section examines the accessibility of EPFL's research output through the lens of open-access publishing. The analysis uses open-access metadata provided directly by OpenAlex for the matched publications included in the analytical scope. OpenAlex applies a broad definition of open access, identifying publications as open when a freely readable full-text version can be found without payment or login. It further distinguishes between gold/diamond access in fully open-access journals, green access through repository copies, hybrid access for openly licensed publications in otherwise paid-access journals, bronze access where publications are free to read, but lack an identifiable open licence and closed access for all other publications.

These categories are used descriptively to assess how EPFL's open-access profile has developed over time and how open-access publications compare with closed-access publications in terms of citation-based impact.[18] In the analyses below, gold publications are reported together with diamond open access. For the broader open/closed comparison, gold/diamond, green and hybrid publications are considered as open access, while bronze publications are grouped with closed publications because they do not provide identifiable open-licence reuse rights.[19] This grouping should be kept in mind when interpreting the following results.

Table 12 decomposes EPFL's publications across the different accessibility categories, showing that open access constitutes the majority publication mode. Of 43,772 publications, 26,998 are classified as open access, corresponding to 61.7% of total output, while 38.3% are closed access, including bronze. Within the open-access category, green open access accounts for the largest share of total output (27.6%), followed by gold open access (19.4%) and hybrid open access (14.7%).

Open-access publications also show higher overall citation impact than closed-access publications. Their PP[top 10%]$_{frac}$ share is 21.3%, compared with 16.3% for closed-access publications, and their MNCS$_{frac}$

[18] Detailed results on the development of each open-access type are reported in the Annex C.5.

[19] Bronze publications are free to read on the publisher's platform, but they carry no identifiable open licence. It is therefore unclear whether free availability is temporary or permanent and reuse rights beyond reading cannot be assumed.

is 1.8 compared with 1.59. Among the open-access categories, green open access stands out most clearly, combining the largest number of publications with the highest citation-impact ($PP[top\ 10\%]_{frac}$: 23.8%, $MNCS_{frac}$: 2.07). Hybrid open access also shows elevated impact values (20.8%, 1.71), while gold open access shows lower citation-impact values (17.6%, 1.43) that are also below those of closed-access publications.

**Table 12.** Open-access profile of EPFL by access category. Shown are total output, closed access including bronze, open access overall and the open-access categories gold including diamond, green and hybrid. The indicators cover publication output, share of total output under full counting, high-impact output, normalised citation impact and international collaboration share under full counting for the full evaluation period.

| | Total | Access status | | Open-access category | | |
|---|---|---|---|---|---|---|
| **Indicator** | **Total** | **Closed access incl. bronze** | **Open access** | **Gold incl. diamond** | **Green** | **Hybrid** |
| $P_{full}$ | 43,772 | 16,774 | 26,998 | 8,507 | 12,074 | 6,417 |
| Share of total | 100.0% | 38.3% | 61.7% | 19.4% | 27.6% | 14.7% |
| $PP[top\ 10\%]_{frac}$ | 19.3% | 16.3% | 21.3% | 17.6% | 23.8% | 20.8% |
| $MNCS_{frac}$ | 1.72 | 1.59 | 1.80 | 1.43 | 2.07 | 1.71 |
| $PP[int\ collab]_{full}$ | 67.7% | 66.5% | 68.4% | 70.3% | 67.0% | 68.6% |

The collaboration indicator ($PP[int\ collab]_{full}$) suggests that differences in international collaboration between open- and closed-access output are small overall, with 68.4% international collaboration for open access compared with 66.5% for closed-access publications. Within the open-access categories, gold open access shows the highest international collaboration share (70.3%). Overall, Table 12 points to a broad open-access profile in which accessibility is associated with higher citation impact, especially for green open access, while differences between categories should be read cautiously because they may also reflect disciplinary publishing cultures, collaboration patterns and repository practices.

Table 13 shows the same indicators for open- and closed-access publications over time,[20] providing a more detailed picture of the open-access development at EPFL. Open-access output increased steadily across the four-year moving intervals, from 8,579 (51%) publications in 2014–2017 to 13,071 (73%) in 2020–2023. Over the same period, closed publication output declined from 8,308 to 4,782 publications. This points to a substantial shift in EPFL's publication portfolio toward more openly accessible research output throughout the evaluation period.

The citation-impact indicators show above-average performance for both open- and closed-access publications, with a consistent overall advantage for open-access output. Over time, the share of highly cited open-access publications declines gradually from 23% in the early intervals to 19% in 2020–2023, while the corresponding share for closed-access publications also eases from 16.6% in 2014–2017 to 15.5% in 2020–2023. Mean normalised citation impact remains higher for open-access publications than for closed-access publications in each interval, with the overall $MNCS_{frac}$ for open access easing from 1.91 in 2014–2017 to 1.63 in 2020–2023. These differences should not be interpreted as a causal effect of open access.

The international collaboration indicator shows a more differentiated pattern. Open-access publications have a consistently high international collaboration share throughout the period, increasing slightly from 67% in 2014–2017 to 69% in 2020–2023. By contrast, the international collaboration share among closed-access publications increases more clearly, from 64% in 2014–2017 to 71% in 2020–2023. One possible interpretation may be that the expansion of open-access output occurred while maintaining an

[20] Detailed results on the development of each open-access type are reported in the Annex C.5.

already highly international publication profile, whereas the shrinking closed-access segment may be increasingly concentrated in internationally collaborative publications.

**Table 13.** Open- and closed-access publication profile of EPFL. Shown are number of publications for open- and closed-access output, together with high-impact output, normalised citation impact and the share of internationally collaborative publications across four-year moving intervals and for the full evaluation period.

| Status | Indicator | 2014-2017 | 2015-2018 | 2016-2019 | 2017-2020 | 2018-2021 | 2019-2022 | 2020-2023 | Overall |
|---|---|---|---|---|---|---|---|---|---|
| Open | $P_{full}$ | 8,579 | 9,115 | 9,862 | 10,893 | 11,826 | 12,573 | 13,071 | 26,998 |
| | Share of total | 51% | 53% | 56% | 61% | 65% | 70% | 73% | 62% |
| | PP[top 10%]$_{frac}$ | 22.7% | 23.0% | 22.8% | 22.2% | 21.5% | 20.0% | 19.4% | 21.3% |
| | MNCS$_{frac}$ | 1.91 | 1.98 | 1.97 | 1.88 | 1.83 | 1.68 | 1.63 | 1.80 |
| | PP[int collab]$_{full}$ | 67% | 68% | 68% | 68% | 69% | 69% | 69% | 68% |
| Closed | $P_{full}$ | 8,308 | 8,157 | 7,770 | 7,103 | 6,324 | 5,422 | 4,782 | 16,774 |
| | Share of total | 49% | 47% | 44% | 39% | 35% | 30% | 27% | 38% |
| | PP[top 10%]$_{frac}$ | 16.6% | 16.6% | 16.6% | 16.4% | 16.2% | 15.6% | 15.5% | 16.3% |
| | MNCS$_{frac}$ | 1.67 | 1.59 | 1.57 | 1.54 | 1.48 | 1.53 | 1.51 | 1.59 |
| | PP[int collab]$_{full}$ | 64% | 65% | 66% | 66% | 68% | 69% | 71% | 67% |

Taken together, these results indicate a clear shift in EPFL's publication profile toward open access, with open-access publications now forming the dominant output mode while maintaining strong citation visibility and high levels of international collaboration.

### 3.4.2. Open access by subfields

In this section, we examine EPFL's open-access adoption across subfields. Figure 10 shows EPFL's fractional publication output within each subfield in the left panel, the share of open-access publications together with the corresponding subfield averages as benchmarks in the middle panel and EPFL's relative open-access adoption compared with these benchmarks in the right panel (see also Figure 15 in Annex C.4 for the corresponding aggregation at the higher field level). Among the subfields shown, the highest open-access share is observed for EPFL publications in Nuclear and High Energy Physics, reaching 83%. This is followed by publications in Condensed Matter Physics (79%) and Cognitive Neuroscience (68%). The lowest open-access share among EPFL's relevant subfields is observed in Civil and Structural Engineering, with roughly 47%. The subfield averages shown in the middle panel of Figure 10 indicate considerable heterogeneity in open-access publishing norms across subfields, underscoring that absolute open-access shares should be interpreted in relation to disciplinary context.

To account for these subfield-specific differences, we also consider EPFL's open-access adoption relative to the corresponding subfield averages as benchmarks (right panel of Figure 10). Values above one indicate that EPFL's open-access share in a given subfield is higher than the respective subfield average, while values below one indicate below-average open-access adoption. In relative terms, the highest relative adoption is observed in Computational Mechanics, with a ratio of 2.55 (more than twice as high relative to the subfield average), while the lowest ratio among the relevant subfields is found in Molecular Biology, with a ratio of 1.26 (roughly 26% higher relative to the subfield average). Even this lowest ratio remains above one, indicating that EPFL's open-access share exceeds the corresponding field average across the displayed subfields. Overall, the results suggest that, although some relevant output remains in closed-access publishing, EPFL's open-access adoption is relatively high across the relevant subfields and consistently exceeds subfield-specific publishing norms.

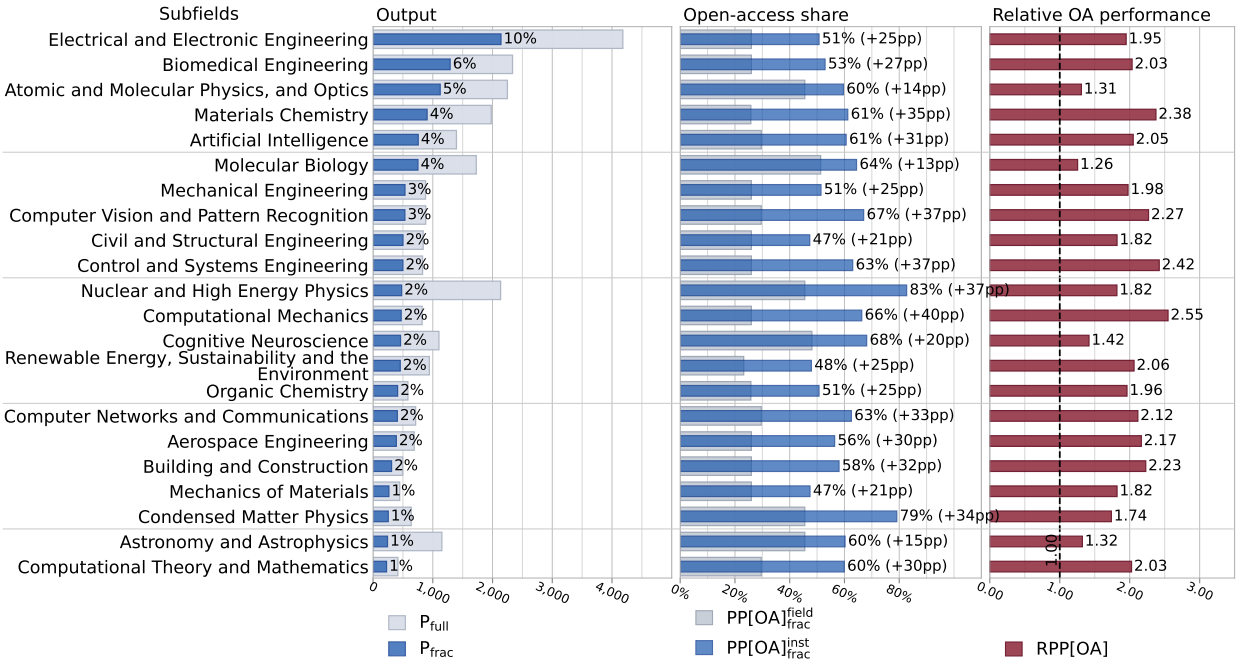



**Figure 10.** Open-access profile of EPFL across subfields. Shown are subfields that account for at least 1% of EPFL's total fractional publication output and have at least 50 fractionally counted publications. **Left** (Output): Publication output by subfield in fractional (dark blue and percentages of total) and full (light blue) counting. **Middle** (Open-access share): Share of EPFL's open-access publications and corresponding field averages. Numbers indicate percentages and percentage differences to field average (in percentage points). **Right** (Relative OA performance): Ratio of EPFL's open-access share to the respective field averages. Dashed line indicates parity with the field average.

### 3.4.3. Open access and geographic citation patterns

Figure 11 complements this subfield view with a geographic comparison of incoming citation origins for open- versus closed-access output. The analysis is conducted for the ETH Domain as a whole, rather than separately for EPFL, in order to provide a common reference frame and avoid overinterpreting small institution-specific differences. The figure compares where citations to ETH Domain open- and closed-access publications originate using a relative measure: $\log_2$(open incoming citation share / closed incoming citation share) for each location. Positive values indicate that a location accounts for a larger share of incoming citations to open-access publications than to closed-access publications; negative values indicate that, using citations to open-access publications as the benchmark, closed-access publications are relatively more represented in incoming citation patterns from that location.

This share-based normalisation is important because open- and closed-access publications differ in both publication output and citation impact. The indicator therefore does not compare absolute citation counts from each location. Instead, it captures how strongly each location is represented among citations to open-access publications relative to citations to closed-access publications, after normalising separately for the total number of incoming citation links to each access type. In this specification, incoming citation links are counted at citation-edge level (distinct cited ETH Domain publications per citing publication) and then aggregated to citing institutions on a full-count basis. Since almost all locations contribute more citation links to open- than to closed-access publications in absolute terms, the figure should be read as an exploratory citation-origin or readership indicator that highlights relative geographic differences within this broader citation pattern.

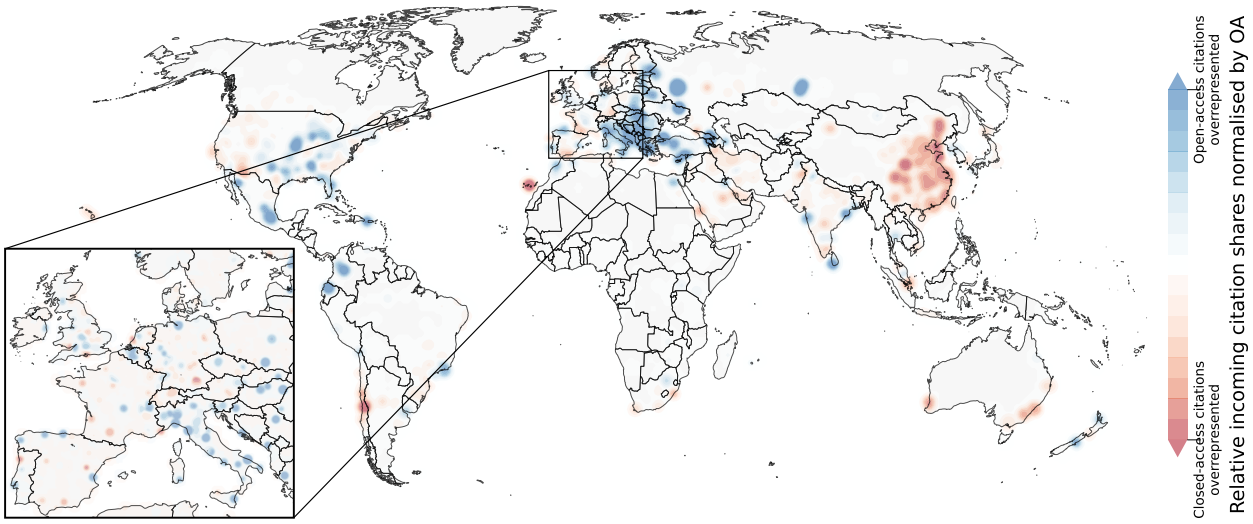



**Figure 11.** Geographic contrast in incoming citation origins for open- versus closed-access ETH Domain publications. The mapped value is $\log_2$(open incoming citation share / closed incoming citation share). Shares are based on incoming citation-link counts (distinct cited ETH Domain publications per citing publication), aggregated to citing institutions (full counting). Values above 0 indicate locations relatively more represented among citations to open-access publications; values below 0 indicate relatively stronger representation among citations to closed-access publications.

Overall, the pattern suggests that both open- and closed-access publications from the ETH Domain receive citations from a broad international base, but that the relative balance between the two access types varies geographically. Open-access publications account for a relatively larger share of incoming citations from many Western science systems, especially in Central and Western Europe and selected areas of North America. By contrast, this relative open-access orientation is less pronounced in Switzerland and parts of East Asia, especially China, where closed-access publications are less underrepresented in incoming citation patterns. This does not imply that researchers from these locations cite closed-access publications more frequently in absolute terms; given the larger output and higher citation impact of open-access ETH Domain publications, open-access publications may still receive more citations overall, including in regions where citations to closed-access publications are relatively over-represented. Rather, the figure indicates that, compared with the global citation-origin distribution for open-access publications, Switzerland and China account for a comparatively larger share of citations to closed-access publications.

## 3.5. Gender distribution of authors

*Main findings*

Female authors account for 29.1% of EPFL's gender-inferred affiliated author base over the evaluation period. This share increases gradually from 27.3% in 2014–2017 to 29.8% in 2020–2023, while the corresponding share among all contributing authors on EPFL's publications rises slightly more strongly from 27.1% to 30.7%. Over the full period, the female share among EPFL's affiliated authors is broadly in line with the share among all contributing authors on the same publications (RPA[F]: 0.98). Subfield-level results show substantial heterogeneity, with comparatively higher female-author shares among EPFL's affiliated authors in publications in Organic Chemistry (33%), Molecular Biology (31%) or Building and Construction (29%), and lower shares in subfields such as Mechanics of Materials (11%), Astronomy and Astrophysics (11%) or Computational Mechanics (13%). Publication-level analyses qualify the near-parity result on the unique-author base. Female EPFL-affiliated authors account for lower shares of repeated authorship appearances than the same-publication benchmark, both under author weighting (22.4% vs. 24.6%; $RP\overline{A}[F]^{pub}$: 0.91) and under equal publication weighting (21.1% vs. 21.9%; $R\overline{PA}[F]^{pub}$: 0.96). This indicates that the near-parity female share in EPFL's unique affiliated author base does not translate into the same level of female representation across publication-level authorship appearances, with the relative gap being more pronounced in larger author teams.

### 3.5.1. Overview and trends

This section examines the gender distribution of EPFL's affiliated author base as reflected in the publication data. The analysis uses probabilistically inferred author-gender information linked to the matched publications included in the analytical scope. Gender is assigned at author level and then used to describe the composition of authors affiliated with EPFL and, where relevant, the wider set of contributing authors. Due to the probabilistic nature of the inference, these indicators should be interpreted with caution. The results therefore do not provide a complete measure of diversity or equality within EPFL.

Table 14 summarises the coverage of inferred gender information used in the author-gender analysis. For EPFL, 15,357 of 18,727 affiliated authors could be assigned to one of the inferred gender categories, while 3,370 authors remain unassigned, corresponding to an unknown share of 18%. Coverage is substantially lower for the wider set of all contributing authors: 98,088 of 178,270 authors could be assigned, while 80,182 remain unassigned, corresponding to an unknown share of 45%. The author-gender indicators should therefore also be interpreted with caution. They describe the identifiable part of the author population rather than the full author base, and this limitation is more pronounced for all contributing authors than for authors affiliated directly with EPFL.

**Table 14.** Coverage of inferred gender information for affiliated and all contributing authors on EPFL's publications.

| | Total authors | Identified | Unknown | Unknown share |
|---|---|---|---|---|
| Affiliated authors ($A_{aff}$) | 18,727 | 15,357 | 3,370 | 18.0% |
| All contributing authors ($A_{all}$) | 178,270 | 98,088 | 80,182 | 45.0% |

The following analyses in this section describe the gender distribution among EPFL-affiliated authors and how it developed over time. All subsequent analyses consider only authors with inferred gender.

Reported shares therefore represent the proportions within the gender-inferred author population, rather than proportions of the full author base.

To put these shares into perspective, we compare affiliated authors with all contributing authors on the same EPFL publication set. This own-publication benchmark keeps the comparison within the same field mix, publication types and collaboration context. The key comparison is the affiliated female share ($PA[F]_{aff}$) versus the publication-level female share ($PA[F]_{all}$), summarised by RPA[F]. Values greater than one indicate that EPFL-affiliated female authors are more strongly represented compared with all authors, affiliated and non-affiliated, on the same publications; values below one indicate lower representation.

Table 15 summarises the gender distribution among authors and compares EPFL's affiliated author base with the wider set of all contributing authors on the same publications as benchmark. Over the full period, EPFL has 15,357 affiliated authors, of whom 4,471 are female and 10,886 are male. This corresponds to a female share of 29.1% among affiliated authors. A very similar full-period female share is observed among all contributing authors on EPFL's publication set, where 29,173 of 98,088 authors are female, corresponding to 29.7%. The relative representation indicator, RPA[F], compares the female share among EPFL affiliated authors with the female share among all contributing authors on the same publications. Over the full period, this value is 0.98, indicating that female authors are represented at about the same level within EPFL's affiliated author base as in the wider co-author population linked to the same publication set.

**Table 15.** Moving-average gender trend and own-publication benchmark for EPFL. Shown are the overlapping four-year intervals and the full-period summary for affiliated and all contributing authors on the same publication set. It includes absolute counts for affiliated and publication-level female and male authors, female shares among identified authors and the relative female representation ratio (RPA[F]), comparing the affiliated female share to the publication-level female share.

| Indicator | 2014-2017 | 2015-2018 | 2016-2019 | 2017-2020 | 2018-2021 | 2019-2022 | 2020-2023 | Overall |
|---|---|---|---|---|---|---|---|---|
| Affiliated authors on own publication set | | | | | | | | |
| $A_{aff}$ | 9,352 | 9,569 | 9,801 | 10,027 | 10,045 | 10,310 | 10,596 | 18,727 |
| $A[F]_{aff}$ | 2,048 | 2,150 | 2,232 | 2,371 | 2,442 | 2,563 | 2,625 | 4,471 |
| $A[M]_{aff}$ | 5,460 | 5,621 | 5,793 | 5,901 | 5,984 | 6,123 | 6,180 | 10,886 |
| $PA[F]_{aff}$ | 27.3% | 27.7% | 27.8% | 28.7% | 29.0% | 29.5% | 29.8% | 29.1% |
| All contributing authors on the same publication set | | | | | | | | |
| $A_{all}$ | 63,149 | 63,341 | 67,368 | 77,632 | 98,203 | 105,482 | 115,792 | 178,270 |
| $A[F]_{all}$ | 9,807 | 10,296 | 11,198 | 12,788 | 15,250 | 16,480 | 17,815 | 29,173 |
| $A[M]_{all}$ | 26,429 | 27,561 | 29,683 | 32,542 | 36,769 | 38,688 | 40,200 | 68,915 |
| $PA[F]_{all}$ | 27.1% | 27.2% | 27.4% | 28.2% | 29.3% | 29.9% | 30.7% | 29.7% |
| Relative female representation | | | | | | | | |
| RPA[F] | 1.01 | 1.02 | 1.02 | 1.02 | 0.99 | 0.99 | 0.97 | 0.98 |

Over the evaluation period, the female share among affiliated authors increases gradually from 27.3% in 2014–2017 to 29.8% in 2020–2023. A similar, slightly stronger increase is visible among all contributing authors, where the female share rises from 27.1% to 30.7%. The relative representation indicator stays close to parity throughout the period, ranging between 0.97 and 1.02 and shifts from 1.01 in 2014–2017 to 0.97 in 2020–2023. Overall, female authors remain considerably less represented (29%) than male

authors within the affiliated author population. At the same time, this share is broadly in line with the benchmark of all contributing authors on the same publications and the gradual, but slightly stronger increase observed in the wider author environment shifts the relative representation indicator from values just above one in the earlier intervals to values just below one in the most recent moving four-year intervals.

These results should be interpreted cautiously for two reasons. First, RPA[F] is based on unique author populations within each reporting interval, so author shares are not additive across time periods. For example, one male author publishing in two sub-periods is counted once in each sub-period and once as male over the full period, while two female authors (e.g., on temporary positions) publishing in one sub-period each are counted once in their respective sub-periods, but are counted as two female authors over the full period. The same logic applies to other partitions of the publication set, such as fields or subfields.

Second and relatedly, RPA[F] does not capture how often individual authors appear across publications. It should therefore not be read as a publication-level measure of female participation or perceived gender composition in the publication output. Repeated authorship, collaboration structures and differences in publication intensity may lead to a different picture when gender distribution is assessed at the publication level; this complementary perspective is addressed in Section 3.5.3.

### 3.5.2. Gender distribution of authors by subfields

This section examines female author representation within EPFL's high-output subfields. The subfield shares should be interpreted as context-specific publication shares rather than as components of the overall gender statistic. The overall statistic counts each author only once within the analysed publication set, whereas the subfield view counts author appearances within subfield-specific publication subsets. The same author may contribute to publications in more than one subfield. The subfield percentages are therefore useful for comparing publication contexts, but they should not be read as a decomposition of EPFL's overall author-gender distribution.

Figure 12 shows that female affiliated-author representation varies considerably across EPFL's publication subfields (see also Figure 16 in Annex C.4 for higher-level aggregation). The female share among EPFL-affiliated authors is highest in Organic Chemistry (33%), Molecular Biology (31%) and Building and Construction (29%). By contrast, lower female shares are observed in Mechanics of Materials (11%), Astronomy and Astrophysics (11%) and Computational Mechanics (13%).

In most subfields, the female share among EPFL-affiliated authors is slightly below the corresponding share among all contributing authors on the same publications. This is reflected in relative representation values below one for most subfields. The largest negative difference is visible in Astronomy and Astrophysics, where female affiliated-author representation is more than half as high as the share among all contributing authors, followed by Mechanics of Materials (roughly 24% lower) and Cognitive Neuroscience (roughly 20% lower). In these cases, women account for a smaller share of EPFL-affiliated authors than of the wider author teams contributing to the same publications. By contrast, the highest female representation among EPFL authors compared to the benchmark is found in Building and Construction, where female affiliated-author representation is roughly 11% higher relative to the benchmark. This is followed by Civil and Structural Engineering (roughly 9% higher) and Nuclear and High Energy Physics (roughly 7% higher).

Figure 12 points overall to substantial subfield-level heterogeneity in female affiliated-author representation. The highest absolute shares appear in selected chemistry-, life-science- and construction-related subfields, while the lowest shares are concentrated in mechanics- and physics-heavy subfields. Relative to the wider author teams on the same publications, most publication contexts show slightly lower female

representation among EPFL-affiliated authors, with a particularly pronounced gap in Astronomy and Astrophysics, where large international consortia bring in many co-authors from outside EPFL.

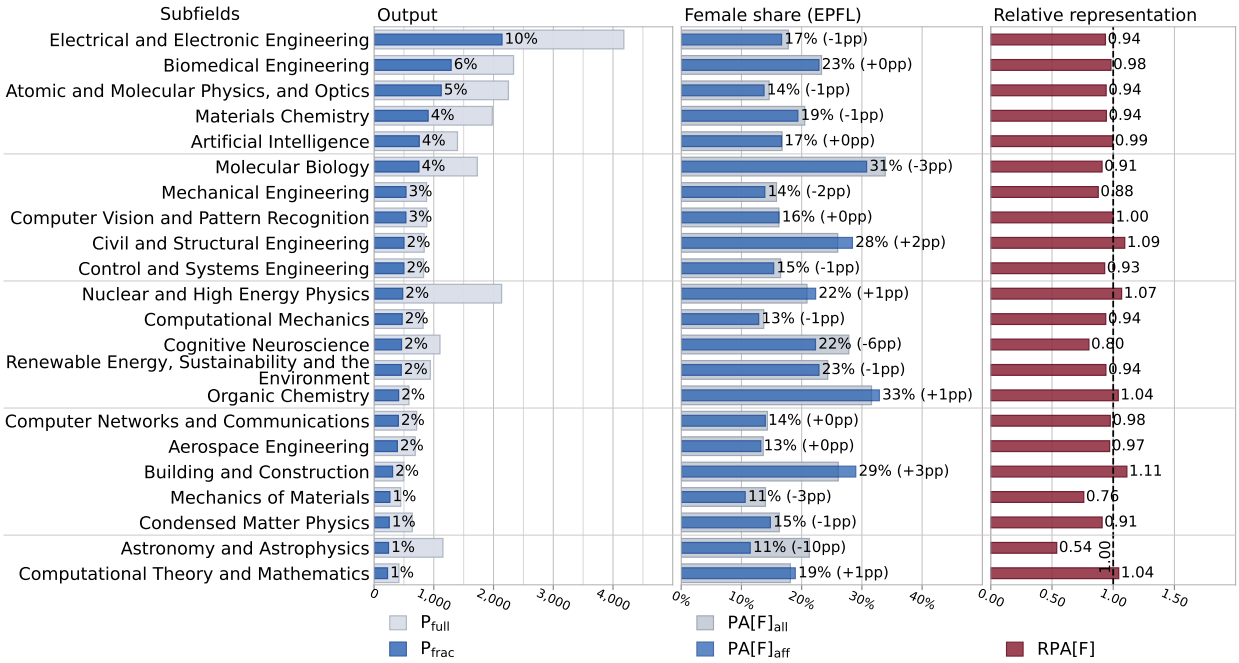



**Figure 12.** Female affiliated-author representation across subfields for EPFL. Shown are subfields that account for at least 1% of EPFL's total fractional publication output and have at least 50 fractionally counted publications. Each row represents one subfield; rows are ordered by fractional publication output. **Left** (Output): Dark blue bars show fractional publication counts, while light blue bars show full counts attributed to EPFL in each subfield. **Middle** (Female share): Dark blue bars show the female share among EPFL-affiliated authors; light blue background bars show the female share among all authors on the same publications. Labels indicate the affiliated-author female share, followed by its percentage-point difference from the all-author female share. **Right** (Relative representation): Bars show the ratio between the affiliated-author female share and the all-author female share. Values below 1 indicate lower female representation among EPFL-affiliated authors than among all contributing authors on the same publications; values above 1 indicate higher representation among EPFL-affiliated authors.

### 3.5.3. Gender distribution of authors per publication

This section extends the preceding gender analysis by shifting the unit of observation from unique authors to publications. Whereas the previous results count each author only once within the analytical scope, the publication-level indicators used here count authors separately for each publication in which they appear. Authors may therefore enter the calculations multiple times, depending on their publication activity. This perspective captures the gender composition of authorship as it appears across publications, rather than the gender composition of the underlying author population.

Table 16 reports average author counts per publication and two publication-level measures of the female author share. Each measure is shown for EPFL-affiliated authors on EPFL's own publication set (subscript "aff") and for all contributing authors on the same publications (subscript "all"). The author-weighted female share, $\overline{PA[F]}^{pub}_{aff}$ for affiliated authors and $\overline{PA[F]}^{pub}_{all}$ for all authors, divides the average number of female authors per publication by the average number of female and male authors per publication combined. Publications with larger author teams therefore have more influence on this measure. The equal-publication-weighted female share, $\overline{\overline{PA[F]}}^{pub}_{aff}$ for affiliated authors and $\overline{\overline{PA[F]}}^{pub}_{all}$ for all authors, first calculates the female share separately within each publication and then averages these publication-level shares. In this second measure, each publication contributes equally, regardless of team size. Publications without any female or male authors in the relevant author group are excluded from the

equal-publication-weighted average; in both measures, authors with unknown gender are excluded from the denominator.

**Table 16.** Per-publication gender trend for EPFL. The table reports overlapping four-year intervals and the full-period summary for affiliated authors and all contributing authors on EPFL's publication set. It shows average author counts per publication, average female-author counts per publication, author-weighted female shares, equal-publication-weight averages of publication-level female shares and the corresponding relative female representation ratios.

| Indicator | 2014-2017 | 2015-2018 | 2016-2019 | 2017-2020 | 2018-2021 | 2019-2022 | 2020-2023 | Overall |
|---|---|---|---|---|---|---|---|---|
| Affiliated authors on own publication set | | | | | | | | |
| $\bar{A}_{aff}^{pub}$ | 2.75 | 2.73 | 2.72 | 2.72 | 2.68 | 2.66 | 2.64 | 2.69 |
| $\bar{A}[F]_{aff}^{pub}$ | 0.40 | 0.42 | 0.43 | 0.45 | 0.47 | 0.49 | 0.50 | 0.45 |
| $P\bar{A}[F]_{aff}^{pub}$ | 20.2% | 20.7% | 21.4% | 22.0% | 22.7% | 23.7% | 24.5% | 22.4% |
| $\overline{PA}[F]_{aff}^{pub}$ | 19.0% | 19.4% | 20.0% | 20.7% | 21.5% | 22.5% | 23.3% | 21.1% |
| All contributing authors on the same publication set | | | | | | | | |
| $\bar{A}_{all}^{pub}$ | 18.30 | 17.22 | 17.13 | 17.41 | 19.27 | 20.47 | 30.05 | 22.79 |
| $\bar{A}[F]_{all}^{pub}$ | 1.08 | 1.13 | 1.20 | 1.33 | 1.52 | 1.67 | 1.95 | 1.47 |
| $P\bar{A}[F]_{all}^{pub}$ | 22.5% | 22.9% | 23.2% | 24.1% | 25.2% | 25.9% | 26.3% | 24.6% |
| $\overline{PA}[F]_{all}^{pub}$ | 20.1% | 20.5% | 21.0% | 21.7% | 22.4% | 23.2% | 23.9% | 21.9% |
| Relative female representation | | | | | | | | |
| $RP\bar{A}[F]^{pub}$ | 0.90 | 0.90 | 0.92 | 0.91 | 0.90 | 0.92 | 0.93 | 0.91 |
| $R\overline{PA}[F]^{pub}$ | 0.94 | 0.94 | 0.95 | 0.95 | 0.96 | 0.97 | 0.97 | 0.96 |

For affiliated authors, the full-period female share is 22.4% under the author-weighted measure ($P\bar{A}[F]_{aff}^{pub}$) and 21.1% under the equal-publication-weighted measure ($\overline{PA[F]}_{aff}^{pub}$). The slightly higher author-weighted value indicates that publications with larger EPFL-affiliated author teams tend to have a somewhat higher female share than the average publication.

Among all contributing authors, the corresponding benchmark shares are 24.6% under author weighting ($P\bar{A}[F]_{all}^{pub}$) and 21.9% under equal publication weighting ($\overline{PA[F]}_{all}^{pub}$). The larger difference between these two values indicates that female representation increases more clearly with team size in the wider author teams than among EPFL-affiliated authors.

The relative female representation ratios compare the affiliated-author share with the all-author benchmark under the same weighting scheme. Over the full period, $RP\bar{A}[F]^{pub}$ is 0.91 under author weighting, while $R\overline{PA}[F]^{pub}$ is 0.96 under equal publication weighting. The ratios below one indicate that female representation among EPFL-affiliated authors is lower than among all contributing authors on the same publications. Since the ratio is lower under author weighting, the relative gap in female representation is more pronounced in larger author teams than in the average publication.

Over time, the average number of female affiliated authors per publication increases from 0.4 in 2014–2017 to 0.5 in 2020–2023. The affiliated-author female share rises from 20.2% to 24.5% under the author-weighted measure ($P\bar{A}[F]_{aff}^{pub}$) and from 19% to 23.3% under the equal-publication-weighted measure ($\overline{PA[F]}_{aff}^{pub}$). Among all contributing authors, the average number of female authors per publication increases from 1.08 to 1.95. Their female share rises from 22.5% to 26.3% under author weighting

($P\overline{A}[F]^{pub}_{all}$) and from 20.1% to 23.9% under equal publication weighting ($\overline{PA[F]}^{pub}_{all}$). The corresponding relative representation ratios change from 0.9 to 0.93 for $RP\overline{A}[F]^{pub}$ and from 0.94 to 0.97 for $R\overline{PA}[F]^{pub}$. This indicates that female participation increased at the publication level for both EPFL-affiliated authors and the wider author teams on the same publications, with the relative representation ratios narrowing only slightly. Female EPFL-affiliated authors therefore remain less represented than the same-publication benchmark throughout the period.

This publication-level perspective qualifies the near-parity result obtained from the preceding unique-author perspective in two important ways. While the unique-author ratio is close to parity (RPA[F]: 0.98), the corresponding publication-level ratios are lower, at 0.91 under author weighting and 0.96 when publications are weighted equally. First, the lower publication-level ratios indicate that female representation among EPFL-affiliated authors is weaker when measured across repeated authorship appearances than when measured across unique authors. Substantively, this means that female affiliated authors account for a smaller share of publication-level authorship instances than of the affiliated author population, relative to the gender composition of all authors contributing to the same publications. One plausible mechanism is that repeated authorship appearances are more strongly shaped by senior, permanent or group-leading positions, which may remain more male-dominated; however, the bibliometric data alone cannot distinguish this from other factors related to career stage, field composition, collaboration structure or publication intensity. Second, the lower author-weighted ratio compared with the equal-publication-weighted ratio suggests that this relative underrepresentation is more pronounced in larger author teams. This indicates that larger teams raise the female share among all contributing authors more strongly than among EPFL-affiliated authors (i.e., female EPFL-affiliated authors are relatively less represented in larger authorship teams than in the average publication, compared with the same-publication benchmark).

# References

[1] Intermediate evaluation of the ETH Domain. https://ethrat.ch/en/news-publications/intermediate-evaluation-of-the-eth-domain/, 2023. Accessed 16 July 2026.

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# Annexes

# A. Data collection

This document describes the methodology underpinning the bibliometric analysis pipeline developed for the ETH Domain institutions. It covers the full journey from raw bibliographic data to analytically ready indicators, with particular attention to openness, transparency, traceability and reproducibility.

This report reflects the data snapshot available from OpenAlex from 20 August 2025 (snapshot downloaded and frozen on 3 September 2025).

The full pipeline, including all scripts and instructions to reproduce the analysis, is publicly available at [public code repository (forthcoming)]. For the sake of brevity, this document focuses on the methodological choices and overall design rather than on the technical details of each processing step; the interested reader is referred to the repository for the complete implementation.

The analysis follows a funnel-shaped data flow. Institutional libraries provide lists of scientific publications (*input*). These are matched against OpenAlex to identify corresponding records (*matched*). Finally, a set of scoping criteria determines which publications are retained for the actual bibliometric analysis (*in scope*). Each of these stages is described in detail below.

## A.1. OpenAlex snapshot

The bibliometric analysis relies on OpenAlex, an open and freely accessible index of scholarly works, authors, institutions, sources and their relationships. At the time of extraction, OpenAlex contained approximately 269 million unique publication records, making it one of the most comprehensive open bibliographic databases available. OpenAlex is maintained by OurResearch and is described in detail in its official documentation (`https://docs.openalex.org`).

Rather than querying the OpenAlex API on a record-by-record basis, the first step of our pipeline is to download a complete copy of the OpenAlex dataset to a local infrastructure using a dedicated download script. A snapshot captures the state of the entire database at a single point in time: All records, relationships and metadata are frozen together, ensuring that no changes introduced by ongoing updates to the live database affect the analysis. This is particularly important for citation counts, which evolve over time. Working from a dated, self-contained copy of the data also means that the exact same analyses can be rerun at any time with identical results and that anyone with access to the same snapshot can independently verify every step of the analysis.

## A.2. Matching publications

Each institutional library provided a list of scientific publications to be considered for the bibliometric analysis. In principle, it would have been possible to derive publication lists entirely from OpenAlex by relying on its institutional affiliation metadata. However, given the relatively recent maturity of OpenAlex's affiliation data, we considered it more robust to start from the lists curated by the institutions themselves. This ensures that the analysis faithfully reflects each institution's own understanding of its research output.

The consequence of this approach is that each publication in the institutional lists must be *matched*,

that is, linked to its corresponding record in the OpenAlex snapshot. This is not always straightforward: Identifiers may be missing or formatted differently, titles may contain typographical variations, and author names may be represented in various ways across systems. After preparing both the OpenAlex snapshot and the institutional lists into standardised, efficient formats, the matching pipeline discovers candidate matches using multiple complementary signals. *Direct identifier matching* uses OpenAlex identifiers or DOIs for exact lookups and resolves the majority of cases. *Fuzzy title matching* compares normalised titles using a locality-sensitive hashing technique, detecting near-identical titles even in the presence of minor typographical differences or encoding variations. *Author overlap* compares normalised author lists, while *journal matching* and *institutional affiliation* serve as supporting signals that increase confidence. These signals are combined and only candidate pairs supported by sufficient converging evidence are retained.

Among the retained candidates, continuous similarity scores are computed for titles, author lists and journal names and a priority score is calculated that accounts for the strength of each signal and whether publication years agree. Retracted publications are excluded at this stage. For each query publication, candidates are then ranked by priority score following a cascading logic: A DOI match takes precedence and a title-based match is accepted only when corroborated by agreement on journal, authors or the uniqueness of the title in the corpus.

At the end of this process, every line in every institutional list has been assigned one of two statuses: matched to a specific OpenAlex record or flagged as unmatched. Publications for which no candidate passes the rules are explicitly recorded as *not found*, ensuring full traceability of all matching decisions. The matched publications then proceed to the scoping stage.

## A.3. Scoping

Not all publications recorded in OpenAlex are suitable for citation-based bibliometric analysis. OpenAlex is intentionally broad in its coverage, including preprints, editorials, book reviews, errata, non-English publications and other items that do not exhibit the citation behaviour expected of peer-reviewed research articles. Including such items would distort citation-based indicators and compromise comparability across institutions and disciplines.

We therefore apply a set of scoping criteria to identify *in-scope publications*, those appropriate for reliable citation analysis. The same criteria are applied uniformly to all six institutions, ensuring a consistent and fair selection regardless of differences in repository practices or metadata completeness. Only publications that satisfy all criteria are retained. Importantly, citation counts are also restricted to the in-scope population: A citation is counted only if the citing work itself is also in scope.

#### Inclusion criteria

Our scoping approach is inspired by the methodology developed by the Centre for Science and Technology Studies (CWTS) for identifying core sources and in-scope publications, notably used in the CWTS Leiden Ranking. A publication from the OpenAlex snapshot is considered *in scope* if and only if it satisfies all of the following conditions:

- **English language.** The publication is written in English, reflecting the standard practice in large-scale bibliometric studies.
- **Has authorship.** The publication has at least one identified author affiliated with at least one institution.
- **Has a research subfield.** The publication is assigned to a research subfield within the OpenAlex

topic classification. This is necessary for field-normalised indicators, which compare a publication's citation performance against that of other works in the same subfield and year.

- **Has referenced works.** The publication includes at least one reference to another work. Publications without references are unlikely to participate meaningfully in citation networks.
- **Published in a core source or conference proceedings.** The publication appears in a source classified as *core* by OpenAlex (corresponding to established international scientific journals) or is identified as a conference proceedings article.
- **Not retracted.** The publication has not been formally withdrawn from the scientific record.
- **Not paratext.** The publication is not classified as paratext (front matter, tables of contents, indexes and similar non-substantive items).

#### Extension to conference proceedings

One important difference with the standard CWTS methodology is the inclusion of conference proceedings. The CWTS definition of core sources focuses primarily on scientific journals. However, for the ETH Domain institutions, where engineering, computer science and other fields with strong conference traditions are well represented, excluding proceedings would systematically undercount a significant portion of the research output. We therefore extend the scoping definition to include conference proceedings articles, provided they satisfy all other inclusion criteria.

#### Scale of scoping

At the time of the snapshot extraction, the OpenAlex database contained approximately 270 million publication records. Of these, roughly 57 million satisfy all inclusion criteria and are considered in scope. This in-scope population forms the universe against which all bibliometric indicators are computed, including citation counts ($MCS_{full}$), field-normalised metrics ($MNCS_{frac}$) and percentile-based indicators such as the proportion of publications in the top 10% most-cited ($PP[top\,10\%]_{frac}$).

### A.4. Bibliometric data warehouse

#### Purpose and design philosophy

All data, transformations and indicators produced by the pipeline are consolidated into a single analytical data warehouse. The primary goals are:

- **Traceability.** For any publication provided as input by an institution, the warehouse records unambiguously whether the publication was not found in OpenAlex, found, but excluded from the analysis (along with the specific reasons for exclusion) or included. This level of traceability represents a significant step toward full analytical transparency.
- **Separation of concerns.** The warehouse is organised into clearly delineated layers, each serving a distinct purpose. This makes the data easier to understand, audit and extend.
- **Reproducibility.** Because the warehouse is built from the OpenAlex snapshot and the institutional input lists through a deterministic, scripted pipeline, it can be reconstructed from scratch at any time. The same pipeline could also be applied to produce equivalent analyses for a different set of institutions.

#### Logical structure

The warehouse comprises 19 tables organised into four logical layers. The full schema is documented separately.

**Core layer.** This layer contains the fundamental bibliometric entities derived from the OpenAlex snapshot: publication records, author identities, institutional affiliations, citation relationships and the hierarchical topic classification (from topics to subfields, fields and domains). With one exception, this layer contains data only for in-scope publications.

The exception is a dedicated status table that records the inclusion status of *every* publication in the OpenAlex snapshot, all 270 million records. For each record, this table stores whether it meets each of the seven scoping criteria individually, as well as the overall inclusion flag. This enables detailed diagnostics (for example, understanding why a particular publication was excluded) and provides a complete audit trail.

**Augmented layer.** This layer enriches the core data with information not available in OpenAlex itself, drawn from two external sources. Approximately 548,000 patent–paper pairs from the Reliance on Science dataset (`https://relianceonscience.org`) link scientific publications to citing patents, but these links are currently retained for traceability, and potential future analyses and are not used in the indicators reported in this study. For approximately 493,000 unique author first names, a probabilistic gender inference from the Genderize.io service (`https://genderize.io`) enables the gender-disaggregated analyses included in this report; results with low confidence or ambiguous input (such as initials only) are classified as unknown rather than forced into a binary category. By placing these enrichments in a separate layer, we make it transparent which data come from OpenAlex and which has been added externally.

**Analytical layer (marts).** This layer contains precomputed tables that support efficient indicator construction. Computing field-normalised metrics requires aggregating citation data across large cohorts defined by publication year, research subfield and publication type. The analytical layer materialises key statistics at the cohort level: the number of publications, mean and median citation counts, standard deviations and percentile thresholds. For each in-scope publication, the layer also stores pre-joined metrics including its raw citation count, field-normalised citation score ($MNCS_{frac}$), percentile rank and whether it falls in the top 10% of its cohort ($PP[top\,10\%]_{frac}$). Fractional and full counting weights at the institutional and country level are also precomputed. By materialising these indicators once, we ensure controlled, auditable results available for reporting without the risk of inconsistent recalculation.

**Reports layer.** This layer is specific to the institutional reporting context. It records the *membership* of each publication in each institution's report, preserving for each institution the original line number from the input file, the matched OpenAlex identifier (if found) and an indication of duplicate entries. This makes it possible to trace any row in the final report back to the exact line in the input file provided by the library. If a publication could not be matched, the corresponding record remains present with an empty identifier. This design guarantees that no input is silently lost or ignored. The layer also stores per-publication author-level gender assignments specific to each report, enabling institution-level gender analyses.

### Efficiency

Despite the scale of the data, the complete data warehouse occupies approximately 76 GB. This compactness is achieved through two main design choices. First, the warehouse retains from the OpenAlex snapshot only the fields strictly required for the bibliometric analysis, discarding large text-heavy fields such as titles and abstracts. Since every publication in the warehouse carries its OpenAlex identifier, any additional metadata can always be retrieved on demand from OpenAlex itself. Second, OpenAlex identifiers are stored as numerical integers rather than full URL strings, which further reduces

storage requirements and accelerates lookups throughout the pipeline.

#### Transparency and reusability

The data warehouse is designed so that the same analyses could be recomputed for a different set of institutions or a different time period, given the appropriate input lists and an OpenAlex snapshot. All analytical choices, from scoping criteria to indicator definitions, are encoded in the warehouse structure and its construction pipeline rather than in ad hoc scripts or manual processes. Doing so ensures a high level of transparency and reproducibility and positions the pipeline as a reusable analytical infrastructure rather than a one-off reporting exercise.

# B. Methodological notes

## B.1. Subfield classification comparison

The bibliometric analyses in this report use the OpenAlex subfield classification. This differs from the Web of Science (WoS) categories used in earlier bibliometric reports, because the two systems follow different classification principles, levels of granularity and naming conventions. In particular, WoS categories are primarily assigned at the journal or source level and then inherited by publications, whereas OpenAlex assigns topics at the level of individual works and maps these topics to subfields, fields and domains. The OpenAlex-based results therefore reflect a publication-level topical classification rather than a direct continuation of journal-based WoS subject categories.

Table 17 provides an empirical comparison of the two systems. It reports frequent observed mappings between the first listed WoS category and the OpenAlex subfield assigned to matched publications. The table is not a formal concordance and does not validate individual assignments. Rather, it provides a practical reference for relating familiar WoS category names to the OpenAlex subfields used in this report.

The comparison shows that some mappings are close, whereas others are less direct. Subfield-level results should therefore be read within the OpenAlex-based analytical framework and not as direct continuations of WoS-based subject categories.

**Table 17.** Most frequent observed mappings between Web of Science primary categories and OpenAlex subfields in the matched publication set. The table reports the 20 most frequent pairs after assigning each publication to its first listed Web of Science category and to the OpenAlex subfield used in this report.

| WoS Primary Category | OpenAlex Subfield | N |
|---|---|---|
| Engineering, Electrical & Electronic | Electrical and Electronic Engineering | 137 |
| Chemistry, Physical | Atomic and Molecular Physics, and Optics | 47 |
| Computer Science, Artificial Intelligence | Computer Vision and Pattern Recognition | 44 |
| Engineering, Electrical & Electronic | Biomedical Engineering | 43 |
| Optics | Atomic and Molecular Physics, and Optics | 39 |
| Chemistry, Multidisciplinary | Organic Chemistry | 38 |
| Automation & Control Systems | Control and Systems Engineering | 36 |
| Chemistry, Physical | Renewable Energy, Sustainability and the Environment | 32 |
| Chemistry, Multidisciplinary | Materials Chemistry | 31 |
| Computer Science, Artificial Intelligence | Artificial Intelligence | 30 |
| Physics, Fluids & Plasmas | Nuclear and High Energy Physics | 29 |
| Construction & Building Technology | Building and Construction | 29 |
| Thermodynamics | Mechanical Engineering | 28 |
| Computer Science, Information Systems | Artificial Intelligence | 28 |
| Optics | Biomedical Engineering | 27 |
| Computer Science, Information Systems | Computer Networks and Communications | 27 |
| Construction & Building Technology | Civil and Structural Engineering | 27 |
| Engineering, Electrical & Electronic | Condensed Matter Physics | 26 |
| Engineering, Civil | Civil and Structural Engineering | 25 |
| Engineering, Electrical & Electronic | Atomic and Molecular Physics, and Optics | 25 |

## B.2. Construction of geographic overrepresentation of citations

The geographic citation maps make the spatial structure of incoming citations visible. They translate citation links into a map of citing-institution locations and add a relative layer showing where citation activity is stronger than expected given those locations' general citation volume. The maps do not locate individual readers or citation decisions. They show the institutional locations attached to publications that cite the analysed publication set, using geocoded institutional affiliation data from OpenAlex and ROR.

To incorporate citing-institution locations, first citation links are identified from citing publications to the focal publication set in the reporting period. For the institutional map, this set consists of publications affiliated with the respective institution or with the ETH Domain. If a citing publication cites several focal publications, each distinct cited publication contributes one link. The links are assigned to all institutions affiliated with the citing publication. The map uses full counting, so links are attributed to each listed institution rather than divided between them. The institution-level counts are matched to ROR coordinates and used as weighted input. Records without coordinates are excluded and post-processing removes ETH-internal sub-units recorded as separate institutions.

A central issue is that scientific institutions are not evenly distributed geographically. Some regions contain many institutions located close to each other, while other areas contain only one or a few identifiable institutions. The map should therefore not be read as if each visible area corresponded to a single institution. Nearby institutions contribute jointly to the same spatial surface and institutions sharing similar coordinates reinforce the same location. An area lights up when there is sufficient weighted citation activity in and around that area, whether from one large citing institution or several smaller nearby ones.

The contours are fitted by placing the geocoded institution weights on a projected map grid and smoothing the grid into a continuous surface (Robinson projection; nominal world grid of 440 x 240 cells; European inset grid of 360 x 210 cells; Gaussian smoothing set to 1.25 for the world map and 2.60 for the inset, with isotropic adjustment for different cell dimensions). Institutions at the same or neighbouring coordinates are therefore aggregated through the weighted grid before the contours are drawn (implemented as a weighted two-dimensional histogram, followed by Gaussian smoothing). This avoids a dense point map and makes broader spatial concentrations easier to see. At the same time, contours are not institutional or administrative boundaries; they summarise nearby weighted locations after smoothing. The world map and the European inset use separate grids and smoothing settings, so the inset clarifies the dense European pattern rather than merely enlarging the global surface.

The filled coloured contours in the main citation map describe absolute incoming citation intensity, weighted by each institution's share of incoming citation links to the focal publication set. This layer naturally gives more visual weight to large research systems and to areas with many active institutions. Its colour scale is deliberately not used to emphasise every low-volume difference. Very low smoothed values are masked (world map: below 0.0005 of the upper reference value; inset: below 0.02 of the local upper reference value), the contour breaks preserve contrast across the visible range (geometric spacing; 18 levels for the world map, 9 for the inset) and the upper end of the scale is capped so that extreme peaks share the highest colour band instead of stretching the scale for all other regions (upper reference: 99th percentile of positive smoothed values, with values above it drawn in the top colour band). This focuses attention on moderate to high citation intensity and reduces overinterpretation of weak background variation.

The black contours add a different interpretation and are drawn on top of this absolute-intensity surface. They compare the observed incoming citation share from each location with the share expected from the same location in a global benchmark of outgoing citation activity. The focal incoming citation share and

the corresponding global benchmark citation share are each smoothed on the same grid before division. The benchmark asks how prominent the citing institutions in a location are in the global citation system during the same period. This is necessary because a map based only on citation volume would largely reproduce the geography of large scientific systems. The relative measure is a lift: the focal incoming citation share divided by the corresponding benchmark share.

The lift contours are calculated from two smoothed surfaces: one for citations to the focal institution and one for the global benchmark. The black contours mark lifts of approximately 1.25-, 1.5- and 2-fold above the baseline (implemented as log2 thresholds of 0.322, 0.585 and 1.000). To avoid the relative map being driven by isolated small institutions with high ratios, but little citation mass, lift is evaluated only where the combined smoothed focal and benchmark surface exceeds a minimum mass threshold (institutional map: 0.0005 of the maximum combined smoothed mass; a small stabilising offset of 1% of the maximum smoothed reference value is used in the ratio). Areas below that threshold are not contoured. Small institutions can still contribute, but only as part of a spatial pattern that passes both the mass condition and the lift threshold. The black lines should therefore be read as above-baseline spatial concentrations, not as point markers for individual high-lift institutions.

The open-access geographic map uses the same general spatial logic, but answers a different question. It is produced for the ETH Domain as a whole and used as a common reference in the institute reports. The comparison separates ETH Domain publications into open-access publications and closed or bronze publications. For each citing institution, links are counted separately for the two groups and normalised by all incoming citation links to each access type. The mapped value is $\log_2$(open incoming citation share / closed incoming citation share). Positive values indicate locations relatively more represented among citations to open-access publications, while negative values indicate locations where closed or bronze publications account for a comparatively larger share. This is a relative citation-origin contrast, not a comparison of absolute citation counts. The smoothing and grid construction are the same as for the citation map (Robinson projection; nominal 440 x 240 world grid, 360 x 210 inset grid; Gaussian smoothing set to 1.25 and 2.60), but the plotted surface is the open-versus-closed log ratio rather than an absolute citation-intensity layer. Low-mass areas are also masked before the ratio is shown (threshold: 0.00001 of the maximum combined open-plus-closed smoothed mass; colour range based on the 98th percentile of valid absolute log-ratio values).

Both maps should therefore be interpreted as spatial summaries of citation origins. They distinguish broad citation volume from relative concentration and make visible areas where citation patterns differ from the global benchmark or from the open-access reference distribution. They do not rank individual institutions, identify causal effects of geography or access status or measure readership directly.

# C. Supplementary results

## C.1. Extended bibliometric indicators by publication type

Table 18 provides an extended set of indicators for EPFL for 2014–2023, with results reported for journal publications, proceedings publications and all publications.

**Table 18.** Extended bibliometric statistics of EPFL for 2014–2023, covering output, open access, collaboration, citation impact and gender-related author indicators, with results reported for journals, proceedings and all publications.

| Indicator | Journals | Proceedings | Overall |
|---|---|---|---|
| | Output | | |
| $P_{full}$ | 37,452 | 6,320 | 43,772 |
| $P_{frac}$ | 16,643 | 3,915 | 20,558 |
| Cov | | | 87.73% |
| | Open Access | | |
| $P[OA]_{full}$ | 24,109 | 2,889 | 26,998 |
| $PP[OA]_{full}$ | 64% | 46% | 62% |
| | Collaborations and Partners | | |
| $PP[collab]_{full}$ | 81% | 66% | 79% |
| $PP[int\ collab]_{full}$ | 70% | 54% | 68% |
| $PP[industry\ collab]_{full}$ | 8% | 10% | 8% |
| InterDisc | 0.35 | 0.30 | 0.34 |
| | Citation Impact | | |
| $TCS_{full}$ | 1,547,393 | 75,186 | 1,622,579 |
| $MCS_{full}$ | 41.32 | 11.90 | 37.07 |
| $C[self\ cits]_{full}$ | 247,422 | 11,696 | 259,118 |
| $PC[self\ cits]_{full}$ | 16% | 16% | 16% |
| $P[uncited]_{full}$ | 1,412 | 1,264 | 2,676 |
| $PP[uncited]_{full}$ | 4% | 21% | 6% |
| $P[top\ 10\%]_{frac}$ | 3,107 | 796 | 3,902 |
| $PP[top\ 10\%]_{frac}$ | 18.7% | 21.7% | 19.3% |
| $MNCS_{frac}$ | 1.69 | 1.84 | 1.72 |
| | Author and Gender Distribution | | |
| $A_{aff}$ | 17,074 | 4,593 | 18,727 |
| $A[F]_{aff}$ | 4,192 | 843 | 4,471 |
| $A[M]_{aff}$ | 9,754 | 3,156 | 10,886 |
| $PA[F]_{aff}$ | 30% | 21% | 29% |
| $PA[F]_{all}$ | 30% | 20% | 30% |
| RPA[F] | 0.99 | 1.05 | 0.98 |

*Note:* Authors with unknown gender represent 3,370 (18.0%) among EPFL-affiliated authors and 80,182 (45.0%) among all authors contributing contributing to EPFL publications.

## C.2. Full-counted citation impact trend

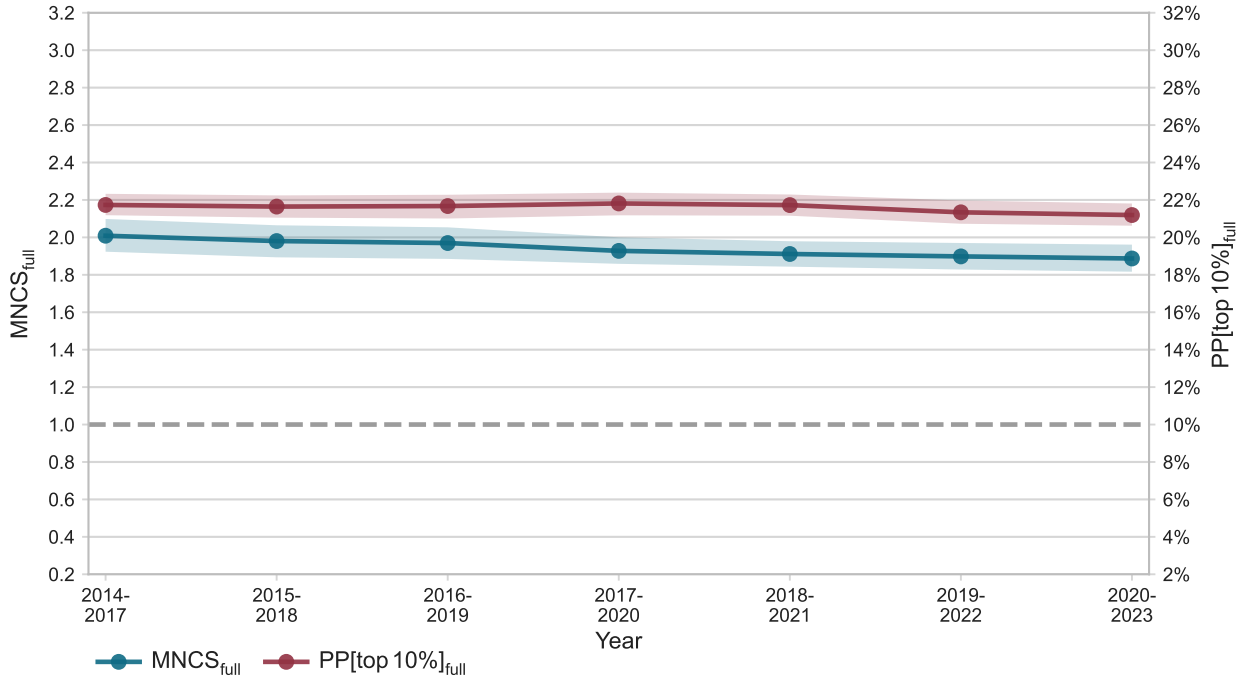



**Figure 13.** Development of EPFL's full-counted impact indicators over time. Shown are field-normalised citation impact ($MNCS_{full}$) and the full-counted output share belonging to the global top 10% of most-cited publications ($PP[top\,10\%]_{full}$) together with bootstrapped 95% stability intervals. The dashed reference line represents the corresponding world averages ($MNCS_{full}$ = 1 and $PP[top\,10\%]_{full}$ = 10%).

## C.3. Overrepresentation of citations

Table 19 lists citing institutions with high relative citation presence among the institutions with the largest absolute incoming citation volumes. Table 20 lists regional clusters in which citations to EPFL are overrepresented. Both tables support the interpretation of Figure 4.

**Table 19.** Citing institutions with the highest relative presence among the top 200 institutions by absolute incoming citation volume. The table reports institution name, country code, citation volume rank, incoming citation count, the proportion of EPFL's incoming citations in which the citing institution is present, the corresponding proportion among all citations worldwide, and the resulting relative presence ratio, which indicates over- or underrepresentation of that institution among EPFL's incoming citations.

| Rank | Institution | CC | Citation Rank (abs.) | $C[inst]^{by\ inst}_{full}$ | $PC[inst]^{by\ inst}_{full}$ | $PC[world]^{by\ inst}_{full}$ | $RPC^{by\ inst}_{full}$ |
|---|---|---|---|---|---|---|---|
| 1 | EPFL | CH | 1 | 127,834 | 7.88% | 0.20% | 39.90 |
| 2 | Laboratoire Physique Nucléaire Hautes Énergies | FR | 175 | 6,149 | 0.38% | 0.01% | 30.78 |
| 3 | Inst Nazionale Fisica Nucleare, Sezione Bologna | IT | 120 | 7,263 | 0.45% | 0.02% | 18.88 |
| 4 | Inst Nazionale Fisica Nucleare, Laboratori Nazionali Frascati | IT | 145 | 6,587 | 0.41% | 0.02% | 16.63 |
| 5 | Inst Nazionale Fisica Nucleare, Sezione Pisa | IT | 166 | 6,260 | 0.39% | 0.02% | 16.10 |
| 6 | Inst Nuclr Phys, Polish Acad Sci | PL | 173 | 6,153 | 0.38% | 0.02% | 16.06 |
| 7 | Inst Natl Physique Nucléaire Physique Particules | FR | 69 | 9,316 | 0.57% | 0.04% | 15.72 |
| 8 | CERN | CH | 38 | 12,073 | 0.74% | 0.05% | 15.30 |
| 9 | Inst Nazionale Fisica Nucleare, Sezione Roma I | IT | 123 | 7,142 | 0.44% | 0.03% | 15.21 |
| 10 | Inst d'Astrophysique Paris | FR | 189 | 5,866 | 0.36% | 0.03% | 12.33 |
| 11 | Inst Rech sur Lois Fondamentales l'Univers | FR | 196 | 5,539 | 0.34% | 0.03% | 11.82 |
| 12 | Inst Astrofísica Canarias | ES | 168 | 6,214 | 0.38% | 0.04% | 10.56 |
| 13 | Fermi Natl Accelerator Lab | US | 192 | 5,713 | 0.35% | 0.04% | 9.72 |
| 14 | PSI | CH | 84 | 8,594 | 0.53% | 0.06% | 9.61 |
| 15 | Rutherford Appleton Lab | GB | 164 | 6,302 | 0.39% | 0.04% | 9.60 |
| 16 | Empa | CH | 188 | 5,868 | 0.36% | 0.04% | 8.79 |
| 17 | SLAC Natl Accelerator Lab | US | 154 | 6,428 | 0.40% | 0.05% | 7.41 |
| 18 | Délégation Paris 7 | FR | 191 | 5,813 | 0.36% | 0.05% | 7.07 |
| 19 | TU Dortmund Univ | DE | 177 | 6,109 | 0.38% | 0.06% | 6.72 |
| 20 | Inst High Energy Phys | CN | 140 | 6,670 | 0.41% | 0.06% | 6.34 |

**Table 20.** Overrepresented geo-citation clusters from the contour-based regional interpretation subset. The table reports cluster region labels, dominant countries, observed and expected surface shares, percentage-point differences, lift ratios and the number of anchor institutions per cluster.

| Cluster region | Dominant countries | Observed share | Expected share | Delta (pp) | Weighted lift | Peak lift | Anchor inst. n | Anchor institutions (top 10) |
|---|---|---|---|---|---|---|---|---|
| France Switzerland Italy cluster | France Switzerland Italy | 20.689% | 11.429% | 9.261 pp | 1.78x | 3.07x | 54 | CERN<br>CEA<br>Univ Geneva<br>ETH Zurich<br>Inst Natl Physique Nucléaire Physique Particules<br>Aix-Marseille Université<br>Univ Lausanne<br>Université Paris-Saclay<br>Sorbonne Université<br>Inst Nazionale Fisica Nucleare, Sezione Bologna |
| USA cluster | USA | 2.338% | 1.506% | 0.832 pp | 1.48x | 1.61x | 11 | Lawrence Berkeley Natl Lab<br>Stanford Univ<br>Univ Calif Berkeley<br>Caltech<br>SLAC Natl Accelerator Lab<br>Univ Califor, Santa Barbara<br>Univ Califor, San Diego<br>Univ Califor, Davis<br>Univ Califor, Los Angeles<br>Univ Califor, Irvine |
| Russia cluster | Russia | 0.973% | 0.457% | 0.515 pp | 1.83x | 1.98x | 1 | Lomonosov Moscow St Univ |
| Japan cluster | Japan | 0.776% | 0.583% | 0.193 pp | 1.30x | 1.38x | 1 | Univ Tokyo |

*Note:* Clusters are derived from connected components of contour-grid cells with lift >=1.25x (log2 lift >=0.322), after excluding low-mass cells where target+benchmark surface mass is <=0.05% of the global maximum. Components use 8-neighbor connectivity and a minimum size of 10 cells (one cell approx. 77km x 77km). Shown are clusters with observed surface share >=0.50%. Anchor institutions are the geocoded institutions in the top-200 by absolute incoming citation volume (excluding the focal institution), assigned to a cluster by grid-cell overlap (or nearest cluster cell within radius 2). Dominant countries are shown by anchor citation volume (top 3), while displayed anchor institutions (top 10) are selected by lowest combined rank of anchor citation volume and overrepresentation (lift ratio). Clusters are ordered by observed surface share (ties by peak lift, then anchor citation volume) and only the top 12 clusters are shown.

## C.4. Aggregation by fields

In this section, we present the field-level counterparts of the previously shown subfield-based Figures 5, 10 and 12. The methodology and indicators remain the same, but results are aggregated from the 252 subfields to the 26 fields to provide a more general overview. In addition, we apply a stricter

inclusion threshold: Only fields with at least a 2% contribution to fractional output share are shown (instead of 1%).

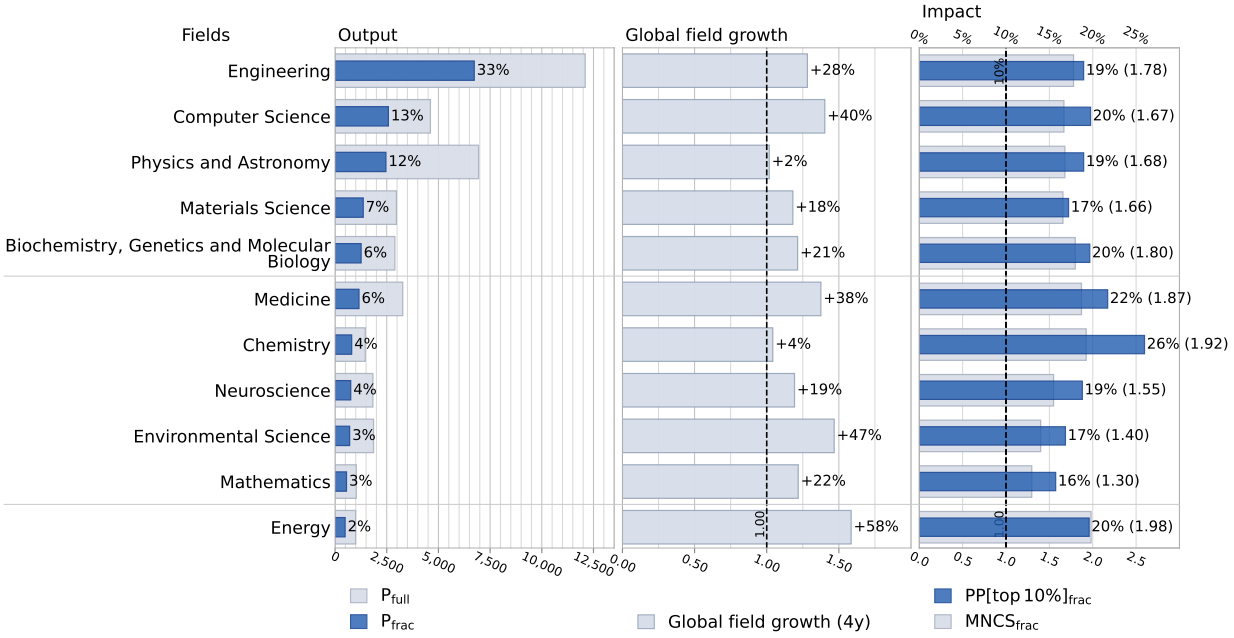



**Figure 14.** Field research profile of EPFL by output, global field growth and citation impact. Shown are fields that account for at least 2% of EPFL's total fractional publication output and have at least 50 fractionally counted publications. **Left** (Output): Publication output in full counting (light blue) and fractional counting (dark blue), with labels indicating each field's share of total fractional output. **Middle** (Global field growth): Global field growth is measured as the ratio of mean annual global publication output in 2020–2023 relative to 2014–2019, using full counting; the dashed line marks a ratio of 1, indicating no absolute growth. **Right** (Impact): Citation impact is reported as the share of publications among the top 10% most-cited publications ($PP[top\ 10\%]_{frac}$, dark blue, in %), followed by the mean normalised citation score ($MNCS_{frac}$, light blue, in brackets); dashed lines indicate the global benchmark values (10% and 1).

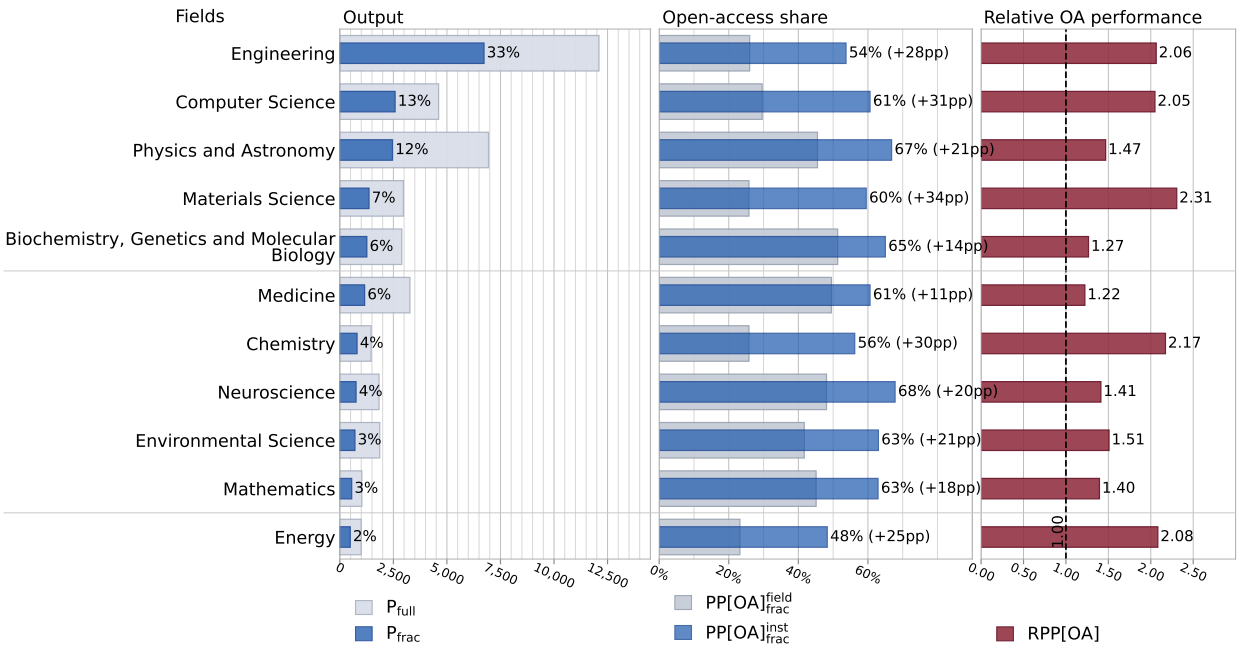



**Figure 15.** Open-access profile of EPFL across fields. Shown are fields that account for at least 2% of EPFL's total fractional publication output and have at least 50 fractionally counted publications. **Left** (Output): Publication output by field in fractional (dark blue and percentages of total) and full (light blue) counting. **Middle** (Open-access share): Share of EPFL's open-access publications and corresponding field averages. Numbers indicate percentages and percentage differences to field average (in percentage points). **Right** (Relative OA performance): Ratio of EPFL open-access share to the respective field averages. Dashed line indicates parity with the field average.

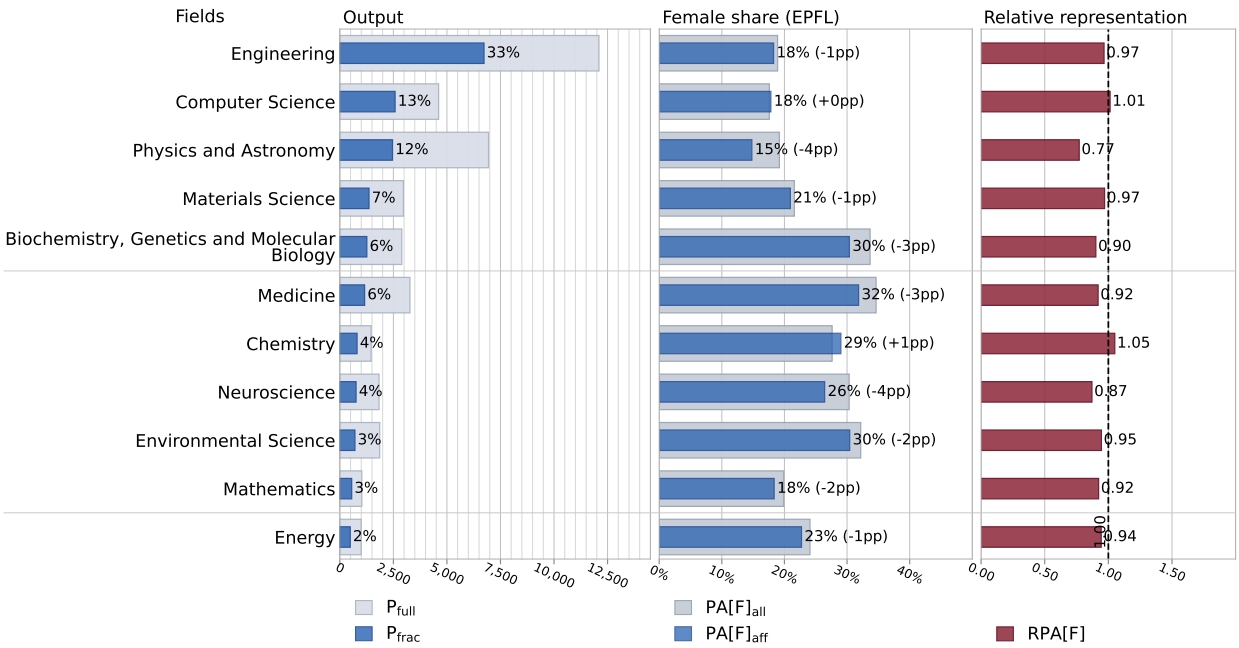



**Figure 16.** Female affiliated-author representation across fields for EPFL. Shown are fields that account for at least 2% of EPFL's total fractional publication output and have at least 50 fractionally counted publications. Each row represents one field, ordered by fractional publication output in the left panel. **Left** (Output): Dark blue bars show fractional publication counts, while light blue bars show full counts attributed to EPFL in each field. **Middle** (Female share): Dark blue bars show the female share among EPFL-affiliated authors; light blue background bars show the female share among all authors on the same publications. Labels indicate the affiliated-author female share, followed by its percentage-point difference from the all-author female share. **Right** (Relative representation): Bars show the ratio between the affiliated-author female share and the all-author female share. Values below 1 indicate lower female representation among EPFL-affiliated authors than among all contributing authors on the same publications; values above 1 indicate higher representation among EPFL-affiliated authors.

## C.5. Open-access trend by access category

**Table 21.** Open-access trend profile of EPFL by access category. Shown are number of publications and shares of total output for closed access including bronze, open access overall and the open-access categories gold including diamond, green and hybrid. The indicators cover high-impact output, normalised citation impact and international collaboration share across four-year moving intervals and for the full evaluation period.

| Access type | Indicator | 2014-2017 | 2015-2018 | 2016-2019 | 2017-2020 | 2018-2021 | 2019-2022 | 2020-2023 | Overall |
|---|---|---|---|---|---|---|---|---|---|
| Closed access incl. bronze | $P_{full}$ | 8,308 | 8,157 | 7,770 | 7,103 | 6,324 | 5,422 | 4,782 | 16,774 |
| | Share of total | 49% | 47% | 44% | 39% | 35% | 30% | 27% | 38% |
| | $PP[top\ 10\%]_{frac}$ | 16.6% | 16.6% | 16.6% | 16.4% | 16.2% | 15.6% | 15.5% | 16.3% |
| | $MNCS_{frac}$ | 1.67 | 1.59 | 1.57 | 1.54 | 1.48 | 1.53 | 1.51 | 1.59 |
| | $PP[int\ collab]_{full}$ | 64% | 65% | 66% | 66% | 68% | 69% | 71% | 67% |
| Open access | $P_{full}$ | 8,579 | 9,115 | 9,862 | 10,893 | 11,826 | 12,573 | 13,071 | 26,998 |
| | Share of total | 51% | 53% | 56% | 61% | 65% | 70% | 73% | 62% |
| | $PP[top\ 10\%]_{frac}$ | 22.7% | 23.0% | 22.8% | 22.2% | 21.5% | 20.0% | 19.4% | 21.3% |
| | $MNCS_{frac}$ | 1.91 | 1.98 | 1.97 | 1.88 | 1.83 | 1.68 | 1.63 | 1.80 |
| | $PP[int\ collab]_{full}$ | 67% | 68% | 68% | 68% | 69% | 69% | 69% | 68% |
| Gold incl. diamond | $P_{full}$ | 2,342 | 2,658 | 2,966 | 3,435 | 3,825 | 4,170 | 4,539 | 8,507 |
| | Share of total | 14% | 15% | 17% | 19% | 21% | 23% | 25% | 19% |
| | $PP[top\ 10\%]_{frac}$ | 18.0% | 19.5% | 19.0% | 18.1% | 18.0% | 16.6% | 16.5% | 17.6% |
| | $MNCS_{frac}$ | 1.47 | 1.54 | 1.51 | 1.42 | 1.42 | 1.38 | 1.40 | 1.43 |
| | $PP[int\ collab]_{full}$ | 68% | 68% | 69% | 69% | 70% | 71% | 72% | 70% |
| Green | $P_{full}$ | 5,113 | 5,158 | 5,386 | 5,484 | 5,303 | 4,907 | 4,134 | 12,074 |
| | Share of total | 30% | 30% | 31% | 30% | 29% | 27% | 23% | 28% |
| | $PP[top\ 10\%]_{frac}$ | 24.1% | 24.0% | 24.5% | 24.2% | 23.3% | 22.5% | 22.1% | 23.8% |
| | $MNCS_{frac}$ | 2.07 | 2.15 | 2.19 | 2.13 | 2.09 | 1.96 | 1.89 | 2.07 |
| | $PP[int\ collab]_{full}$ | 66% | 66% | 66% | 67% | 67% | 69% | 69% | 67% |
| Hybrid | $P_{full}$ | 1,124 | 1,299 | 1,510 | 1,974 | 2,698 | 3,496 | 4,398 | 6,417 |
| | Share of total | 7% | 8% | 9% | 11% | 15% | 19% | 25% | 15% |
| | $PP[top\ 10\%]_{frac}$ | 25.4% | 25.8% | 22.9% | 22.8% | 22.3% | 19.9% | 19.3% | 20.8% |
| | $MNCS_{frac}$ | 1.99 | 2.16 | 2.01 | 1.89 | 1.82 | 1.61 | 1.58 | 1.71 |
| | $PP[int\ collab]_{full}$ | 75% | 74% | 74% | 71% | 68% | 66% | 66% | 69% |